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149-D
Refrigerant receiver
QUEP

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\149-DRefrigerant receiver7 Per.pvdb

Date of Analysis : Sep 16,2026 12:03pm

PV Elite 28 U1, March 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2400.000 x 2400.000 mm. ID	
Vessel Design Length, Tangent to Tangent	7600.00 mm.
Specified Datum Line Distance	0.00 mm.
Internal Design Temperature	130 °C
Internal Design Pressure	22.100 bars
External Design Temperature	0 °C
Maximum Allowable Working Pressure	23 bars
Shop Test Pressure	33.998 bars
Field Test Pressure	28.730 bars
Required Minimum Design Metal Temperature	-33.0 °C
Warmest Computed Minimum Design Metal Temperature	-33.0 °C
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Head	SA-516 70	...	...	K02700	+IT
Flange	SA-266 2	...	...	K03506	+IT
Nozzle	SA-266 2	...	...	K03506	+IT
Nozzle	SA-105	...	...	K03504	+IT
Nozzle Flg	SA-105	...	...	K03504	+IT
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Left Dish	22.254	0.00	23.000	3.0000	No	No
S1	22.254	0.00	24.000	3.0000	N/A	No
S2	22.254	0.00	24.000	3.0000	N/A	No
S3	22.254	0.00	24.000	3.0000	N/A	No
S4	22.254	0.00	24.000	3.0000	N/A	No
Right Dish	22.254	0.00	23.000	3.0000	No	No
40" Scrub Flg	22.132	0.00	26.000	3.0000	N/A	No

Liquid Level: 2400.00 mm. Dens.: 0.001 kg./cm3 Sp. Gr.: 0.590

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
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Ellipse	50.0	50.0	28.0	23.0	22.4	...	1.00	1.00
Cylinder	3050.0	3000.0	...	24.0	22.6	...	1.00	1.00
Cylinder	4400.0	1350.0	...	24.0	22.6	...	1.00	1.00
Cylinder	4550.0	150.0	...	24.0	22.6	...	1.00	1.00
Cylinder	7550.0	3000.0	...	24.0	22.6	...	1.00	1.00
Ellipse	7600.0	50.0	28.0	23.0	22.4	...	1.00	1.00
Body Flg	5300.0	0.0	110.0	110.0	101.7	83.5	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Lift Lug	WRC-107/537	0.709	n/a	Passed
Lift Lug	WRC-107/537	0.898	n/a	Passed
N1_12"	WRC-297	0.920	Shell	Passed
N2_6"	WRC-297	0.920	Shell	Passed
N3_6"	WRC-297	0.928	Shell	Passed
N9_3"	WRC-297	0.928	Shell	Passed
N7_4"	WRC-297	0.920	Shell	Passed
D1_3"	WRC-297	0.928	Shell	Passed
N4_10"	WRC-297	0.928	Shell	Passed
Scrubber_40"	WRC-297	0.964	Shell	Passed

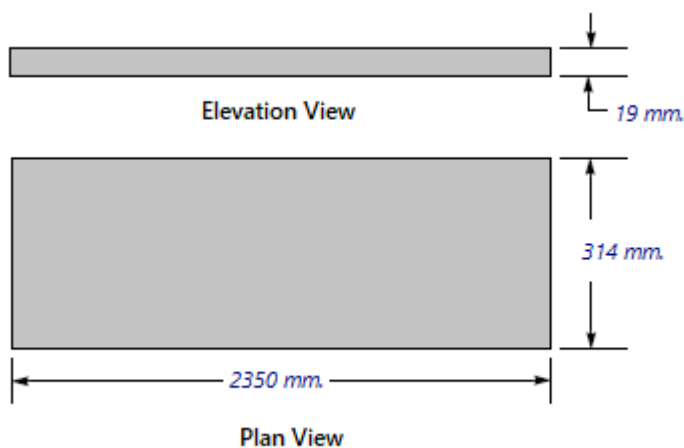
Foundation/Support Summary

Saddle Parameters:

Saddle Width	228.600 mm.
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1905.000 mm.
Wear Pad Width	315.000 mm.
Wear Pad Thickness	24.000 mm.
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	157.000 mm.

Baseplate Length	2350.000 mm.
Baseplate Thickness	19.050 mm.
Baseplate Width	314.000 mm.
Number of Ribs (including outside ribs)	5
Rib Thickness	15.875 mm.
Web Thickness	15.875 mm.
Height of Center Web	637.000 mm.
Number of Bolts in Baseplate	8

Baseplate Sketch



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Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	54456.34	Kgf
Maximum Transverse Saddle Shear Load	3019.02	Kgf
Maximum Longitudinal Saddle Shear Load	2579.03	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	58802.36	Kgf
Maximum Transverse Saddle Shear Load	819.31	Kgf
Maximum Longitudinal Saddle Shear Load	360.35	Kgf

Weights:

Fabricated - Bare w/o removable internals	24643.2	kg.
Shop Test - Fabricated + water (full)	87150.2	kg.
Shipping - Fab. + rem. intls.+ shipping app.	24643.2	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	28857.7	kg.
Empty - Fab. + intls. + details + wghts.	28857.7	kg.
Operating - Empty + operating liquid (no ca)	71533.9	kg.
Field Test - Empty weight + test medium (full)	91364.8	kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
M1_24 inch	23	...	...	...		OK	Passed
K4_2"	24	...	...	OK	7.80	OK	No Calc[*]
K3_2"	24	...	...	OK	7.80	OK	No Calc[*]
K5_2"	24	...	...	OK	7.80	OK	No Calc[*]
K6_2"	24	...	...	OK	7.80	OK	No Calc[*]
N1_12"	23	...	...	OK	11.33	OK	Passed
N2_6"	24	...	...	OK	9.22	OK	Passed
N3_6"	24	...	...	OK	9.22	OK	Passed
N9_3"	24	...	...	OK	7.80	OK	Passed
N7_4"	24	...	...	OK	8.02	OK	Passed
D1_3"	24	...	...	OK	7.80	OK	Passed
N4_10"	24	...	...	OK	11.10	OK	Passed
Scrubber_40"	24	...	...	OK	11.33	OK	Passed
M2_24"	23	...	...	...		OK	Passed
K1_3"	23	...	...	OK	8.73	OK	Passed
K2_3"	23	...	...	OK	8.73	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 23 Nozzle : M224"
 Minimum MAWP Shells/Flanges : 23 Element : Right Dish
 Minimum MAPnc Shells/Flanges : 26 Element : Right Dish

Computed Vessel M.A.W.P. : 23 bars

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
10	M1_24 inch	0.000	0.000	1183.200
20	K4_2"	350.000	180.000	126.400
20	K3_2"	350.000	0.000	126.400
20	K5_2"	1200.000	0.000	126.400
20	K6_2"	1500.000	0.000	126.400
20	N1_12"	2400.000	0.000	573.600
20	N2_6"	800.000	0.000	276.800
20	N3_6"	800.000	180.000	276.800
20	N9_3"	1350.000	180.000	124.400
50	N7_4"	7200.000	0.000	175.200
50	D1_3"	4850.000	180.000	124.400
50	N4_10"	6500.000	180.000	472.000
50	Scrubber_40"	5300.000	0.000	2044.000
60	M2_24"	0.000	0.000	1183.200
60	K1_3"	0.000	0.000	164.400
60	K2_3"	0.000	180.000	164.400

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction



If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
K4_2"	2.000 in	Actual	LWN	3.323	16.800	...	...	280.37	300
K3_2"	2.000 in	Actual	LWN	3.323	16.800	...	...	502.39	300
K5_2"	2.000 in	Actual	LWN	3.323	16.800	...	...	502.39	300
K6_2"	2.000 in	Actual	LWN	3.323	16.800	...	...	502.39	300
N9_3"	3.000 in	Actual	WNF	3.000	10.000	...	...	226.05	600
D1_3"	3.000 in	Actual	WNF	3.000	10.000	...	...	226.05	600
K1_3"	3.000 in	Actual	LWN	4.622	20.600	...	...	215.61	300
K2_3"	3.000 in	LWN	LWN	4.622	20.600	...	...	268.16	300
N7_4"	4.000 in	Actual	WNF	4.000	10.000	...	...	474.28	300
N2_6"	6.000 in	Actual	WNF	6.000	10.000	...	...	475.62	300
N3_6"	6.000 in	Actual	WNF	6.000	10.000	...	...	375.45	300
N4_10"	10.000 in	Actual	WNF	10.000	12.000	...	...	375.45	300
N1_12"	12.000 in	Actual	WNF	12.000	12.000	...	...	482.92	300
M1_24 inch	24.000 in	Actual	WNF	24.000	12.000	...	...	294.43	300
M2_24"	24.000 in	Actual	WNF	24.000	12.000	...	...	294.43	300
Scrubber_40"	40.000 in	Actual	None	41.260	16.000	...	...	499.14	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K4_2"	SA-105	24.000	10.000	...	...	...
K3_2"	SA-105	24.000	10.000	...	...	...
K5_2"	SA-105	24.000	10.000	...	...	...
K6_2"	SA-105	24.000	10.000	...	...	...
N9_3"	SA-266 2	24.000	10.000	...	...	...
D1_3"	SA-266 2	24.000	10.000	...	...	...
K1_3"	SA-105	23.000	10.000	...	...	...
K2_3"	SA-105	23.000	10.000	...	...	...
N7_4"	SA-266 2	24.000	10.000	...	...	...
N2_6"	SA-266 2	24.000	10.000	...	...	...
N3_6"	SA-266 2	24.000	10.000	...	...	...
N4_10"	SA-266 2	24.000	10.000	...	...	...
N1_12"	SA-266 2	24.000	10.000	...	...	...
M1_24 inch	SA-266 2	23.000	10.000	...	...	...
M2_24"	SA-266 2	23.000	10.000	...	...	...
Scrubber_40	SA-266 2	24.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
K4_2"	350.000	180.0	206.38	50.00	S1
K3_2"	350.000	0.0	477.65	0.00	S1



K5_2"	1200.000	0.0	477.65	0.00	S1
K6_2"	1500.000	0.0	477.65	0.00	S1
N9_3"	1350.000	180.0	201.45	0.00	S1
D1_3"	4850.000	180.0	201.45	0.00	S4
K1_3"	...	0.0	171.55	0.00	Right Dish
K2_3"	...	180.0	200.02	0.00	Right Dish
N7_4"	7200.000	0.0	449.20	0.00	S4
N2_6"	800.000	0.0	449.20	0.00	S1
N3_6"	800.000	180.0	201.45	150.00	S1
N4_10"	6500.000	180.0	201.45	150.00	S4
N1_12"	2400.000	0.0	449.20	0.00	S1
M1_24 inch	...	0.0	250.00	0.00	Left Dish
M2_24"	...	0.0	250.00	0.00	Right Dish
Scrubber_40"	5300.000	0.0	300.00	0.00	S4

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 28.0mm. THK X 2400.0mm. ID X 50.0mm.	SA-516 70
2	CYLINDER: 24.0mm. THK X 2400.0mm. ID X 3000.0mm.	SA-516 70
1	CYLINDER: 24.0mm. THK X 2400.0mm. ID X 1350.0mm.	SA-516 70
1	WELD NECK FLANGE: 110.0mm. THK X 1215.0mm. OD	SA-266 2
2	SADDLE: 228.6mm X 120 DEG	SA-516 70
1	PLATFORM: 1000mm X 2448mm WIDE	
1	LADDER: 4100.0mm	...
4	LIFTING LUGS:	SA-516 70
1	INSULATION: 150mm X 60mm THK	Cold Insulation
1	INSULATION: 3000mm X 60mm THK	Cold Insulation
1	INSULATION: 650mm X 60mm THK	Cold Insulation
1	INSULATION: 134mm X 60mm THK	Cold Insulation
2	CLASS 300 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	GASKET: 1082.0mm. OD X 1032.0mm. ID	...
32	BODY FLANGE BOLTS: 31.8mm. DIA	SA-193 B7
32	NUTS FOR BODY FLANGE BOLTS: 31.8mm. DIA	...
1	CYLINDRICAL SEGMENT 24.0mm. THK X 3048.0mm. X 150.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 24.0mm. THK X 4491.7mm. X 150.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 24.0mm. THK X 0.0mm. X 150.0mm.	SA-516 70
1	NAMEPLATE	...

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MDMT Summary:

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Minimum Design Metal Temperature Results Summary (°C):

Description Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
40" Scrub Flg[11]	!	-33	-52		0.657	16.000	1.00	Yes
Left Dish [10]	!	-33	-33		0.994	23.000	1.00	Yes
Left Dish [7]	!	-33	-44		0.805	28.000	1.00	Yes
S1 [8]	!	-33	-35		0.958	24.000	1.00	Yes
S2 [8]	!	-33	-35		0.958	24.000	1.00	Yes
S3 [8]	!	-33	-35		0.958	24.000	1.00	Yes
S4 [8]	!	-33	-35		0.958	24.000	1.00	Yes
Right Dish [10]	!	-33	-33		0.994	23.000	1.00	Yes
Right Dish [7]	!	-33	-44		0.805	28.000	1.00	Yes
M124 inch [1]	!	-33	-40		0.870	23.000	1.00	Yes
Nozzle Flg [5]	!	-18	-61					
K42" [1]	!	-33	-104		0.034	16.800	1.00	Yes
Nozzle Flg [5]	B	-18	-48					
K32" [1]	B	-13	-104	-29	0.033	16.800	1.00	Yes
Nozzle Flg [5]	B	-18	-48					
K52" [1]	B	-13	-104	-29	0.033	16.800	1.00	Yes
Nozzle Flg [5]	B	-18	-48					
K62" [1]	B	-13	-104	-29	0.033	16.800	1.00	Yes
Nozzle Flg [5]	B	-18	-48					
N112" [1]	!	-33	-37		0.927	24.000	1.00	Yes
Nozzle Flg [5]	!	-18	-61					
N26" [1]	!	-33	-37		0.927	24.000	1.00	Yes
Nozzle Flg [5]	!	-18	-61					
N36" [1]	!	-33	-37		0.933	24.000	1.00	Yes
Nozzle Flg [5]	B	-18	-48					
N93" [1]	!	-33	-37		0.933	24.000	1.00	Yes
Nozzle Flg [5]	B	-18	-104					
N74" [1]	!	-33	-37		0.927	24.000	1.00	Yes
Nozzle Flg [5]	B	-18	-48					
D13" [1]	!	-33	-37		0.933	24.000	1.00	Yes
Nozzle Flg [5]	B	-18	-104					
N410" [1]	!	-33	-37		0.933	24.000	1.00	Yes
Nozzle Flg [5]	B	-18	-48					
Scrubber40" [1]	!	-33	-76		0.433	48.000	1.00	Yes
M224" [1]	!	-33	-40		0.870	23.000	1.00	Yes
Nozzle Flg [5]	B	-18	-48					
K13" [1]	B	-7	-104	-29	0.038	20.600	1.00	Yes
Nozzle Flg [5]	B	-18	-48					
K23" [1]	!	-33	-35		0.968	23.000	1.00	Yes
Nozzle Flg [5]	B	-18	-48					
Nozl. Bolting[22]		-48			1.000	64.000	1.00	Yes
Bolting [21]		-48						

Warmest MDMT: -7 -33

Required Minimum Design Metal Temperature -33.0 °C

Warmest Computed Minimum Design Metal Temperature -33.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3

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MDMT Summary:

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- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (barsUnicode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

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Design Internal Pressure (for Hydrotest)	22.1	bars
Design Internal Temperature	130.0	°C
Projection of Nozzle from Vessel Top	500	mm.
Projection of Nozzle from Vessel Bottom	350	mm.
Minimum Design Metal Temperature	-33.0	°C
Type of Construction	Welded	
Special Service	Lethal	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS

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Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW
Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.000
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	9695 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.
Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.100
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	C
Component Elevation Ratio z/h	1.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No

Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	Left Dish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	2400 mm.
Element Thickness	23 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	28 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	22.1 bars
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Allowable Stress, Ambient	138 N./mm2
Allowable Stress, Operating	138 N./mm2
Allowable Stress, Hydrotest	234 N./mm2
Material Density	0.00775 kg./cm3
P Number Thickness	32.004 mm.
Yield Stress, Operating	234.4 N./mm2
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	10
Detail Type	Liquid
Detail ID	Liquid 10
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	2400 mm.
Liquid Density	0.0005891 kg./cm3

Element From Node	10
Detail Type	Nozzle
Detail ID	M124 inch
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	10
Detail Type	Weight
Detail ID	Davit24"
Dist. from "FROM" Node / Offset dist	-600 mm.
Miscellaneous Weight	50 Kg
Offset from Element Centerline	300 mm.

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Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	S1
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	2400 mm.
Element Thickness	24 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	0 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	22.1 bars
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	1520 mm.
Width of Saddle	228.6 mm.
Height of Saddle at Bottom	1905 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	315 mm.
Thickness of Wear Plate	24 mm.
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	20
Detail Type	Platform
Detail ID	PLAT:[TOP]
Dist. from "FROM" Node / Offset dist	0 mm.
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	76000 cm2
Platform Weight	3316.1 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm2
Platform Width	2448 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	85 mm.
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	4100 mm.
Unit Weight of Ladder	0.037 Kg/mm.
Platform Length (top head platform)	7600 mm.

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	2400 mm.
Liquid Density	0.0005891 kg/cm3



Element From Node 20
 Detail Type Nozzle
 Detail ID K42"
 Dist. from "FROM" Node / Offset dist 300 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 180.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-105 +IT

Element From Node 20
 Detail Type Nozzle
 Detail ID K32"
 Dist. from "FROM" Node / Offset dist 300 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-105

Element From Node 20
 Detail Type Nozzle
 Detail ID K52"
 Dist. from "FROM" Node / Offset dist 1150 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-105

Element From Node 20
 Detail Type Nozzle
 Detail ID K62"
 Dist. from "FROM" Node / Offset dist 1450 mm.
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-105

Element From Node 20
 Detail Type Nozzle
 Detail ID N112"
 Dist. from "FROM" Node / Offset dist 2350 mm.
 Nozzle Diameter 12 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-266 2 +IT

Element From Node 20



Detail Type	Nozzle
Detail ID	N26"
Dist. from "FROM" Node / Offset dist	750 mm.
Nozzle Diameter	6 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	20
Detail Type	Nozzle
Detail ID	N36"
Dist. from "FROM" Node / Offset dist	750 mm.
Nozzle Diameter	6 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	20
Detail Type	Nozzle
Detail ID	N93"
Dist. from "FROM" Node / Offset dist	1300 mm.
Nozzle Diameter	3 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	20
Detail Type	Weight
Detail ID	N1Feed PP
Dist. from "FROM" Node / Offset dist	2350 mm.
Miscellaneous Weight	166 Kgf
Offset from Element Centerline	1200 mm.

Element From Node	20
Detail Type	Weight
Detail ID	N1Support
Dist. from "FROM" Node / Offset dist	2350 mm.
Miscellaneous Weight	5 Kgf
Offset from Element Centerline	1200 mm.

Element From Node	20
Detail Type	Weight
Detail ID	N2Hood BF
Dist. from "FROM" Node / Offset dist	700 mm.
Miscellaneous Weight	10 Kgf
Offset from Element Centerline	1200 mm.

Element From Node	20
Detail Type	Weight
Detail ID	N3VB
Dist. from "FROM" Node / Offset dist	700 mm.
Miscellaneous Weight	20 Kgf
Offset from Element Centerline	1050 mm.

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Element#	3/7
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	S2
Distance "FROM" to "TO"	1350 mm.
Inside Diameter	2400 mm.
Element Thickness	24 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	0 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	22.1 bars
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	2400 mm.
Liquid Density	0.0005891 kg./cm3
Element From Node	30
Detail Type	Weight
Detail ID	Clip Wt
Dist. from "FROM" Node / Offset dist	550 mm.
Miscellaneous Weight	142 Kg
Offset from Element Centerline	0 mm.

Element#	4/7
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	S3
Distance "FROM" to "TO"	150 mm.
Inside Diameter	2400 mm.
Element Thickness	24 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	0 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	22.1 bars
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	40
Detail Type	Liquid
Detail ID	Liquid 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	2400 mm.
Liquid Density	0.0005891 kg./cm3
Element From Node	40

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Detail Type	Insulation
Detail ID	Cold Insulation
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	150 mm.
Thickness of Insulation	60 mm.
Density	0.00032 kg./cm3

Element#	5/7
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	S4
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	2400 mm.
Element Thickness	24 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	0 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	22.1 bars
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Saddle
Detail ID	Right Saddle
Dist. from "FROM" Node / Offset dist	1480 mm.
Width of Saddle	228.6 mm.
Height of Saddle at Bottom	1905 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	315 mm.
Thickness of Wear Plate	24 mm.
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.30000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	2400 mm.
Liquid Density	0.0005891 kg./cm3

Element From Node	50
Detail Type	Insulation
Detail ID	Cold Insulation
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	3000 mm.
Thickness of Insulation	60 mm.
Density	0.00032 kg./cm3

Element From Node	50
Detail Type	Nozzle
Detail ID	N74"
Dist. from "FROM" Node / Offset dist	2650 mm.

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Nozzle Diameter	4 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	50
Detail Type	Nozzle
Detail ID	D13"
Dist. from "FROM" Node / Offset dist	300 mm.
Nozzle Diameter	3 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	50
Detail Type	Nozzle
Detail ID	N410"
Dist. from "FROM" Node / Offset dist	1950 mm.
Nozzle Diameter	10 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2 +IT

Element From Node	50
Detail Type	Nozzle
Detail ID	Scrubber40"
Dist. from "FROM" Node / Offset dist	750 mm.
Nozzle Diameter	40 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	None
Nozzle Matl	SA-266 2 +IT

Element From Node	50
Detail Type	Weight
Detail ID	N5VB
Dist. from "FROM" Node / Offset dist	1950 mm.
Miscellaneous Weight	40 Kgf
Offset from Element Centerline	1050 mm.

Element From Node	50
Detail Type	Weight
Detail ID	Baffle PL
Dist. from "FROM" Node / Offset dist	1050 mm.
Miscellaneous Weight	570.01 Kgf
Offset from Element Centerline	0 mm.

Element From Node	50
Detail Type	Weight
Detail ID	Scrub Fab wt
Dist. from "FROM" Node / Offset dist	750 mm.
Miscellaneous Weight	4044 Kgf

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Offset from Element Centerline 3400 mm.

Element From Node 50

Detail Type Weight

Detail ID Scrub Ope wt

Dist. from "FROM" Node / Offset dist 750 mm.

Miscellaneous Weight 20201 Kg

Offset from Element Centerline 3400 mm.

Element From Node 50

Detail Type Weight

Detail ID Scrub Test wt

Dist. from "FROM" Node / Offset dist 750 mm.

Miscellaneous Weight 24417 Kg

Offset from Element Centerline 3400 mm.

Element# 6/7

Element From Node 60

Element To Node 70

Element Type Elliptical

Description Right Dish

Distance "FROM" to "TO" 50 mm.

Inside Diameter 2400 mm.

Element Thickness 23 mm.

Internal Corrosion Allowance 3 mm.

Nominal Thickness 28 mm.

External Corrosion Allowance 0 mm.

Design Internal Pressure 22.1 bars

Design Temperature Internal Pressure 130 °C

Design External Pressure 0 bars

Design Temperature External Pressure 0 °C

Effective Diameter Multiplier 1.2

Material Name SA-516 70 +IT

Efficiency, Longitudinal Seam 1.0

Efficiency, Circumferential Seam 1.0

Elliptical Head Factor 2.0

Weld is pre-Heated No

Element From Node 60

Detail Type Liquid

Detail ID Liquid 60

Dist. from "FROM" Node / Offset dist 0 mm.

Height/Length of Liquid 2400 mm.

Liquid Density 0.0005891 kg./cm3

Element From Node 60

Detail Type Insulation

Detail ID Cold Insulation

Dist. from "FROM" Node / Offset dist 0 mm.

Height/Length of Insulation 650 mm.

Thickness of Insulation 60 mm.

Density 0.00032 kg./cm3

Element From Node 60

Detail Type Nozzle

Detail ID M224"

Dist. from "FROM" Node / Offset dist 0 mm.

Nozzle Diameter 24 in.

Nozzle Schedule None

Nozzle Class 300

Layout Angle 0.0

Blind Flange (Y/N) Y

Weight of Nozzle (Used if > 0) 0 Kg

Grade of Attached Flange GR 1.1

Nozzle Matl SA-266 2 +IT

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Element From Node	60
Detail Type	Nozzle
Detail ID	K13"
Dist. from "FROM" Node / Offset dist	670 mm.
Nozzle Diameter	3 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	60
Detail Type	Nozzle
Detail ID	K23"
Dist. from "FROM" Node / Offset dist	960 mm.
Nozzle Diameter	3 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	25.622 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	60
Detail Type	Weight
Detail ID	M2Davit24"
Dist. from "FROM" Node / Offset dist	600 mm.
Miscellaneous Weight	50 Kgf
Offset from Element Centerline	300 mm.

Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Flange
Description	40" Scrub Flg
Distance "FROM" to "TO"	134 mm.
Flange Inside Diameter	1000 mm.
Element Thickness	110 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	110 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	22.1 bars
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2 +IT
Allowable Stress, Ambient	138 N./mm2
Allowable Stress, Operating	138 N./mm2
Allowable Stress, Hydrotest	225 N./mm2
Material Density	0.00775 kg./cm3
P Number Thickness	32.004 mm.
Yield Stress, Operating	222.2 N./mm2
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	70

Detail Type	Liquid
Detail ID	Liquid 70
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1000 mm.
Liquid Density	0.0005891 kg./cm3
Element From Node	70
Detail Type	Insulation
Detail ID	Cold insulation
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	134 mm.
Thickness of Insulation	60 mm.
Density	0.00032 kg./cm3

XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Left Dish		50	...	50	...
S1		3050	...	3000	...
S2		4400	...	1350	...
S3		4550	...	150	...
S4		7550	...	3000	...
Right Dish		7600	...	50	...
40" Scrub Flg		5300	...	...	134

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

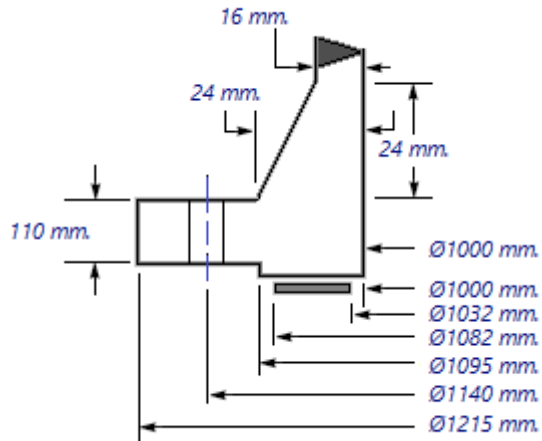
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Flange Input Data Values

Description: Scr 40" Body :

40" Scrub Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	22.13	bars
Design Temperature		130	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	1000.000	mm.
Flange Outside Diameter	A	1215.000	mm.
Flange Thickness	t	110.0000	mm.
Thickness of Hub at Small End	go	16.0000	mm.
Thickness of Hub at Large End	gl	24.0000	mm.
Length of Hub	h	24.0000	mm.
Flange Material		SA-266 2 +IT	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1140.000	mm.
Nominal Bolt Diameter	a	31.7500	mm.
Type of Threads	TEMA Thread Series		
Number of Bolts		32	
Flange Face Outside Diameter	Fod	1095.000	mm.
Flange Face Inside Diameter	Fid	1000.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1082.000	mm.
Gasket Inside Diameter	Gi	1032.000	mm.
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm.
Raised face depth	Fd	0.0000	mm.
Axial Force on Flange		19750.35	Kgf
Bending Moment on Flange		2600.09	Kg-m.



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (22.13 \cdot (1000.0/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 22.13) + C_a$$

$$= 11.1461 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.0) / (503.0 + 0.6 \cdot 13.0)$$

$$= 35.119 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (500.0 + 0.6 \cdot 16.0)$$

$$= 43.325 \text{ bars}$$

Corroded Flange ID,	$B_{cor} = B + 2 \cdot c_i$	1006.000 mm.
Corroded Large Hub,	$g1_{cor} = g1 - c_i$	21.000 mm.
Corroded Small Hub,	$g0_{cor} = g0 - c_i$	13.000 mm.
Code R Dimension,	$R = ((C - B_{cor})/2) - g1_{cor}$	46.000 mm.
Gasket Contact Width,	$N = (G_o - G_i)/2$	25.000 mm.
Basic Gasket Width,	$b_o = N/2$	12.500 mm.
Effective Gasket Width,	$b = C_b \cdot \sqrt{b_o}$	8.909 mm.
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	1064.181 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1064.1814^2 \cdot 22.132$$

$$= 200745.250 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.9093 \cdot \pi \cdot 1064.1814 \cdot 3.0 \cdot 22.13$$

$$= 40335.086 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1006.0^2 \cdot 22.1323$$

$$= 179394.844 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 200745 - 179395$$

$$= 21350.412 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max(W_{o,1} + W_{o,2} + H'p, 0) \\ &= \max(200745 + 40335 + 0, 0) \\ &= 241080.344 \text{ Kgf} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\ &= 68.95 \cdot 8.9093 \cdot \pi \cdot 1064.181 + 0.0 \cdot 0.0 \cdot 0.0 \\ &= 209417.625 \text{ Kgf} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a) \\ &= \max((241080 + 19750 + 4 \cdot 2600 / 1064.18) / 172, 209418 / 172) \\ &= 154.291 \text{ cm}^2 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned} &= 2a + 6t / (m + 0.5) \\ &= 2 \cdot 31.75 + 6 \cdot 110.0 / (3.0 + 0.5) \\ &= 252.071 \text{ mm.} \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned} &= C \cdot \sin(\pi / n) \\ &= 1140.0 \cdot \sin(\pi / 32) \\ &= 111.740 \text{ mm.} \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$\begin{aligned} &= \max(\sqrt{ B_s / (2a + t) }, 1) \\ &= \max(\sqrt{ 111.74 / (2 \cdot 31.75 + 110.0) }, 1) \\ &= 1.0000 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	154.291	191.793	
Radial Distance between Hub and Bolts:	44.450	46.000	
Radial Distance between Bolts and Edge:	31.750	37.500	
Circ. Spacing between the Bolts:	71.450	111.740	252.071

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned} &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\ &= 191.793 \cdot 172.0 / (68.95 \cdot \pi (1082.0 + 1032.0)) \\ &= 7.204 \text{ mm.} \end{aligned}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned} &= S_a (A_m + A_b) / 2 \\ &= 172.0 (154.2914 + 191.7932) / 2 \\ &= 303489.84 \text{ Kgf} \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned} &= W_o - H \\ &= 241080 - 200745 \\ &= 40335.09 \text{ Kgf} \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned} &= (C - G) / 2 \\ &= (1140.0 - 1064.1814) / 2 \\ &= 37.9093 \text{ mm.} \end{aligned}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B)/2 + hg)$$

$$= 0.5((1140.0 - 1006.0)/2 + 37.9093)$$

$$= 52.4546 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1)/2$$

$$= (1140.0 - 1006.0 - 21.0)/2$$

$$= 56.5000 \text{ mm.}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 104.5, Bb = 110.0, Cc = 24.0 \text{ and } Ddg = 17.0 \text{ mm.}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 104.5 \cdot 110.0^3 \cdot (1/3 - 0.210 \cdot (110.0/104.5) \cdot (1 - 1/12 \cdot (110.0/104.5)^4))$$

$$= 1876.273 \text{ cm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 24.0 \cdot 17.0^3 \cdot (1/3 - 0.105 \cdot (17.0/24.0) \cdot (1 - 1/192 \cdot (17.0/24.0)^4))$$

$$= 3.055 \text{ cm}^4$$

Flange Inertia Term [lp,term1]:

$$= Kab + Kcd = 1144.973 + 1.864$$

$$= 1879.328 \text{ cm}^4$$

Flange Inertia Term [lp,term2]:

$$= Ar \cdot t^3 (1/3 - 0.21 (t/AR) (1 - 1/12 (t/AR)^4))$$

$$= 104.5 \cdot 110.0^3 (1/3 - 0.21 (110.0/104.5) (1 - 1/12 (110.0/104.5)^4))$$

$$= 1876.273 \text{ cm}^4$$

Flange Inertia [lp]:

$$= \max(lp,term1, lp,term2)$$

$$= \max(1879.3, 1876.3)$$

$$= 1879.3 \text{ cm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 3.38 \cdot 13.0^2 \cdot 114.359 \cdot 1006.0 / 0.392$$

$$= 1464.350 \text{ cm}^4$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	179395.	56.5000	1.0000	10136.
Face Pressure, Mt	21350.	52.4546	1.0000	1120.
Gasket Load, Mg	40335.	37.9093	1.0000	1529.
Gasket Seating, Matm	303490.	37.9093	1.0000	11505.
Moment Load, Moe	...	...	...	1499.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me [I / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$$

$$= 4 \cdot 2600 [1464.3 / (0.384 \cdot 1879.3 + 1464.3)] \cdot (56.5 / (1140.0 - 2 \cdot 56.5)) +$$

$$19750 \cdot 56.5$$

$$= 1499.199 \text{ Kg-m.}$$

Total Moment for Operation, Mo 14284. Kg-m.

Total Moment for Gasket seating, Mg 11505. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 114.359 mm.

Hub Ratio, h/h0 = HL / H0 0.210

Thickness Ratio, $g1/g0 = (g1Cor/goCor)$ 1.615

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.897

Factor V per 4.16.5 0.392

Factor f per 4.16.5 1.696

Stress factor from Table 4.16.4 [K]:

= A / B

= 1215.0/1006.0

= 1.208

Stress factor from Table 4.16.4 [Y]:

= $(0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$

= $(0.66845 + (5.7169(1.208^2 \cdot \log_{10}(1.208))) / (1.208^2 - 1)) / (1.208 - 1)$

= 10.392

Stress factor from Table 4.16.4 [T]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$

= $(1.208^2(1 + 8.55246 \cdot \log_{10}(1.208)) - 1) / ((1.0472 + 1.9448 \cdot 1.208^2) \cdot (1.208 - 1))$

= 1.836

Stress factor from Table 4.16.4 [U]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$

= $(1.208^2(1 + 8.55246 \cdot \log_{10}(1.208)) - 1) / (1.36136(1.208^2 - 1) \cdot (1.208 - 1))$

= 11.419

Stress factor from Table 4.16.4 [Z]:

= $(K^2 + 1) / (K^2 - 1)$

= $(1.208^2 + 1) / (1.208^2 - 1)$

= 5.360

Stress factor from Table 4.16.4 for integral flanges [d]:

= $U \cdot g0^2 \cdot h0 / V$

= $11.419 \cdot 13.0^2 \cdot 114.359 / 0.392$

= 562711.625 mm.³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0

= 0.897/114.359

= 0.008 1/mm.

Stress factor from Table 4.16.4 [L]:

= $(t \cdot e + 1) / T + t^3/d$

= $(110.0 \cdot 0.008 + 1) / 1.836 + 110.0^3/562712$

= 3.380

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= $(f \cdot Mop / Bcor) / (L \cdot g1^2)$

= $(1.6964 \cdot 14284 / 1006.0) / (3.3798 \cdot 21.0^2)$

= 158.48 N./mm²

Longitudinal Hub Stress, Seating [SHa]:

= $(f \cdot Matm / Bcor) / (L \cdot g1^2)$

= $(1.6964 \cdot 11505 / 1006.0) / (3.3798 \cdot 21.0^2)$

= 127.65 N./mm²

Radial Flange Stress, Operating [SRo]:

= $(\beta \cdot Mop / Bcor) / (L \cdot t^2)$

= $(2.1499 \cdot 14284 / 1006.0) / (3.3798 \cdot 110.0^2)$

= 7.32 N./mm²

Radial Flange Stress, Seating [SRa]:

= $(\beta \cdot Matm / Bcor) / (L \cdot t^2)$

= $(2.1499 \cdot 11505 / 1006.0) / (3.3798 \cdot 110.0^2)$



$$= 5.90 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$$

$$= (10.3916 \cdot 14284 / (110.0^2 \cdot 1006.0)) - 5.3604 \cdot 7$$

$$= 80.35 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (10.3916 \cdot 11505 / (110.0^2 \cdot 1006.0)) - 5.3604 \cdot 6$$

$$= 64.72 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (158 + \max(7, 80)) / 2$$

$$= 119.41 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRa, STa)) / 2$$

$$= (128 + \max(6, 65)) / 2$$

$$= 96.18 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= (Wm1 + Fa + 4Me/G) / Ab$$

$$= 270604 / 191.7932$$

$$= 138.37 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (209418 / 191.7932)$$

$$= 107.08 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm2

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	158.48	207.00	127.65	207.00
Radial Flange	7.32	138.00	5.90	138.00
Tangential Flange	80.35	138.00	64.72	138.00
Maximum Average	119.41	138.00	96.18	138.00
Bolting	138.37	172.00	107.08	172.00

Minimum Required Flange Thickness 101.651 mm.

Estimated M.A.W.P. (Operating) 26 bars

Estimated Finished Weight of Flange at given Thk. 330.8 kg.

Estimated Unfinished Weight of Forging at given Thk 388.5 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 11505.3 / 1.0 \cdot 9806.637 \cdot 0.392 / (2.845 \cdot 201334 \cdot 13.0^2 \cdot 114.359 \cdot 0.3)$$

$$= 0.694 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 14284.3 / 1.0 \cdot 9806.637 \cdot 0.392 / (2.845 \cdot 194922 \cdot 13.0^2 \cdot 114.359 \cdot 0.3)$$

$$= 0.890 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 11505.3 / 1.0 \cdot 9806.637 \cdot 0.392 / (3.38 \cdot 201334 \cdot 13.0^2 \cdot 114.359 \cdot 0.3)$$

$$= 0.584 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 14284.3 / 1.0 \cdot 9806.637 \cdot 0.392 / (3.38 \cdot 194922 \cdot 13.0^2 \cdot 114.359 \cdot 0.3)$$

$$= 0.749 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	151.66	207.00	119.26	207.00
Radial Flange	9.61	138.00	7.55	138.00
Tangential Flange	69.58	138.00	54.72	138.00
Maximum Average	110.62	138.00	86.99	138.00
Bolting	138.37	172.00	107.08	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 95.529 mm.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Left Dish		22.254	28	3	2400	138	1
S1		22.254	...	3	2400	138	2
S2		22.254	...	3	2400	138	3
S3		22.254	...	3	2400	138	4
S4		22.254	...	3	2400	138	5
Right Dish		22.254	28	3	2400	138	6
40" Scrub Flg		22.132	110	3	1000	138	7

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Left Dish		22.1	22.8253	26.3979	23	22.3678	1
S1		22.1	23.6842	27.2706	24	22.5908	2
S2		22.1	23.6842	27.2706	24	22.5908	3
S3		22.1	23.6842	27.2706	24	22.5908	4
S4		22.1	23.6842	27.2706	24	22.5908	5
Right Dish		22.1	22.8248	26.3973	23	22.3682	6
40" Scrub Flg		22.1	25.9495	28.1719	110	101.651	7

Minimum 23 26

MAWP: 23 bars, limited by: Nozzle Reinforcement.

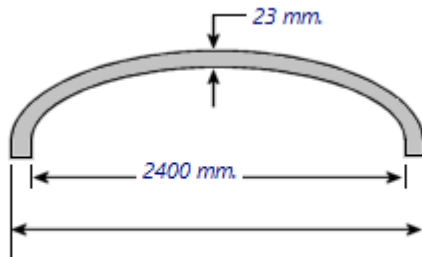
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Left Dish

Elliptical Head Design Information:					
Design Pressure	P	22.254 bars	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2400.000 mm.	Actual Stress		133.6 N./mm^2
Spec. Min. Thk t _e , or s _a		23.0000 mm.	Spec. Nominal Thk.		28.000 mm.
Surface Area		69243.2 cm^2	SG of Contents		0.590
Element Volume		2036116.9 cm3	Weight of Contents		1200.7 Kg
Empty Weight		1613.9 Kg	Operating Weight		2814.5 Kg
			Comp. All. Ope. Stress		125.1 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 30.000 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.000 mm.

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (22.254 \cdot 2406.0 \cdot 0.997) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 22.254)$$

$$= 19.3678 + 3.0000 = 22.3678 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.154 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 20.0) / (0.997 \cdot 2406.0 + 0.2 \cdot 20.0)$$

$$= 23 - 0 = 23 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 23.0) / (1.0 \cdot 2400.0 + 0.2 \cdot 23.0)$$

$$= 26 \text{ bars}$$

149-D

Refrigerant receiver
QUEP

PV Elite 28 U1 Licensee: SL User

FileName : 149-D_Refrigerant receiver_7 Per

Element: Left Dish

Step: 5 12:03pm Sep 16,2026

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Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned} &= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t) \\ &= (22.254 \cdot (0.997 \cdot 2406.0 + 0.2 \cdot 20.0)) / (2 \cdot 1.0 \cdot 20.0) \\ &= 133.645 \text{ N./mm}^2 \end{aligned}$$

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (22.254 \cdot 1203.0) / (138.0 \cdot 1.0 - 0.6 \cdot 22.254) + 3.0 \\ &= 22.590 \text{ mm.} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.154 bars

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 25.0) / (1203.0 + 0.6 \cdot 25.0) \\ &= 28 - 0 = 28 \text{ bars} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6 \\ &= (2 + (2406.0 / (2 \cdot 603.0)))^2 / 6 \\ &= 0.996687 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.976 %

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

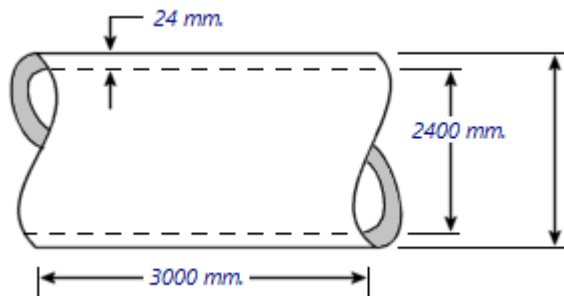
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 30.000 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.000 mm.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: S1

Cylindrical Shell Design Information:					
Design Pressure	P	22.254 bars	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2400.000 mm.	Actual Stress		128.8 N./mm^2
Spec. Min. Thk t _e , or sa		24.0000 mm.	Spec. Nominal Thk.		... mm.
Surface Area		230718.6 cm^2	SG of Contents		0.590
Element Volume		13574115.0 cm3	Weight of Contents		8004.4 Kg
Empty Weight		4249.6 Kg	Operating Weight		12254.0 Kg
			Comp. All. Ope. Stress		126.4 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (22.254 \cdot 1203.0) / (138.0 \cdot 1.0 - 0.6 \cdot 22.254)$$

$$= 19.5908 + 3.0000 = 22.5908 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.154 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 21.0) / (1203.0 + 0.6 \cdot 21.0)$$

$$= 24 - 0 = 24 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 24.0) / (1200.0 + 0.6 \cdot 24.0)$$

$$= 27 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (22.254 \cdot (1203.0 + 0.6 \cdot 21.0)) / (1.0 \cdot 21.0)$$

$$= 128.826 \text{ N./mm}^2$$



% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.990 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

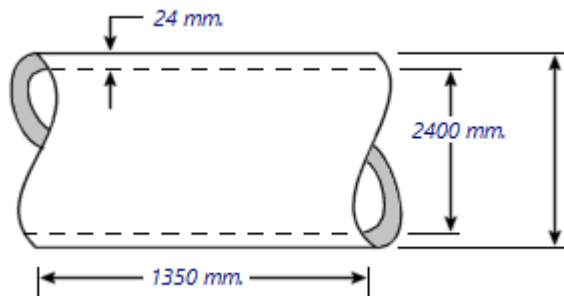
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: S2

Cylindrical Shell Design Information:					
Design Pressure	P	22.254 bars	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1350.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2400.000 mm.	Actual Stress		128.8 N./mm^2
Spec. Min. Thk t _e , or sa		24.0000 mm.	Spec. Nominal Thk.		... mm.
Surface Area		103823.4 cm^2	SG of Contents		0.590
Element Volume		6108350.0 cm3	Weight of Contents		3602.0 Kg
Empty Weight		1912.3 Kg	Operating Weight		5514.3 Kg
			Comp. All. Ope. Stress		126.4 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (22.254 \cdot 1203.0) / (138.0 \cdot 1.0 - 0.6 \cdot 22.254)$$

$$= 19.5908 + 3.0000 = 22.5908 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.154 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 21.0) / (1203.0 + 0.6 \cdot 21.0)$$

$$= 24 - 0 = 24 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 24.0) / (1200.0 + 0.6 \cdot 24.0)$$

$$= 27 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (22.254 \cdot (1203.0 + 0.6 \cdot 21.0)) / (1.0 \cdot 21.0)$$

$$= 128.826 \text{ N./mm}^2$$



% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.990 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

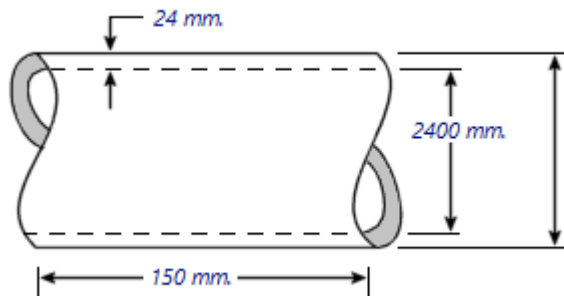
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: S3

Cylindrical Shell Design Information:					
Design Pressure	P	22.254 bars	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		150.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2400.000 mm.	Actual Stress		128.8 N./mm^2
Spec. Min. Thk t _e , or s _a		24.0000 mm.	Spec. Nominal Thk.		... mm.
Surface Area		11535.9 cm^2	SG of Contents		0.590
Element Volume		678705.6 cm3	Weight of Contents		400.2 Kg
Empty Weight		212.5 Kg	Operating Weight		612.7 Kg
			Comp. All. Ope. Stress		126.4 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (22.254 \cdot 1203.0) / (138.0 \cdot 1.0 - 0.6 \cdot 22.254)$$

$$= 19.5908 + 3.0000 = 22.5908 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.154 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 21.0) / (1203.0 + 0.6 \cdot 21.0)$$

$$= 24 - 0 = 24 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 24.0) / (1200.0 + 0.6 \cdot 24.0)$$

$$= 27 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (22.254 \cdot (1203.0 + 0.6 \cdot 21.0)) / (1.0 \cdot 21.0)$$

$$= 128.826 \text{ N./mm}^2$$



% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.990 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

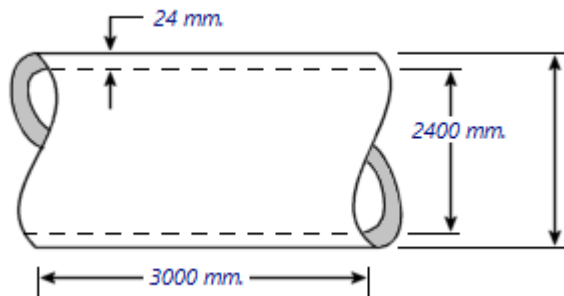
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: S4

Cylindrical Shell Design Information:					
Design Pressure	P	22.254 bars	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2400.000 mm.	Actual Stress		128.8 N./mm^2
Spec. Min. Thk t _e , or s _a		24.0000 mm.	Spec. Nominal Thk.		... mm.
Surface Area		230718.6 cm^2	SG of Contents		0.590
Element Volume		13574111.0 cm3	Weight of Contents		8004.4 Kg
Empty Weight		4249.6 Kg	Operating Weight		12254.0 Kg
			Comp. All. Ope. Stress		126.4 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (22.254 \cdot 1203.0) / (138.0 \cdot 1.0 - 0.6 \cdot 22.254)$$

$$= 19.5908 + 3.0000 = 22.5908 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.154 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 21.0) / (1203.0 + 0.6 \cdot 21.0)$$

$$= 24 - 0 = 24 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 24.0) / (1200.0 + 0.6 \cdot 24.0)$$

$$= 27 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (22.254 \cdot (1203.0 + 0.6 \cdot 21.0)) / (1.0 \cdot 21.0)$$

$$= 128.826 \text{ N./mm}^2$$



% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.990 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

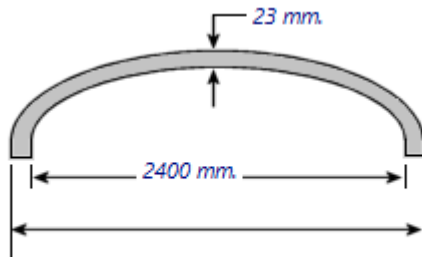
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Right Dish

Elliptical Head Design Information:					
Design Pressure	P	22.254 bars	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	2400.000 mm.	Actual Stress		133.6 N./mm^2
Spec. Min. Thk t _e , or s _a		23.0000 mm.	Spec. Nominal Thk.		28.000 mm.
Surface Area		69243.2 cm^2	SG of Contents		0.590
Element Volume		2036116.9 cm3	Weight of Contents		1200.7 Kg
Empty Weight		1613.9 Kg	Operating Weight		2814.6 Kg
			Comp. All. Ope. Stress		126.4 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 30.000 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.000 mm.

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (22.254 \cdot 2406.0 \cdot 0.997) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 22.254)$$

$$= 19.3682 + 3.0000 = 22.3682 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.154 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 20.0) / (0.997 \cdot 2406.0 + 0.2 \cdot 20.0)$$

$$= 23 - 0 = 23 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 23.0) / (1.0 \cdot 2400.0 + 0.2 \cdot 23.0)$$

$$= 26 \text{ bars}$$

149-D

Refrigerant receiver
QUEP

PV Elite 28 U1 Licensee: SL User

FileName : 149-D_Refrigerant receiver_7 Per

Element: Right Dish

Step: 10 12:03pm Sep 16,2026

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HEXAGON

Elite® v28

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned} &= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t) \\ &= (22.254 \cdot (0.997 \cdot 2406.0 + 0.2 \cdot 20.0)) / (2 \cdot 1.0 \cdot 20.0) \\ &= 133.645 \text{ N./mm}^2 \end{aligned}$$

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (22.254 \cdot 1203.0) / (138.0 \cdot 1.0 - 0.6 \cdot 22.254) + 3.0 \\ &= 22.591 \text{ mm.} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 0.154 bars](#)

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 25.0) / (1203.0 + 0.6 \cdot 25.0) \\ &= 28 - 0 = 28 \text{ bars} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6 \\ &= (2 + (2406.0 / (2 \cdot 603.0)))^2 / 6 \\ &= 0.996687 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.976 %

MDMT Calculations in the Knuckle Portion:

[Note: This Element/Detail was specified as being Impact Tested.](#)

MDMT Calculations in the Head Straight Flange:

[Note: This Element/Detail was specified as being Impact Tested.](#)

[Note:](#)

[Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 30.000 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.000 mm.

Material treatment options: +IT

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	29.503 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	28.730 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	33.998 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	24.964 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	31.603 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	31.603 bars
App 27-4	= M.A.W.P.	22.695 bars

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Left Dish	179.0	234.0	0.765	34.23
S1	173.2	234.0	0.740	34.23
S2	173.2	234.0	0.740	34.23
S3	173.2	234.0	0.740	34.23
S4	173.2	234.0	0.740	34.23
Right Dish	179.0	234.0	0.765	34.23

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1_24 inch	Nozzle	138.00	138.00	1.000
K4_2"	Nozzle	138.00	138.00	1.000
K3_2"	Nozzle	138.00	138.00	1.000
K5_2"	Nozzle	138.00	138.00	1.000
K6_2"	Nozzle	138.00	138.00	1.000
N1_12"	Nozzle	138.00	138.00	1.000
N2_6"	Nozzle	138.00	138.00	1.000
N3_6"	Nozzle	138.00	138.00	1.000
N9_3"	Nozzle	138.00	138.00	1.000
N7_4"	Nozzle	138.00	138.00	1.000
D1_3"	Nozzle	138.00	138.00	1.000
N4_10"	Nozzle	138.00	138.00	1.000
Scrubber_40"	Nozzle	138.00	138.00	1.000
M2_24"	Nozzle	138.00	138.00	1.000
K1_3"	Nozzle	138.00	138.00	1.000
K2_3"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Left Dish	138.00	138.00	1.000
S1	138.00	138.00	1.000
S2	138.00	138.00	1.000
S3	138.00	138.00	1.000
S4	138.00	138.00	1.000
Right Dish	138.00	138.00	1.000
40" Scrub Flg	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
M1_24 inch	85.59	225.00	0.380
K4_2"	7.23	225.00	0.032
K3_2"	7.23	225.00	0.032
K5_2"	7.23	225.00	0.032
K6_2"	7.23	225.00	0.032
N1_12"	42.11	225.00	0.187
N2_6"	24.72	225.00	0.110
N3_6"	24.72	225.00	0.110
N9_3"	11.67	225.00	0.052
N7_4"	16.02	225.00	0.071
D1_3"	11.67	225.00	0.052
N4_10"	34.86	225.00	0.155
Scrubber_40"	110.75	225.00	0.492
M2_24"	85.59	225.00	0.380
K1_3"	8.39	225.00	0.037
K2_3"	8.39	225.00	0.037

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	29.503 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	28.730 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	34.032 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	24.964 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	31.603 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	31.603 bars
App 27-4	= M.A.W.P.	22.695 bars

UG-99(b) Note 35, Test Pressure Calculation:
= Test Factor • Design Pressure • Stress Ratio
= 1.3 • 22.1 • 1.0
= 28.730 bars

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Left Dish	174.2	234.0	0.745	29.01
S1	168.0	234.0	0.718	29.01
S2	168.0	234.0	0.718	29.01
S3	168.0	234.0	0.718	29.01
S4	168.0	234.0	0.718	29.01
Right Dish	174.2	234.0	0.745	29.01

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1_24 inch	Nozzle	138.00	138.00	1.000
K4_2"	Nozzle	138.00	138.00	1.000
K3_2"	Nozzle	138.00	138.00	1.000
K5_2"	Nozzle	138.00	138.00	1.000
K6_2"	Nozzle	138.00	138.00	1.000
N1_12"	Nozzle	138.00	138.00	1.000
N2_6"	Nozzle	138.00	138.00	1.000
N3_6"	Nozzle	138.00	138.00	1.000
N9_3"	Nozzle	138.00	138.00	1.000
N7_4"	Nozzle	138.00	138.00	1.000
D1_3"	Nozzle	138.00	138.00	1.000
N4_10"	Nozzle	138.00	138.00	1.000
Scrubber_40"	Nozzle	138.00	138.00	1.000
M2_24"	Nozzle	138.00	138.00	1.000
K1_3"	Nozzle	138.00	138.00	1.000
K2_3"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Left Dish	138.00	138.00	1.000
S1	138.00	138.00	1.000
S2	138.00	138.00	1.000
S3	138.00	138.00	1.000
S4	138.00	138.00	1.000
Right Dish	138.00	138.00	1.000

40" Scrub Flg	138.00	138.00	1.000
---------------	--------	--------	-------

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm2):

Description	Ambient	Operating	Ratio
M1_24 inch	72.54	225.00	0.322
K4_2"	6.13	225.00	0.027
K3_2"	6.13	225.00	0.027
K5_2"	6.13	225.00	0.027
K6_2"	6.13	225.00	0.027
N1_12"	35.69	225.00	0.159
N2_6"	20.95	225.00	0.093
N3_6"	20.95	225.00	0.093
N9_3"	9.89	225.00	0.044
N7_4"	13.58	225.00	0.060
D1_3"	9.89	225.00	0.044
N4_10"	29.55	225.00	0.131
Scrubber_40"	93.87	225.00	0.417
M2_24"	72.54	225.00	0.322
K1_3"	7.11	225.00	0.032
K2_3"	7.11	225.00	0.032

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	2446	20	0.0011356	89.1145
20	30	8000	2448	21	0.00030528	30.5211
30	40	8000	2448	21	0.00030528	30.5211
40	50	8000	2448	21	0.00030528	30.5211
50	60	8000	2448	21	0.00030528	30.5211
60	70	No Calc	2446	20	0.0011356	89.1145
70	80	No Calc	...	107	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	23	No Calc	...	8.0957
20	30	24	No Calc	...	3.49078
30	40	24	No Calc	...	3.49078
40	50	24	No Calc	...	3.49078
50	60	24	No Calc	...	3.49078
60	70	23	No Calc	...	8.0957
70	80	110	83.4644	...	No Calc

Minimum

3

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	8000	No Calc	No Calc	No Calc
30	40	8000	No Calc	No Calc	No Calc
40	50	8000	No Calc	No Calc	No Calc
50	60	8000	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	1613.86	2036116	1440.95	2050857	112.97
20	30	4249.57	13574115	3722.97	13642071	297.47
30	40	1912.31	6108350	1675.34	6138930	133.862
40	50	212.478	678706	186.149	682103	14.8735
50	60	4249.58	13574111	3722.98	13642067	297.47
60	70	1613.95	2036116	1441.02	2050857	112.976
70	80	330.799	106302	320.982	106529	23.156
Total		14182	38113816.00	12510	38313416.00	992

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Liqd	1200.67	-200	...	...	Liquid 10
10	Nozl	896.902	-600	...	...	M1_24 inch
10	Wght	50	-600	...	300	Davit_24"
20	Sadl	759.776	1520	1528.5	...	Left Saddle
20	Plat	3548.28	3800	1300.2	...	PLAT:[TOP]
20	Liqd	8004.44	1500	...	...	Liquid 20
20	Nozl	12.8739	300	-1225.4	...	K4_2"
20	Nozl	19.4202	300	1225.4	...	K3_2"
20	Nozl	19.4202	1150	1225.4	...	K5_2"
20	Nozl	19.4202	1450	1225.4	...	K6_2"
20	Nozl	172.233	2350	1352.4	...	N1_12"
20	Nozl	64.5104	750	1276.2	...	N2_6"
20	Nozl	90.9614	750	-1276.2	...	N3_6"
20	Nozl	27.773	1300	-1238.1	...	N9_3"
20	Wght	166	2350	1200	...	N1_Feed PP
20	Wght	5	2350	-1200	...	N1_Support
20	Wght	10	700	1200	...	N2_Hood BF
20	Wght	20	700	-1050	...	N3_VB
30	Liqd	3602	675	...	...	Liquid 30
30	Wght	141.999	550	...	...	Clip Wt
40	Liqd	400.222	75	...	...	Liquid 40
40	Insl	22.7184	75	...	...	Cold Insulation
50	Sadl	759.776	1480	1528.5	...	Right Saddle
50	Liqd	8004.44	1500	...	...	Liquid 50
50	Insl	454.368	1500	...	...	Cold Insulation
50	Nozl	42.1241	2650	1250.8	...	N7_4"
50	Nozl	27.773	300	-1238.1	...	D1_3"
50	Nozl	158.515	1950	-1327	...	N4_10"
50	Nozl	710.194	750	1708	...	Scrubber_40"
50	Wght	40	1950	-1050	...	N5_VB
50	Wght	570.01	1050	...	...	Baffle PL
50	Wght	4043.97	750	3400	...	Scrub Fab wt
50	Wght	20201.1	750	3400	...	Scrub Ope wt
50	Wght	24416.5	750	3400	...	Scrub Test wt
60	Liqd	1200.67	250	...	...	Liquid 60
60	Insl	179.998	325	...	...	Cold Insulation
60	Nozl	896.902	650	...	...	M2_24"
60	Nozl	18.8329	547.77	670	...	K1_3"
60	Nozl	27.4153	410	-960	...	K2_3"
60	Wght	50	410	...	300	M2_Davit_24"
70	Liqd	62.6847	67	...	...	Liquid 70
70	Insl	9.20285	...	67	...	Cold insulation

Total Weight of Each Detail Type:

Saddles	1519.6
Platforms	3548.3
Liquid	22475.1
Insulation	666.3
Nozzles	3205.3
Weights	49714.6

Sum of the Detail Weights 81129.1 kg.

Note:

Because some nozzles were attached to a head or flange, their weights include the attached head or flange.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	14821.4	14821.4	14821.4	14821.4	14821.4	14821.4
Saddles	1519.6	1519.6	1519.6	1519.6	1519.6	1519.6
Nozzles	3205.3	3205.3	3205.3	3205.3	3205.3	3205.3
Wld Weights	5097.0	5097.0	5097.0	5097.0	5097.0	5097.0
Platforms	...	...	...	3548.3	3548.3	3548.3
Insulation	...	...	...	666.3	666.3	666.3
Test Weights	...	24416.5	...	...	...	...
Ope Weights	...	...	...	...	...	20201.1
Ope. Liquid	...	...	...	...	...	22475.1
Test Liquid	...	38090.6	...	...	...	...
Totals	24643.2	87150.2	24643.2	28857.7	28857.7	71533.9

Field Installation Options:

Platforms installed before lifting.
Insulation installed before lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	24643.2 kg.
Shop Test Wt.	- Fabricated weight + water (full)	87150.2 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	24643.2 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	28857.7 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	28857.7 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	71533.9 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	69755.2 kg.
Field Test Wt.	- Empty weight + test medium (full)	91364.8 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm^2	Inside Surface Area cm^2
10	20	69243.2	66207.5
20	30	230719	226195
30	40	103823	101788
40	50	11535.9	11309.7
50	60	230719	226195
60	70	69243.2	66207.5
70	80	8729.18	4209.73



Total	724012.0	702111.4 cm2
Total	779.3	755.7 ft²

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
M1_24 inch	45.74	51.00	130	300	GR 1.1	...	...	...	...
K4_2"	45.74	51.00	130	300	GR 1.1	...	...	...	...
K3_2"	45.74	51.00	130	300	GR 1.1	...	...	...	...
K5_2"	45.74	51.00	130	300	GR 1.1	...	...	...	...
K6_2"	45.74	51.00	130	300	GR 1.1	...	...	...	...
N1_12"	45.74	51.00	130	300	GR 1.1	22.141	45.740	45.740	45.740
N2_6"	45.74	51.00	130	300	GR 1.1	33.805	34.805	45.740	34.805
N3_6"	45.74	51.00	130	300	GR 1.1	33.805	34.805	45.740	34.805
N9_3"	91.42	102.00	130	600	GR 1.1	53.197	83.934	91.421	83.934
N7_4"	45.74	51.00	130	300	GR 1.1	43.209	25.401	45.740	25.401
D1_3"	91.42	102.00	130	600	GR 1.1	...	...	...	...
N4_10"	45.74	51.00	130	300	GR 1.1	35.943	32.668	45.740	32.668
M2_24"	45.74	51.00	130	300	GR 1.1	...	...	...	...
K1_3"	45.74	51.00	130	300	GR 1.1	...	...	...	...
K2_3"	45.74	51.00	130	300	GR 1.1	...	...	...	...
Min Rating	25.401	51.000, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE KgF	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
N1_12"	357.251	0.50	1816	2295	22.695	45.740	Pass
N2_6"	196.088	0.50	454	1147	22.695	34.805	Pass
N3_6"	196.088	0.50	454	1147	22.695	34.805	Pass
N9_3"	111.125	0.50	129	612	22.695	83.934	Pass
N7_4"	138.176	0.50	202	765	22.695	25.401	Pass
N4_10"	302.387	0.50	1816	2295	22.695	32.668	Pass

Evaluating Flange on Nozzle [N112"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(1816) + 4 \cdot \text{abs}(2295) \cdot 357.251 \leq \pi \cdot 357.251^3 \cdot [(45.740 - 22.695) + .500 \cdot 45.740]$$

$$= 0.32342E+05 \leq 0.67072E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N112" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(45.740, 0.500 \cdot 45.740 + 45.740 - 32342.4 / (\pi \cdot 357.251^3))$$

$$= \min(45.740, 46.470)$$

$$= 45.740 \text{ bars}$$

Evaluating Flange on Nozzle [N26"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(454) + 4 \cdot \text{abs}(1147) \cdot 196.088 \leq \pi \cdot 196.088^3 \cdot [(45.740 - 22.695) + .500 \cdot 45.740]$$

$$= 0.81657E+04 \leq 0.11091E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N26" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$



$$= \min(45.740, 0.500 \cdot 45.740 + 45.740 - 8165.7/(\pi \cdot 196.088^3))$$

$$= \min(45.740, 34.805)$$

$$= 34.805 \text{ bars}$$

Evaluating Flange on Nozzle [N36]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(454) + 4 \cdot \text{abs}(1147) 196.088 \leq \pi \cdot 196.088^3 [(45.740 - 22.695) + .500 \cdot 45.740]$$

$$= 0.81657E+04 \leq 0.11091E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N36" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(45.740, 0.500 \cdot 45.740 + 45.740 - 8165.7/(\pi \cdot 196.088^3))$$

$$= \min(45.740, 34.805)$$

$$= 34.805 \text{ bars}$$

Evaluating Flange on Nozzle [N93]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(129) + 4 \cdot \text{abs}(612) 111.125 \leq \pi \cdot 111.125^3 [(91.421 - 22.695) + .500 \cdot 91.421]$$

$$= 0.23387E+04 \leq 0.50310E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N93" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(91.421, 0.500 \cdot 91.421 + 91.421 - 2338.7/(\pi \cdot 111.125^3))$$

$$= \min(91.421, 83.934)$$

$$= 83.934 \text{ bars}$$

Evaluating Flange on Nozzle [N74]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(202) + 4 \cdot \text{abs}(765) 138.176 \leq \pi \cdot 138.176^3 [(45.740 - 22.695) + .500 \cdot 45.740]$$

$$= 0.36520E+04 \leq 0.38808E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N74" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(45.740, 0.500 \cdot 45.740 + 45.740 - 3652.0/(\pi \cdot 138.176^3))$$

$$= \min(45.740, 25.401)$$

$$= 25.401 \text{ bars}$$

Evaluating Flange on Nozzle [N410]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(1816) + 4 \cdot \text{abs}(2295) 302.387 \leq \pi \cdot 302.387^3 [(45.740 - 22.695) + .500 \cdot 45.740]$$

$$= 0.31839E+05 \leq 0.40673E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N410" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(45.740, 0.500 \cdot 45.740 + 45.740 - 31838.9/(\pi \cdot 302.387^3))$$

$$= \min(45.740, 32.668)$$

$$= 32.668 \text{ bars}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.000
Basic Wind Speed	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	9695 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1+1.7 \cdot 3.4 \cdot 0.219 \cdot 0.942)/(1+1.7 \cdot 3.4 \cdot 0.219)))$
 $= \min(0.85, 0.895)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.508
Structure Height to Diameter ratio	1.468

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:
 Exposure Category: C from Table 26.11-1
 α = 9.5: Zg = 274320.0 mm.

Effective Height [z]:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1905.0 + 9695.0 = 11599.998 \text{ mm.}$
 $= 38.058 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z [Kz]:
 Because z (38.058 ft.) > 15 ft.
 $= 2.01(z / Zg)^{2/\alpha}$
 $= 2.01(38.058/900.0)^{2/9.5}$
 $= 1.033$

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
 $= 1, \text{ per Table 26.6-1}$

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0.0 • 0.0 • 0.0)²

= 1.0

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 1.033 • 1.0 • 1.0 • 1.0 • 60.399²)

= 16.0 psf [78.12] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16.0 • 0.85 • 0.508 • 14.462

= 99.9 lbs. [45.3] Kgf

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Left Dish	11600.0	0.000	0.000	0.000	1.033	1.000	78.120
S1	11600.0	0.000	0.000	0.000	1.033	1.000	78.120
S2	11600.0	0.000	0.000	0.000	1.033	1.000	78.120
S3	11600.0	0.000	0.000	0.000	1.033	1.000	78.120
S4	11600.0	0.000	0.000	0.000	1.033	1.000	78.120
Right Dish	11600.0	0.000	0.000	0.000	1.033	1.000	78.120
40" Scrub Flg	9762.0	0.000	0.000	0.000	0.996	1.000	78.120

Platform Load Calculations

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf	Cf
PLAT:[TOP]	76000.00	11600.00	78.12	593.71	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
Davit24"	0.00	11600.00	78.12	0.00
N1Feed PP	0.00	12800.00	78.12	0.00
N1Support	0.00	9695.00	78.12	0.00
N2Hood BF	0.00	12800.00	78.12	0.00
N3VB	0.00	9695.00	78.12	0.00
Clip Wt	0.00	11600.00	78.12	0.00
N5VB	0.00	9695.00	78.12	0.00
Baffle PL	0.00	11600.00	78.12	0.00
Scrub Fab wt	233277.00	15000.00	78.12	1822.36
Scrub Ope wt	0.00	15000.00	78.12	0.00
Scrub Test wt	0.00	15000.00	78.12	0.00
M2Davit24"	0.00	11600.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	11600	2935.2	13436	78.12	45.3043
20	30	11600	2937.6	88128	78.12	890.869
30	40	11600	2937.6	39657.6	78.12	133.72
40	50	11600	3081.6	4622.4	78.12	15.5861



50	60	11600	3081.6	92448	78.12	2134.09
60	70	11600	3079.2	15304.3	78.12	51.6041
70	80	9762	1344	1800.96	78.12	6.0726



Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.100
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	C
Component Elevation Ratio z/h	1.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\
 SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\
 SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hn^x$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(10.1870.75)$
 $= 0.114$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33.0)$
 $= 0.030$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.114, 0.03) \text{ per 12.8.2}$
 $= 0.030$

Loads on Building/Structure Supported Equipment:

$$\begin{aligned}
 &\text{Seismic Force per equation 13.3-1, [Fp]:} \\
 &= 0.4 \cdot Ap \cdot Sds \cdot W(1 + 2 \cdot (z/h)) / (R / Ie) \\
 &= 0.4 \cdot 1.0 \cdot 0.157 \cdot 70014(1 + 2 \cdot 1.0) / (3.0/1.25) \\
 &= 5489.125 \text{ Kgf}
 \end{aligned}$$

Maximum value of Fp per equation 13.3-2:
 $= 1.6 \cdot Sds \cdot I \cdot W$
 $= 1.6 \cdot 0.157 \cdot 1.25 \cdot 70014 = 21956.5 \text{ Kgf}$

Minimum value of Fp per equation 13.3-3:
 $= 0.3 \cdot 0.16 \cdot 1.25 \cdot 70014$
 $= 4116.844 \text{ Kgf}$

Total Base Shear $V = Fp$, [V]:
 $= 5489.125 \text{ Kgf}$

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.157 \cdot 70014$

= 2195.6 Kgf

Final Base Shear, V = 6038.04 Kgf
Final Vertical Load, Ev = 2415.21 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	1200	8751.79	754.755
20	30	1200	8751.79	754.755
20	30	1200	8751.79	754.755
30	40	1200	8751.79	754.755
40	50	1200	8751.79	754.755
50	0	1200	8751.79	754.755
50	60	1200	8751.79	754.755
60	70	1200	8751.79	754.755

The Earthquake Weight does not include the weight of the saddles.

The Earthquake Loads calculated and printed in the Earthquake Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

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Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.
Insulation is installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	3800.000 mm.
Center of Gravity of the Platforms	3850.000 mm.
Center of Gravity of the Liquid	3804.371 mm.
Center of Gravity of the Insulation	6478.963 mm.
Center of Gravity of the Nozzles	4074.078 mm.
Center of Gravity of the Added Weights (Operating)	5267.713 mm.
Center of Gravity of the Added Weights (Empty)	5139.749 mm.
Center of Gravity of Bare Shell New and Cold	3833.449 mm.
Center of Gravity of Bare Shell Corroded	3836.793 mm.
Vessel CG in the Operating Condition	4378.639 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	4128.629 mm.
Vessel CG in the Test Condition	3932.625 mm.

Rigging Analysis Results:

- Lifting weight based on the user selected calculated empty weight

Total Effective Length of Vessel for this analysis	7600.00 mm.
Total vessel weight (No Liquid) Twt	28857.73 Kgf
Impact weight multiplication factor Imp	1.50
Design lifting weight, DWT = Imp * Twt	43286.60 Kgf
Distance to the Lug on the Left End	1000.00 mm.
Distance to the Lug on the Right End	6600.00 mm.
Design Reaction force at the Left End Lug	19103.08 Kgf
Design Reaction force at the Right End Lug	24183.52 Kgf
CG distance during field lifting operation	4128.63 mm.
CG distance from the left end lug/trunnion	3128.63 mm.
CG distance from the nearer lifting lug/trunnion	2471.37 mm.
Compressive Allowable Increase Factor	1.00

Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	1.11	4400.00	108.09 (UG-23)
Shear	0.79	6350.00	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kgf	Shear Stress N./mm ²
0.00	0.0	0.0	9378.8	0.5
50.00	1743.0	0.2	10100.2	0.5
3050.00	8619.2	0.8	2608.8	0.1
4400.00	12384.7	1.1	2802.0	0.2
4550.00	11991.4	1.1	8212.9	0.4
7550.00	1454.5	0.1	721.4	0.0

Unity Check (Actual Stress / Allowable Stress):



Maximum Unity Check is 0.0102 at elevation 4400.0 mm. - Must be ≤ 1

Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ...----

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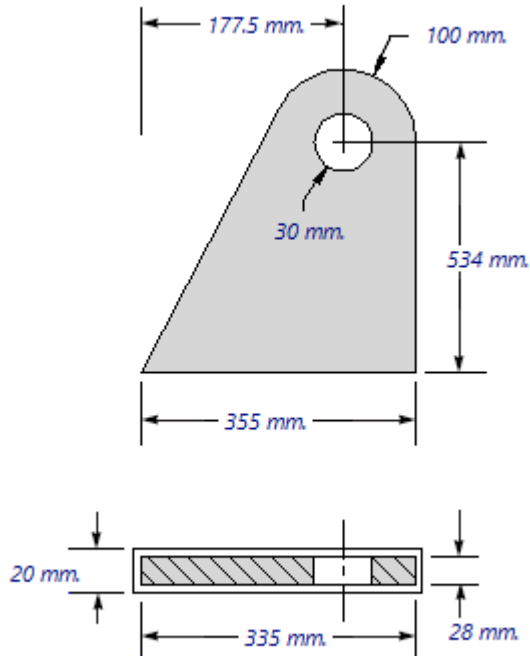


Lifting Lug Calculations: Lug(s) on Left End of Vessel

Input Values:

Lifting Lug Material	SA-516 70
Lifting Lug Yield Stress	Yield 262.00 N./mm2
Total Height of Lifting Lug	w 355.0000 mm.
Thickness of Lifting Lug	t 28.0000 mm.
Diameter of Hole in Lifting Lug	dh 30.0000 mm.
Radius of Semi-Circular Arc of Lifting Lug	r 100.0000 mm.
Height of Lug from Edge to Center of Hole	h 177.5000 mm.
Offset from Vessel Surface to Center of Hole	off 534.0000 mm.
Lug Fillet Weld Size	tw 20.0000 mm.
Length of weld along side of Lifting Lug	wl 335.0000 mm.
Length of Weld along Bottom of Lifting Lug	wb 20.0000 mm.
Thickness of Collar (if any)	tc 0.0000 mm.
Diameter of Collar (if any)	dc 0.0000 mm.
Impact Factor	Impfac 1.50
Sling Angle from Horizontal	90.0 degrees
Number of Lugs in Group	2

Lifting Lug Orientation to Vessel: Perpendicular
Lift Orientation : Horizontal Lift



PV Elite does not compute weak axis bending forces on the lugs. It is assumed that a spreader bar is used.

Computed Results:

Total vessel weight (no liquid)	28857.73 Kg
Design reaction force at the left end lug	19103.08 Kg
Design reaction force at the right end lug	24183.52 Kg
Force along vessel axis	Fax 0.00 Kg
Force normal to vessel	Fn 9551.54 Kg

149-D

Refrigerant receiver
QUEP

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FileName : 149-D_Refrigerant receiver_7_Per

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Lifting Lug Calcs: Left End

Step: 19 12:03pm Sep 16,2026



HEXAGON

Force tangential to vessel Ft 0.00 Kgf

Converting the weld leg dimension (tw) to the weld throat dimension.

Weld Group Inertia Calculations:

Weld group inertia about the longitudinal axis	I11	10584.556 cm ⁴
Weld group centroid distance in the long. direction	Y11	181.640 mm.
Dist. of weld group centroid from lug bottom	Y11b	167.500 mm.
Weld group inertia about the circumferential axis	I1c	438.254 cm ⁴
Weld group centroid distance in the circ. direction	Y1c	28.140 mm.

Note: The Impact Factor is applied to the Forces acting on the Lug.

Primary Shear Stress in the Welds due to Shear Loads [Ssll]:

$$= \sqrt{(Fax^2 + Ft^2 + Fn^2) / ((2 \cdot (wl + wb)) \cdot tw)}$$

$$= \sqrt{(0^2 + 0^2 + 9552^2) / ((2 \cdot (335.0 + 20.0)) \cdot 14.14)}$$

$$= 9.33 \text{ N./mm}^2$$

Shear Stress in the Welds due to Bending Loads [Sblf]:

$$= (Fax \cdot \text{off} \cdot Y11/I11) + (Ft \cdot \text{off} \cdot Y1c/I1c)$$

$$= (0 \cdot 534.0 \cdot 181.64/10584.556) +$$

$$(0 \cdot 534.0 \cdot 28.14/438.254)$$

$$= 0.00 \text{ N./mm}^2$$

Total Shear Stress for Combined Loads [St]:

$$= Ssll + Sblf$$

$$= 9.33 + 0.0$$

$$= 9.33 \text{ N./mm}^2$$

Allowable Shear Stress for Combined Loads [Sta]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac (AISC Shear Allowable)}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Shear Stress in Lug above Hole [Shs]:

$$= \sqrt{(Pl^2 + Fax^2) / Sha}$$

$$= \sqrt{(9552^2 + 0^2) / 47.6}$$

$$= 19.68 \text{ N./mm}^2$$

Allowable Shear Stress in Lug above Hole [Sas]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Pin Hole Bearing Width [Pbw]:

$$= \text{Lug Thickness} + 2 \cdot \text{Collar Thickness}$$

$$= 28.0 + 2 \cdot 0.0$$

$$= 28.000 \text{ mm.}$$

Pin Hole Bearing Stress [Pbs]:

$$= \sqrt{(Pl^2 + Faxial^2) / (Pbw \cdot dh)}$$

$$= \sqrt{(9552^2 + 0^2) / (28.0 \cdot 30.0)}$$

$$= 111.51 \text{ N./mm}^2$$

Allowable Bearing Stress [Pba]:

$$= \min(0.75 \cdot \text{Yield} \cdot \text{Occfac}, 0.9 \cdot \text{Yield}) \text{ AISC Bearing All.}$$

$$= \min(0.75 \cdot 262 \cdot 1.0, 235.8)$$

$$= 196.50 \text{ N./mm}^2$$

Bending Stress at the Base of the Lug [Fbs]:

$$= Ft \cdot \text{off} / (w \cdot t^2/6) + Fax \cdot \text{off} / (w^2 \cdot t/6)$$

$$= 0 \cdot 534.0 / (355.0 \cdot 28.0^2/6) +$$

$$0 \cdot 534.0 / (355.0^2 \cdot 28.0/6)$$

$$= 0.00 \text{ N./mm}^2$$

Tensile Stress at the Base of the Lug [Fa]:

$$= F_n / (w \cdot t)$$

$$= 9552 / (355.0 \cdot 28.0)$$

$$= 9.42 \text{ N./mm}^2$$

Total Combined Stress at the Base of the Lug:

$$= F_{bs} + F_a$$

$$= 0.0 + 9.4$$

$$= 9.42 \text{ N./mm}^2$$

Lug Allowable Stress for Bending and Tension:

$$= \min(0.66 \cdot \text{Yield} \cdot \text{Occfac}, 0.75 \cdot \text{Yield})$$

$$= \min(0.66 \cdot 262 \cdot 1.0, 196.5)$$

$$= 172.92 \text{ N./mm}^2$$

Required Shackle Pin Diameter [Spd]:

$$= \sqrt{((2(F_n^2 + F_{ax}^2)^{1/2} / (\pi \cdot \text{Sta})))}$$

$$= \sqrt{(2(9552^2 + 0^2)^{1/2} / (\pi \cdot 105))}$$

$$= 23.8542 \text{ mm.}$$

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).



The ratio Beta1/Beta2 for rectangular attachments was outside the range of values in tables 7 & 8 and was set to be an acceptable (maximum or minimum) value. This generally results in one dimension being altered so that the aspect ratio does not exceed 4 to 1.

Attachment dimensions (C11P, C22P, etc.) may be modified accounting for attachment welds where applicable.

Input Echo, WRC107/537 Item 1, Description: Lift Lug

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	2400.000 mm.
Vessel Thickness	Tv	24.000 mm.

Note: Lifting lug calculations are performed at ambient temperature conditions.

Attachment Type	Type	Rectangular
Parameter C11	C11	20.00 mm.
Parameter C22	C22	80.00 mm.
Design Internal Pressure	Dp	0.000 bars
Include Pressure Thrust		No

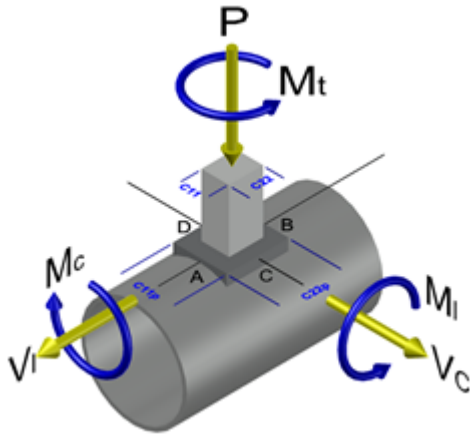
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-9551.5	Kgf
Longitudinal Shear (SUS)	Vl	-0.0	Kgf
Circumferential Shear (SUS)	Vc	0.0	Kgf
Circumferential Moment (SUS)	Mc	0.0	Kg-m.
Longitudinal Moment (SUS)	Ml	-0.0	Kg-m.
Torsional Moment (SUS)	Mt	0.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	-9551.5	Kgf
Circumferential Shear	VC	0.0	Kgf
Longitudinal Shear	VL	-0.0	Kgf
Circumferential Moment	MC	0.0	Kg-m.
Longitudinal Moment	ML	-0.0	Kg-m.
Torsional Moment	MT	0.0	Kg-m.

Dimensionless Parameters used : Gamma = 50.50

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.024	4C	9.799	(A,B)
N(PHI) / (P/Rm)	0.024	3C	9.991	(C,D)
M(PHI) / (P)	0.015	2C1	0.235	(A,B)
M(PHI) / (P)	0.015	1C !	0.263	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.013	3A	0.118	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.016	1A !	0.104	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.021	3B	1.056	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.018	1B !	0.063	(A,B,C,D)
N(x) / (P/Rm)	0.020	3C	10.106	(A,B)
N(x) / (P/Rm)	0.020	4C	9.851	(C,D)
M(x) / (P)	0.021	1C1	0.268	(A,B)
M(x) / (P)	0.021	2C	0.217	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.013	4A	0.123	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.021	2A !	0.062	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.021	4B	0.250	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.025	2B !	0.103	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb.	P	31.6	31.6	31.6	31.6	32.2	32.2	32.2	32.2

149-D

Refrigerant receiver
QUEP

Elite® v28



HEXAGON

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FileName : 149-D_Refrigerant receiver_7 Per

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Lifting Lug Calcs: Left End

Step: 19 12:03pm Sep 16,2026

Circ. Bend. P	229.3	-229.3	229.3	-229.3	256.4	-256.4	256.4	-256.4
Circ. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML	0.0	0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML	0.0	-0.0	-0.0	0.0	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.	260.8	-197.7	260.8	-197.7	288.6	-224.2	288.6	-224.2
Long. Memb. P	32.5	32.5	32.5	32.5	31.7	31.7	31.7	31.7
Long. Bend. P	261.1	-261.1	261.1	-261.1	212.2	-212.2	212.2	-212.2
Long. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML	0.0	0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	0.0	-0.0	-0.0	0.0	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	293.6	-228.5	293.6	-228.5	243.9	-180.4	243.9	-180.4
Shear VC	0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear VL	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Shear MT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Str. Int.	293.6	228.5	293.6	228.5	288.6	224.2	288.6	224.2

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		31.6	31.6	31.6	31.6	32.2	32.2	32.2	32.2
Circ. Q (SUS)		229.3	-229.3	229.3	-229.3	256.4	-256.4	256.4	-256.4
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		32.5	32.5	32.5	32.5	31.7	31.7	31.7	31.7
Long. Q (SUS)		261.1	-261.1	261.1	-261.1	212.2	-212.2	212.2	-212.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		32.5	32.5	32.5	32.5	32.2	32.2	32.2	32.2
Pm+Pl+Q (Total)		293.6	228.5	293.6	228.5	288.6	224.2	288.6	224.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	32.54	207.00	Passed
Pm+Pl+Q (TOTAL)	293.59	413.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.



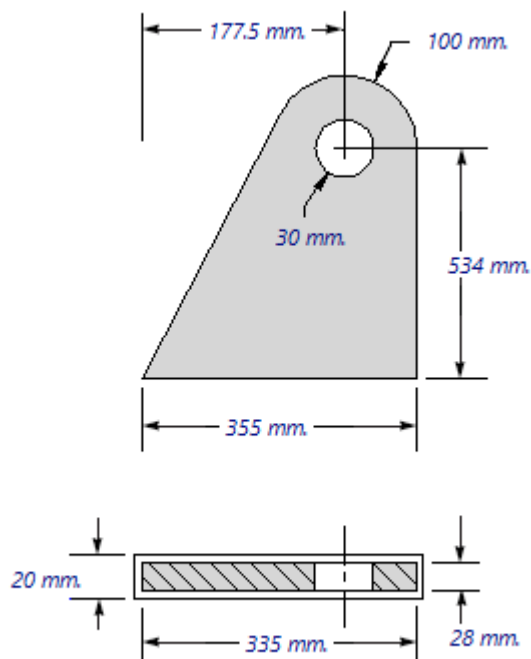


Lifting Lug Calculations: Lug(s) on Right End of Vessel

Input Values:

Lifting Lug Material		SA-516 70
Lifting Lug Yield Stress	Yield	262.00 N./mm ²
Total Height of Lifting Lug	w	355.0000 mm.
Thickness of Lifting Lug	t	28.0000 mm.
Diameter of Hole in Lifting Lug	dh	30.0000 mm.
Radius of Semi-Circular Arc of Lifting Lug	r	100.0000 mm.
Height of Lug from Edge to Center of Hole	h	177.5000 mm.
Offset from Vessel Surface to Center of Hole	off	534.0000 mm.
Lug Fillet Weld Size	tw	20.0000 mm.
Length of weld along side of Lifting Lug	wl	335.0000 mm.
Length of Weld along Bottom of Lifting Lug	wb	20.0000 mm.
Thickness of Collar (if any)	tc	0.0000 mm.
Diameter of Collar (if any)	dc	0.0000 mm.
Impact Factor	Impfac	1.50
Number of Lugs in Group		2

Lifting Lug Orientation to Vessel: Perpendicular
Lift Orientation : Horizontal Lift



PV Elite does not compute weak axis bending forces on the lugs. It is assumed that a spreader bar is used.

Computed Results:

Total vessel weight (no liquid)		28857.73 Kg
Design reaction force at the left end lug		19103.08 Kg
Design reaction force at the right end lug		24183.52 Kg
Force along vessel axis	Fax	0.00 Kg
Force normal to vessel	Fn	12091.76 Kg
Force tangential to vessel	Ft	0.00 Kg



Sling angle from horizontal

0.0 deg

Converting the weld leg dimension (tw) to the weld throat dimension.

Weld Group Inertia Calculations:

Weld group inertia about the longitudinal axis	Ill	10584.556 cm4
Weld group centroid distance in the long. direction	Yll	181.640 mm.
Dist. of weld group centroid from lug bottom	Yllb	167.500 mm.
Weld group inertia about the circumferential axis	Ilc	438.254 cm4
Weld group centroid distance in the circ. direction	Ylc	28.140 mm.

Note: The Impact Factor is applied to the Forces acting on the Lug.

Primary Shear Stress in the Welds due to Shear Loads [Ssll]:

$$= \sqrt{(Fax^2 + Ft^2 + Fn^2)} / ((2 \cdot (wl + wb)) \cdot tw)$$

$$= \sqrt{(0^2 + 0^2 + 12092^2)} / ((2 \cdot (335.0 + 20.0)) \cdot 14.14)$$

$$= 11.81 \text{ N./mm}^2$$

Shear Stress in the Welds due to Bending Loads [Sblf]:

$$= (Fax \cdot \text{off} \cdot Yll/III) + (Ft \cdot \text{off} \cdot Ylc/Ilc)$$

$$= (0 \cdot 534.0 \cdot 181.64/10584.556) +$$

$$(0 \cdot 534.0 \cdot 28.14/438.254)$$

$$= 0.00 \text{ N./mm}^2$$

Total Shear Stress for Combined Loads [St]:

$$= Ssll + Sblf$$

$$= 11.812 + 0.0$$

$$= 11.81 \text{ N./mm}^2$$

Allowable Shear Stress for Combined Loads [Sta]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac} \text{ (AISC Shear Allowable)}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Shear Stress in Lug above Hole [Shs]:

$$= \sqrt{(Pl^2 + Fax^2)} / Sha$$

$$= \sqrt{(12092^2 + 0^2)} / 47.6$$

$$= 24.91 \text{ N./mm}^2$$

Allowable Shear Stress in Lug above Hole [Sas]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Pin Hole Bearing Width [Pbw]:

$$= \text{Lug Thickness} + 2 \cdot \text{Collar Thickness}$$

$$= 28.0 + 2 \cdot 0.0$$

$$= 28.000 \text{ mm.}$$

Pin Hole Bearing Stress [Pbs]:

$$= \sqrt{(Pl^2 + Faxial^2)} / (Pbw \cdot dh)$$

$$= \sqrt{(12092^2 + 0^2)} / (28.0 \cdot 30.0)$$

$$= 141.17 \text{ N./mm}^2$$

Allowable Bearing Stress [Pba]:

$$= \min(0.75 \cdot \text{Yield} \cdot \text{Occfac}, 0.9 \cdot \text{Yield}) \text{ AISC Bearing All.}$$

$$= \min(0.75 \cdot 262 \cdot 1.0, 235.8)$$

$$= 196.50 \text{ N./mm}^2$$

Bending Stress at the Base of the Lug [Fbs]:

$$= Ft \cdot \text{off} / (w \cdot t^2/6) + Fax \cdot \text{off} / (w^2 \cdot t/6)$$

$$= 0 \cdot 534.0 / (355.0 \cdot 28.0^2/6) +$$

$$0 \cdot 534.0 / (355.0^2 \cdot 28.0/6)$$

$$= 0.00 \text{ N./mm}^2$$



Tensile Stress at the Base of the Lug [Fa]:

$$= F_n / (w \cdot t)$$

$$= 12092 / (355.0 \cdot 28.0)$$

$$= 11.93 \text{ N./mm}^2$$

Total Combined Stress at the Base of the Lug:

$$= F_{bs} + F_a$$

$$= 0.0 + 11.9$$

$$= 11.93 \text{ N./mm}^2$$

Lug Allowable Stress for Bending and Tension:

$$= \min(0.66 \cdot \text{Yield} \cdot \text{Occfac}, 0.75 \cdot \text{Yield})$$

$$= \min(0.66 \cdot 262 \cdot 1.0, 196.5)$$

$$= 172.92 \text{ N./mm}^2$$

Required Shackle Pin Diameter [Spd]:

$$= \sqrt{((2(F_n^2 + F_{ax}^2)^{1/2} / (\pi \cdot \text{Sta})))}$$

$$= \sqrt{(2(12092^2 + 0^2)^{1/2} / (\pi \cdot 105))}$$

$$= 26.8394 \text{ mm.}$$

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).



The ratio Beta1/Beta2 for rectangular attachments was outside the range of values in tables 7 & 8 and was set to be an acceptable (maximum or minimum) value. This generally results in one dimension being altered so that the aspect ratio does not exceed 4 to 1.

Attachment dimensions (C11P, C22P, etc.) may be modified accounting for attachment welds where applicable.

Input Echo, WRC107/537 Item 1, Description: Lift Lug

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	2400.000 mm.
Vessel Thickness	Tv	24.000 mm.

Note: Lifting lug calculations are performed at ambient temperature conditions.

Attachment Type	Type	Rectangular
Parameter C11	C11	20.00 mm.
Parameter C22	C22	80.00 mm.
Design Internal Pressure	Dp	0.000 bars
Include Pressure Thrust		No

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-12091.8	Kgf
Longitudinal Shear (SUS)	Vl	-0.0	Kgf
Circumferential Shear (SUS)	Vc	0.0	Kgf
Circumferential Moment (SUS)	Mc	0.0	Kg-m.
Longitudinal Moment (SUS)	Ml	-0.0	Kg-m.
Torsional Moment (SUS)	Mt	0.0	Kg-m.

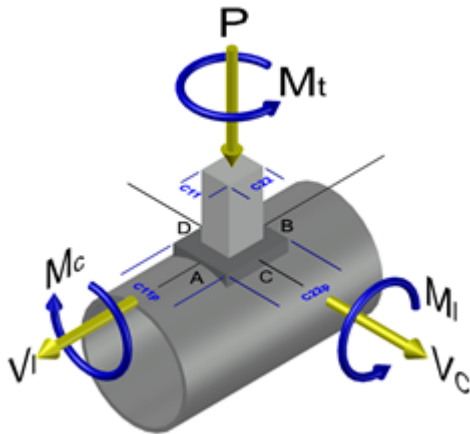
Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the



results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	-12091.8	Kgf
Circumferential Shear	VC	0.0	Kgf
Longitudinal Shear	VL	-0.0	Kgf
Circumferential Moment	MC	0.0	Kg-m.
Longitudinal Moment	ML	-0.0	Kg-m.
Torsional Moment	MT	0.0	Kg-m.

Dimensionless Parameters used : Gamma = 50.50

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
$N(\Phi) / (P/Rm)$	0.024	4C	9.799	(A,B)
$N(\Phi) / (P/Rm)$	0.024	3C	9.991	(C,D)
$M(\Phi) / (P)$	0.015	2C1	0.235	(A,B)
$M(\Phi) / (P)$	0.015	1C !	0.263	(C,D)
$N(\Phi) / (MC/(Rm^{**2} * Beta))$	0.013	3A	0.118	(A,B,C,D)
$M(\Phi) / (MC/(Rm^{**2} * Beta))$	0.016	1A !	0.104	(A,B,C,D)
$N(\Phi) / (ML/(Rm^{**2} * Beta))$	0.021	3B	1.056	(A,B,C,D)
$M(\Phi) / (ML/(Rm^{**2} * Beta))$	0.018	1B !	0.063	(A,B,C,D)
$N(x) / (P/Rm)$	0.020	3C	10.106	(A,B)
$N(x) / (P/Rm)$	0.020	4C	9.851	(C,D)
$M(x) / (P)$	0.021	1C1	0.268	(A,B)
$M(x) / (P)$	0.021	2C	0.217	(C,D)
$N(x) / (MC/(Rm^{**2} * Beta))$	0.013	4A	0.123	(A,B,C,D)
$M(x) / (MC/(Rm^{**2} * Beta))$	0.021	2A !	0.062	(A,B,C,D)
$N(x) / (ML/(Rm^{**2} * Beta))$	0.021	4B	0.250	(A,B,C,D)
$M(x) / (ML/(Rm^{**2} * Beta))$	0.025	2B !	0.103	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl

149-D

Refrigerant receiver
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FileName : 149-D_Refrigerant receiver_7 Per

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Lifting Lug Calcs: Right End

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Circ. Memb. P	39.9	39.9	39.9	39.9	40.7	40.7	40.7	40.7
Circ. Bend. P	290.2	-290.2	290.2	-290.2	324.6	-324.6	324.6	-324.6
Circ. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML	0.0	0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML	0.0	-0.0	-0.0	0.0	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.	330.2	-250.3	330.2	-250.3	365.3	-283.9	365.3	-283.9
Long. Memb. P	41.2	41.2	41.2	41.2	40.2	40.2	40.2	40.2
Long. Bend. P	330.5	-330.5	330.5	-330.5	268.6	-268.6	268.6	-268.6
Long. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML	0.0	0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	0.0	-0.0	-0.0	0.0	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	371.7	-289.3	371.7	-289.3	308.7	-228.4	308.7	-228.4
Shear VC	0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear VL	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Shear MT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Str. Int.	371.7	289.3	371.7	289.3	365.3	283.9	365.3	283.9

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N/mm2)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		39.9	39.9	39.9	39.9	40.7	40.7	40.7	40.7
Circ. Q (SUS)		290.2	-290.2	290.2	-290.2	324.6	-324.6	324.6	-324.6
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		41.2	41.2	41.2	41.2	40.2	40.2	40.2	40.2
Long. Q (SUS)		330.5	-330.5	330.5	-330.5	268.6	-268.6	268.6	-268.6
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		41.2	41.2	41.2	41.2	40.7	40.7	40.7	40.7
Pm+Pl+Q (Total)		371.7	289.3	371.7	289.3	365.3	283.9	365.3	283.9

Vessel Stress Summation Comparison (N/mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	41.20	207.00	Passed
Pm+Pl+Q (TOTAL)	371.68	413.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.



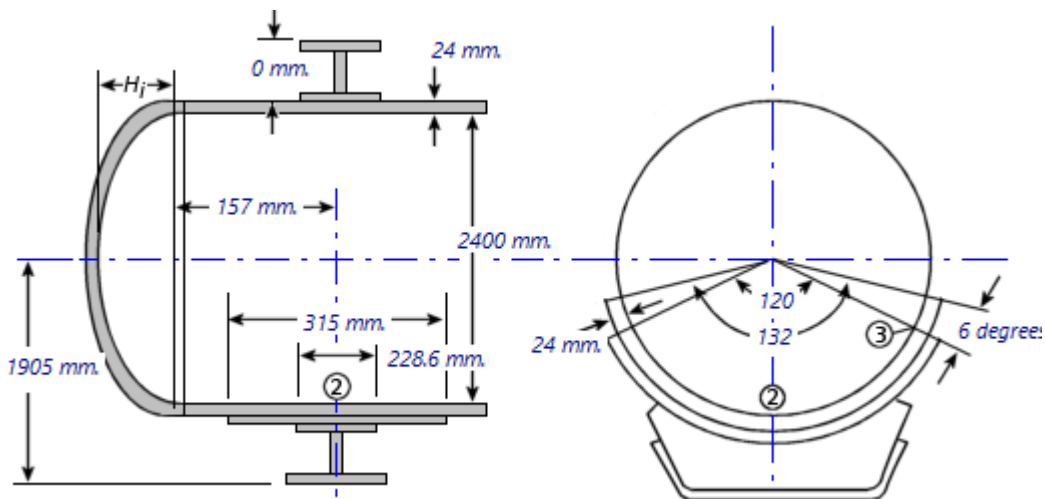


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1213.50	mm.
Shell Thickness used in this Case	t	24.000	mm.
Stiffened Vessel Length per 4.15.6	L	7600.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	157.00	mm.
Saddle Width	b1 or b	228.60	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	315.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	613.00	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1905.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		35050.77	Kgf
Pressure used in Analysis	P	22.169	bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		58.31	138.00
Long. Stress at Bottom of Midspan		69.80	138.00
Long. Stress at Top of Saddles		63.89	138.00
Long. Stress at Bottom of Saddles		64.22	138.00
Tangential Shear in Shell		11.87	110.40
Tangential Shear in Head		12.46	110.40
Circ. Stress at Horn of Saddle		10.38	172.50

Addl. Stress in Head as Stiffener	136.54	172.50
Circ. Compressive Stress in Shell	1.85	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(3277.2/2 + 0) \cdot 1905.0/2161.5996$
 $= 4332.3 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(720.7, 0) \cdot 1905.0/4460.0$
 $= 307.8 \text{ Kg}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(6038.04, 0) \cdot 1905.0/4460.0$
 $= 2579.0 \text{ Kg}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(6038/2 + 0) \cdot 1905.0/2161.5996$
 $= 7981.9 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 27069 + \max(308, 4332, 2579, 7982)$
 $= 35050.8 \text{ Kg}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.0 \cdot 766.08(2.448 + \pi/4(2.448 \cdot 1.2)^2)$
 $= 7067.591 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	35810.55	Kgf
Transverse Shear Load	3019.02	Kgf
Longitudinal Shear Load	6038.04	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(2400.0/2 + 24.0 + 24.0)\sin(120.0/2), 2 \cdot 1213.5)$
 $= 2161.600 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0109			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is less than $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -35051 \cdot 157.0 [1 - (1 - 157.0/7600.0 + (1213.5^2 - 613.0^2)/(2 \cdot 157.0 \cdot 7600.0))/(1 + (4 \cdot 613.0)/(3 \cdot 7600.0))]$
 $= 1646.7 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 35051 \cdot 7600/4((1 + 2(1214^2 - 613^2)/(7600^2))/(1 + (4 \cdot 613)/$



$$(3 \cdot 7600) - 4 \cdot 157 / 7600 \\ = 56911.7 \text{ Kg-m.}$$

$$\text{Longitudinal Stress at Top of Shell (4.15.4) } [\sigma_1]: \\ = P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 22.169 \cdot 1213.5 / (2 \cdot 21.0) - 56911.7 / (\pi \cdot 1213.5^2 \cdot 21.0) \\ = 58.31 \text{ N./mm}^2$$

$$\text{Longitudinal Stress at Bottom of Shell (4.15.5) } [\sigma_2]: \\ = P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 22.169 \cdot 1213.5 / (2 \cdot 21.0) + 56911.7 / (\pi \cdot 1213.5^2 \cdot 21.0) \\ = 69.80 \text{ N./mm}^2$$

$$\text{Longitudinal Stress at Top of Shell at Support (4.15.6) } [\sigma_3]: \\ = P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t) \\ = 22.169 \cdot 1213.5 / (2 \cdot 21.0) - 1646.7 / (\pi \cdot 1213.5^2 \cdot 21.0) \\ = 63.89 \text{ N./mm}^2$$

$$\text{Longitudinal Stress at bottom of Shell at Support (4.15.7) } [\sigma_4]: \\ = P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t) \\ = 22.169 \cdot 1213.5 / (2 \cdot 21.0) + 1646.7 / (\pi \cdot 1213.5^2 \cdot 21.0) \\ = 64.22 \text{ N./mm}^2$$

$$\text{Maximum Shear Force in the Saddle (4.15.3) } [T]: \\ = Q(L-2a) / (L + (4 \cdot h_m / 3)) \\ = 35051(7600.0 - 2 \cdot 157.0) / (7600.0 + (4 \cdot 613.0 / 3)) \\ = 30339.8 \text{ Kgf}$$

$$\text{Shear Stress in the shell no rings, stiffened (4.15.13) } [\tau_3]: \\ = K_3 \cdot Q / (R_m \cdot t) \\ = 0.8799 \cdot 35051 / (1213.5001 \cdot 21.0) \\ = 11.87 \text{ N./mm}^2$$

$$\text{Shear Stress in the head, shell stiffened (4.15.14) } [\tau_3^*]: \\ = K_3 \cdot Q / (R_m \cdot t_h) \\ = 0.8799 \cdot 35051 / (1213.5001 \cdot 20.0) \\ = 12.46 \text{ N./mm}^2$$

$$\text{Membrane stress in the head as a stiffener (4.15.16) } [\sigma_5]: \\ = K_4 \cdot Q / (R_m \cdot t_h) + ((P \cdot R_i) / (2 \cdot t_h)) \cdot (R_i / h_m) \\ = 0.4011 \cdot 35051 / (1213.5 \cdot 20.0) + \\ ((22.17 \cdot 1203.0) / (2 \cdot 20.0)) \cdot (1203.0 / 613.0) \\ = 136.54 \text{ N./mm}^2$$

$$\text{Decay Length (4.15.20) } [x_1, x_2]: \\ = 0.78 \cdot \sqrt{(R_m \cdot t)} \\ = 0.78 \cdot \sqrt{(1213.5 \cdot 21.0)} \\ = 124.516 \text{ mm.}$$

$$\text{Effective reinforcing plate width (4.15.24) } [B_1]: \\ = \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a) \\ = \min(228.6 + 1.56 \cdot \sqrt{(1213.5 \cdot 21.0)}, 2 \cdot 157.0) \\ = 314.00 \text{ mm.}$$

$$\text{Wear Plate/Shell Stress ratio (4.15.26) } [\eta]: \\ = \min(S_r / S, 1) \\ = \min(137.997 / 137.997, 1) \\ = 1.0000$$

$$\text{Circumferential Stress at Saddle Base with Wear Plate (4.15.25) } [\sigma_6, r]: \\ = -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r)) \\ = -0.7603 \cdot 35051 \cdot 0.1 / (314.0(21.0 + 1.0 \cdot 24.0)) \\ = -1.85 \text{ N./mm}^2$$

$$\text{Circ. Comp. Stress at Horn of Saddle, } L < 8R_m \text{ (4.15.29) } [\sigma_7, r^*]: \\ = -Q / (4(t + \eta \cdot t_r)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot t_r)^2) \\ = -35051 / (4(21.0 + 1.0 \cdot 24.0)314.0) -$$

149-D

Refrigerant receiver
QUEP

PV Elite 28 U1

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Saddle Calcs: Operating Case: Step: 21 12:03pm Sep 16,2026



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$$12 \cdot 0.013 \cdot 35051 \cdot 1213.5 / (7600.0 (21.0 + 1.0 \cdot 24.0)^2) \\ = -10.38 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:
= 4460.0 mm.

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:
= $\alpha \cdot Ls$ (Design Temperature - Ambient Temperature)
= $0.000012 \cdot 4460.0$ (130.0 - 21.1)
= 5.972 mm.

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm.
Baseplate Thickness	Bpthk	19.0500	mm.
Baseplate Width	Bpwid	314.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.8750	mm.
Web Thickness	Webtk	15.8750	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	234.4	N./mm ²
Height of Web at Center	Hw,c	637.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
= $((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$
= $((1.56 \cdot \sqrt{(1203.0 \cdot 21.0)} + 315.0) 21.0^3) / 12$
= 43.446 cm⁴

Resolved Inertia for the Shell [I,shell]:
= ShellInertia + AShell(C1 - Y)²
= 43.446 + 118.22(231.098 - 10.5)²
= 57573.203 cm⁴

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² cm ⁴
Shell	563.0	21.0	10.5	118.2	124130.8	0.576E+05
Wearplate	315.0	24.0	33.0	75.6	249480.0	0.297E+05
Web	15.9	637.9	364.0	101.3	3686138.0	0.522E+05
BasePlate	314.0	19.0	692.5	59.8	4142175.5	0.127E+06
Totals	...	...	...	354.9	8201924.5	0.267E+06

Distance to Centroid [C1]:
= AY / A
= 8201925/354.911
= 231.098 mm.

Angle [β]:
= 180 - Saddle Angle/2
= 180 - 120.0/2
= 120.0

Saddle Splitting Coefficient [K1]:
= $(1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
= $(1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$
= 0.2035

Saddle Splitting Force [Fh]:
= K1 • Q
= 0.204 • 35050.77
= 7133.5918 Kgf

Tension Stress, St = (Fh/As) = 2.9557 N./mm²
Allowed Stress, Sa = 0.6 * Yield Str = 140.6383 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1905.0 - 1203.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 910.127 \text{ mm.}$$

$$\text{Bending Moment, } M = F_h \cdot d = 6492.6016 \text{ Kg-m.}$$

$$\text{Bending Stress, } S_b = (M \cdot C_1 / I) = 5.5139 \text{ N./mm}^2$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 156.2648 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(35051 + 760) \cdot 314.0 / (4 \cdot 2350.0 \cdot 156.265))}$$

$$= 15.007 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1200.0 + 24.0 + 24.0)$$

$$= 2161.599 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2161.5994 / (5 - 1)$$

$$= 540.400 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 540.3998 \cdot 314.0/2$$

$$= 848.428 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 35810.547 / (2350.0 \cdot 314.0)$$

$$= 4.853 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 848.4 \cdot 4.85$$

$$= 4117.449 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 33.77 + 42.894$$

$$= 76.664 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 4117.4 / 76.6643$$

$$= 5.267 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	15.9	228.6	...	36.3	...	0.158E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 228.6 - 15.875$$

$$= 212.725 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

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$$= AY / A$$

$$= 0.0/76.664$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228.6/2$$

$$= 114.300 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1580.4/76.664)}$$

$$= 45.403 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1905.0 - \cos(120.0/2) (1200.0 + 24.0 + 24.0) - 19.05$$

$$= 1261.950 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/234.4)}$$

$$= 129.764$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1261.95/45.403$$

$$= 27.794$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 6038.0 / (2 \cdot 2350.0) \cdot 540.4 \cdot 1261.95/2$$

$$= 438.060 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($27.7945 < 129.7641$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (27.79)^2 / (2 \cdot 129.76^2)) 234 / (5/3 + 3 \cdot (27.79) / (8 \cdot 129.76) - (27.79^3) / (8 \cdot 129.76^3))$$

$$= 131.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 5.27/131.19 + (438.06 \cdot 114.3/15803786) / 156.26$$

$$= 0.239$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.9	212.7	0.0	33.8	0.0	0.158E+04
Web	540.4	15.9	0.0	85.8	0.0	18.0
Totals	...	...	...	119.6	...	0.160E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/119.559$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228.6/2$$

$$= 114.300 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1905.0 - \cos(1248.0/2) (19.05 + 0.0 + 0.0) - 0.0$$

= 637.950 mm.

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{1597.9/119.559}$

= 36.558 mm.

Slenderness ratio [KL/r]:

= KL/r

= $1 \cdot 637.95/36.558$

= 17.450

Unit Force [Force,u]:

= $Fl / (2 \cdot \text{Baseplate Length})$

= $6038.037 / (2 \cdot 2350.0)$

= 1.285 Kg/mm.

Moment at base of inner Rib [Mbase,c]:

= Unit Force $\cdot e \cdot L$

= $1.285 \cdot 540.4 \cdot 637.95$

= 442.903 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= $442.903/139795.812$

= 31.069 N./mm²

Compressive Allowable, KL/r < Cc (17.4505 < 129.7641) per AISC E2-1 [Sca]:

= $(1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$

= $(1 - (17.45)^2 / (2 \cdot 129.76^2))^{2/3} /$

$(5/3 + 3 \cdot (17.45) / (8 \cdot 129.76) - (17.45^3) / (8 \cdot 129.76^3))$

= 135.3 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $Sc/Sca + (Mbase,c \cdot C1/I) / Sba$

= $6.26/135.3 + (442.9 \cdot 114.3/1597.865) / 156.26$

= 0.245

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm2
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	75.0 mm.
Nominal Bolt Diameter	Bnd	24.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm2
Area Available in a Single Bolt	BlArea	2.2577 cm2
Saddle Load Q0 (Weight)	Q0	27828.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	2579.0 Kg
Maximum Transverse Force	Ft	3019.0 Kg
Maximum Longitudinal Force	Fl	6038.0 Kg
Saddle Bolted to Steel Foundation		Yes

Friction force due to thermal growth, [Ff]:

= 13942.3 Kg

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= $\max(Ff, Fl) / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $13942.3 / (2.258 \cdot 8)$

= 75.7 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$



$$= 3019.0 / (2.258 \cdot 8)$$

$$= 16.4 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 $= 6038.04 / (82.8 \cdot 8.0)$
 $= 0.8939 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot F_t + \text{Sum of X Moments}$
 $= 1905.0 \cdot 3019.02 + 0.0$
 $= 5751.34 \text{ Kg-m.}$

Eccentricity (e):
 $= \text{Rmom} / \text{QO}$
 $= 5751.34 / 27828.64$
 $= 206.67 \text{ mm.} < \text{Bplen}/6 \text{ --> No Uplift in Transverse direction}$

Bolt Area due to Transverse Load [Bltareart]:
 $= 0$ (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 $= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 $= \max[0.0, 0.8939, 0.0]$
 $= 0.8939 \text{ cm}^2$

Note: Saddle and wear plate material are both specified as impact tested.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1213.50	mm.
Shell Thickness used in this Case	t	24.000	mm.
Stiffened Vessel Length per 4.15.6	L	7600.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	157.00	mm.
Saddle Width	b1 or b	228.60	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	315.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm2
Inside or Mean Depth of Head	Hi or hm	613.00	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm2
Head Allowable Stress used in Calculation		138.00	N./mm2
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1905.00	mm.
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		53696.56	Kgf
Pressure used in Analysis	P	22.169	bars



Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	55.26	138.00
Long. Stress at Bottom of Midspan	72.86	138.00
Long. Stress at Top of Saddles	63.80	138.00
Long. Stress at Bottom of Saddles	64.31	138.00
Tangential Shear in Shell	18.18	110.40
Tangential Shear in Head	19.09	110.40
Circ. Stress at Horn of Saddle	15.90	172.50
Addl. Stress in Head as Stiffener	139.56	172.50
Circ. Compressive Stress in Shell	2.83	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(3277.2/2 + 0) \cdot 1905.0/2161.5994$$

$$= 4332.3 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(720.7, 0) \cdot 1905.0/4460.0$$

$$= 307.8 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(6038.04, 0) \cdot 1905.0/4460.0$$

$$= 2579.0 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(6038/2 + 0) \cdot 1905.0/2161.5994$$

$$= 7981.9 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 45715 + \max(308, 4332, 2579, 7982)$$

$$= 53696.6 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.0 \cdot 766.08(2.448 + \pi/4(2.448 \cdot 1.2)^2)$$

$$= 7067.590 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	54456.34	Kgf
Transverse Shear Load	3019.02	Kgf
Longitudinal Shear Load	6038.04	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(2400.0/2 + 24.0 + 24.0)\sin(120.0/2), 2 \cdot 1213.5)$$

$$= 2161.599 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0109			

The suffix 'p' denotes the values for a wear plate if it exists.



Note: Dimension a is less than Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -53697 \cdot 157.0 [1 - (1 - 157.0/7600.0 + (1213.5^2 - 613.0^2)/(2 \cdot 157.0 \cdot 7600.0))/(1 + (4 \cdot 613.0)/(3 \cdot 7600.0))]$$

$$= 2522.7 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 53697 \cdot 7600/4 ((1 + 2(1213.5^2 - 613^2)/(7600^2))/(1 + (4 \cdot 613)/(3 \cdot 7600)) - 4 \cdot 157/7600)$$

$$= 87186.7 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 22.169 \cdot 1213.5/(2 \cdot 21.0) - 87186.7/(\pi \cdot 1213.5^2 \cdot 21.0)$$

$$= 55.26 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 22.169 \cdot 1213.5/(2 \cdot 21.0) + 87186.7/(\pi \cdot 1213.5^2 \cdot 21.0)$$

$$= 72.86 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot Rm/(2t) - M1/(\pi \cdot Rm^2 \cdot t)$$

$$= 22.169 \cdot 1213.5/(2 \cdot 21.0) - 2522.7/(\pi \cdot 1213.5^2 \cdot 21.0)$$

$$= 63.80 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot Rm/(2t) + M1/(\pi \cdot Rm^2 \cdot t)$$

$$= 22.169 \cdot 1213.5/(2 \cdot 21.0) + 2522.7/(\pi \cdot 1213.5^2 \cdot 21.0)$$

$$= 64.31 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 53697(7600.0 - 2 \cdot 157.0)/(7600.0 + (4 \cdot 613.0/3))$$

$$= 46479.5 \text{ Kgf}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K3 \cdot Q / (Rm \cdot t)$$

$$= 0.8799 \cdot 53697/(1213.5 \cdot 21.0)$$

$$= 18.18 \text{ N./mm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_3^*]:

$$= K3 \cdot Q / (Rm \cdot th)$$

$$= 0.8799 \cdot 53697/(1213.5 \cdot 20.0)$$

$$= 19.09 \text{ N./mm}^2$$

Membrane stress in the head as a stiffener (4.15.16) [σ_5]:

$$= K4 \cdot Q/(Rm \cdot th) + ((P \cdot Ri)/(2 \cdot th)) \cdot (Ri/hm)$$

$$= 0.4011 \cdot 53697/(1213.5 \cdot 20.0) + ((22.17 \cdot 1203.0)/(2 \cdot 20.0)) \cdot (1203.0/613.0)$$

$$= 139.56 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1213.5 \cdot 21.0)}$$

$$= 124.516 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(228.6 + 1.56 \cdot \sqrt{(1213.5 \cdot 21.0)}, 2 \cdot 157.0)$$

$$= 314.00 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

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$$= \min(S_r/S, 1)$$

$$= \min(137.997/137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) $[\sigma_6, r]$:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 53697 \cdot 0.1 / (314.0(21.0 + 1.0 \cdot 24.0))$$

$$= -2.83 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) $[\sigma_7, r^*]$:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -53697 / (4(21.0 + 1.0 \cdot 24.0)314.0) -$$

$$12 \cdot 0.013 \cdot 53697 \cdot 1213.5 / (7600.0(21.0 + 1.0 \cdot 24.0)^2)$$

$$= -15.90 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm.
Baseplate Thickness	Bpthk	19.0500	mm.
Baseplate Width	Bpwid	314.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.8750	mm.
Web Thickness	Webtk	15.8750	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	234.4	N./mm ²
Height of Web at Center	Hw,c	637.0	mm.
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1203.0 \cdot 21.0)} + 315.0)^3) / 12$$

$$= 43.446 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 43.446 + 118.22(231.098 - 10.5)^2$$

$$= 57573.211 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	563.0	21.0	10.5	118.2	124130.8	0.576E+05
Wearplate	315.0	24.0	33.0	75.6	249480.0	0.297E+05
Web	15.9	637.9	364.0	101.3	3686138.8	0.522E+05
BasePlate	314.0	19.0	692.5	59.8	4142176.0	0.127E+06
Totals	...	...	...	354.9	8201926.0	0.267E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 8201926/354.911$$

$$= 231.098 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$

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$$= 0.204 \cdot 53696.562$$

$$= 10928.4150 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 4.5280 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 140.6383 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1905.0 - 1203.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 910.127 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 9946.4424 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 8.4471 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 156.2648 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{3(53697 + 760) \cdot 314.0 / (4 \cdot 2350.0 \cdot 156.265)}$$

$$= 18.506 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1200.0 + 24.0 + 24.0)$$

$$= 2161.599 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2161.5991 / (5 - 1)$$

$$= 540.400 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 540.3998 \cdot 314.0 / 2$$

$$= 848.428 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 54456.34 / (2350.0 \cdot 314.0)$$

$$= 7.380 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 848.4 \cdot 7.38$$

$$= 6261.319 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 33.77 + 42.894$$

$$= 76.664 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 6261.3 / 76.6643$$

$$= 8.009 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	15.9	228.6	...	36.3	...	0.158E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
= 228.6 - 15.875
= 212.725 mm.

Distance to Centroid from Datum [ytot]:
= AY / A
= 0.0/76.664
= 0.000 mm.

Distance to Centroid [C1]:
= Saddle Width / 2
= 228.6/2
= 114.300 mm.

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{1580.4/76.664}$
= 45.403 mm.

Length of Outer Rib [L]:
= Saddle Height - $\cos(\theta/2) \cdot (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
= 1905.0 - $\cos(120.0/2) \cdot (1200.0 + 24.0 + 24.0) - 19.05$
= 1261.950 mm.

Intermediate Term [Cc]:
= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/234.4}$
= 129.764

Slenderness ratio [KL/r]:
= KL/r
= 1 * 1261.95/45.403
= 27.794

Bending Moment [Rm]:
= FI / (2 * Bplen) * e * L / 2
= 6038.0 / (2 * 2350.0) * 540.4 * 1261.95/2
= 438.060 Kg-m.

Compressive Allowable, KL/r < Cc (27.7945 < 129.7641) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (27.79)^2 / (2 \cdot 129.76^2)) 234 / (5/3 + 3 \cdot (27.79) / (8 \cdot 129.76) - (27.79^3) / (8 \cdot 129.76^3))$
= 131.2 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
= Sc/Sca + (Rm * C1 / I) / Sba
= 8.01/131.19 + (438.06 * 114.3/15803786) / 156.26
= 0.260

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.9	212.7	0.0	33.8	0.0	0.158E+04
Web	540.4	15.9	0.0	85.8	0.0	18.0
Totals	...	...	...	119.6	...	0.160E+04

Distance to Centroid from Datum [ytot]:
= AY / A
= 0.0/119.559
= 0.000 mm.

Distance to Centroid [C1]:
= Saddle Width / 2
= 228.6/2

= 114.300 mm.

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 1905.0 - cos(1248.0/2)(19.05 + 0.0 + 0.0) - 0.0

= 637.950 mm.

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{1597.9/119.559}$

= 36.558 mm.

Slenderness ratio [KL/r]:

= KL/r

= 1 • 637.95/36.558

= 17.450

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 6038.037/(2 • 2350.0)

= 1.285 Kg/mm.

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 1.285 • 540.4 • 637.95

= 442.903 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 442.903/139795.812

= 31.069 N./mm²

Compressive Allowable, KL/r < Cc (17.4505 < 129.7641) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (17.45)^2 / (2 \cdot 129.76^2)) 234 / (5/3 + 3 \cdot (17.45) / (8 \cdot 129.76) - (17.45)^3 / (8 \cdot 129.76^3))$

= 135.3 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 9.51/135.3 + (442.9 • 114.3/1597.865)/156.26

= 0.269

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm2
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	75.0 mm.
Nominal Bolt Diameter	Bnd	24.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm2
Area Available in a Single Bolt	BlArea	2.2577 cm2
Saddle Load Q0 (Weight)	Q0	46474.4 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	2579.0 Kg
Maximum Transverse Force	Ft	3019.0 Kg
Maximum Longitudinal Force	Fl	6038.0 Kg
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= FI / (Bolt Area • Number of Bolts)

= 6038.0/(2.258 • 8)

= 32.8 N./mm². Must be less than 82.8 N./mm².



Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3019.0 / (2.258 \cdot 8)$$

$$= 16.4 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 6038.04 / (82.8 \cdot 8.0)$$

$$= 0.8939 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1905.0 \cdot 3019.02 + 0.0$$

$$= 5751.34 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 5751.34 / 46474.43$$

$$= 123.75 \text{ mm.} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.8939, 0.0]$$

$$= 0.8939 \text{ cm}^2$$

Note: Saddle and wear plate material are both specified as impact tested.

Review notes about nozzle loadings and supports in the Warnings report.

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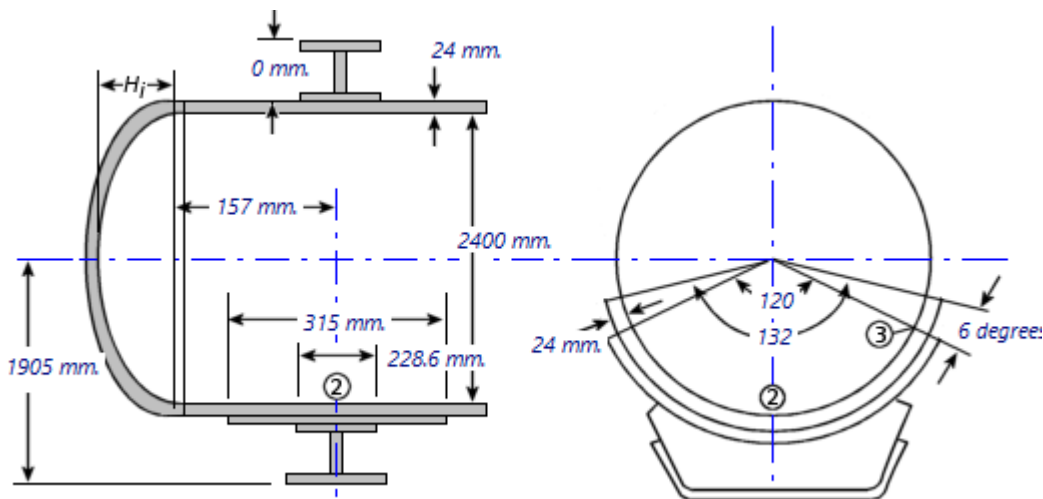


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1212.00	mm.
Shell Thickness used in this Case	t	24.000	mm.
Stiffened Vessel Length per 4.15.6	L	7600.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	157.00	mm.
Saddle Width	b1 or b	228.60	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	315.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	611.50	mm.
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1905.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		36488.94	Kgf
Pressure used in Analysis	P	28.849	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	67.60	234.00
Long. Stress at Bottom of Midspan	78.09	234.00
Long. Stress at Top of Saddles	72.70	234.00
Long. Stress at Bottom of Saddles	73.00	234.00
Tangential Shear in Shell	10.82	138.58
Tangential Shear in Head	11.30	124.94
Circ. Stress at Horn of Saddle	9.86	350.99

Addl. Stress in Head as Stiffener	152.84	234.00
Circ. Compressive Stress in Shell	1.81	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (1638.6/2 + 0) \cdot 1905.0/2161.5996$
 $= 2166.2 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(360.35, 0) \cdot 1905.0/4460.0$
 $= 153.9 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 34323 + \max(154, 2166, 0, 0)$
 $= 36488.9 \text{ Kgf}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.0 \cdot 766.08 (2.448 + \pi/4 (2.448 \cdot 1.2)^2)$
 $= 7067.591 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	37248.71	Kgf
Transverse Shear Load	819.31	Kgf
Longitudinal Shear Load	360.35	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(2400.0/2 + 24.0 + 24.0) \sin(120.0/2), 2 \cdot 1212.0)$
 $= 2161.600 \text{ mm.}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0109			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is less than $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -36489 \cdot 157.0 [1 - (1 - 157.0/7600.0 + (1212.0^2 - 611.5^2)/(2 \cdot 157.0 \cdot 7600.0)) / (1 + (4 \cdot 611.5)/(3 \cdot 7600.0))]$
 $= 1712.1 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 \{ (1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L \}$
 $= 36489 \cdot 7600/4 \{ (1 + 2(1212^2 - 612^2)/(7600^2)) / (1 + (4 \cdot 612)/(3 \cdot 7600)) - 4 \cdot 157/7600 \}$
 $= 59258.3 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 28.849 \cdot 1212.0/(2 \cdot 24.0) - 59258.3/(\pi \cdot 1212.0^2 \cdot 24.0)$
 $= 67.60 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$



$$= 28.849 \cdot 1212.0 / (2 \cdot 24.0) + 59258.3 / (\pi \cdot 1212.0^2 \cdot 24.0)$$

$$= 78.09 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 28.849 \cdot 1212.0 / (2 \cdot 24.0) - 1712.1 / (\pi \cdot 1212.0^2 \cdot 24.0)$$

$$= 72.70 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 28.849 \cdot 1212.0 / (2 \cdot 24.0) + 1712.1 / (\pi \cdot 1212.0^2 \cdot 24.0)$$

$$= 73.00 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 36489(7600.0 - 2 \cdot 157.0) / (7600.0 + (4 \cdot 611.5 / 3))$$

$$= 31592.1 \text{ Kgf}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K_3 \cdot Q / (R_m \cdot t)$$

$$= 0.8799 \cdot 36489 / (1212.0 \cdot 24.0)$$

$$= 10.82 \text{ N./mm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_{3*}]:

$$= K_3 \cdot Q / (R_m \cdot t_h)$$

$$= 0.8799 \cdot 36489 / (1212.0 \cdot 23.0)$$

$$= 11.30 \text{ N./mm}^2$$

Membrane stress in the head as a stiffener (4.15.16) [σ_5]:

$$= K_4 \cdot Q / (R_m \cdot t_h) + ((P \cdot R_i) / (2 \cdot t_h)) \cdot (R_i / h_m)$$

$$= 0.4011 \cdot 36489 / (1212.0 \cdot 23.0) + ((28.85 \cdot 1200.0) / (2 \cdot 23.0)) \cdot (1200.0 / 611.5)$$

$$= 152.84 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1212.0 \cdot 24.0)}$$

$$= 133.031 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(228.6 + 1.56 \cdot \sqrt{(1212.0 \cdot 24.0)}, 2 \cdot 157.0)$$

$$= 314.00 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 36489 \cdot 0.1 / (314.0(24.0 + 1.0 \cdot 24.0))$$

$$= -1.81 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [$\sigma_7, r*$]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -36489 / (4(24.0 + 1.0 \cdot 24.0)314.0) - 12 \cdot 0.013 \cdot 36489 \cdot 1212.0 / (7600.0(24.0 + 1.0 \cdot 24.0)^2)$$

$$= -9.86 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm.
Baseplate Thickness	Bpthk	19.0500	mm.
Baseplate Width	Bpwid	314.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.8750	mm.
Web Thickness	Webtk	15.8750	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm2

149-D

Refrigerant receiver
QUEP

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Saddle Calcs: Test Case: Step: 22 12:03pm Sep 16,2026



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Height of Web at Center Hw,c 637.0 mm.
Friction Coefficient mu 0.000

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1200.0 \cdot 24.0)} + 315.0)^3) / 12$$

$$= 66.786 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 66.786 + 139.138 (221.264 - 12.0)^2$$

$$= 60997.418 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	579.7	24.0	12.0	139.1	166965.3	0.610E+05
Wearplate	315.0	24.0	36.0	75.6	272160.0	0.260E+05
Web	15.9	637.9	367.0	101.3	3716520.0	0.558E+05
BasePlate	314.0	19.0	695.5	59.8	4160120.5	0.135E+06
Totals	...	...	...	375.8	8315766.0	0.277E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 8315766 / 375.829$$

$$= 221.264 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 36488.938$$

$$= 7426.2900 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 3.0769 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1905.0 - 1200.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 912.608 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 6777.4243 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 5.3021 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3 \cdot (Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3 \cdot (36489 + 760) \cdot 314.0) / (4 \cdot 2350.0 \cdot 173.33)}$$

$$= 14.533 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1200.0 + 24.0 + 24.0)$$



$$= 2161.599 \text{ mm.}$$

$$\begin{aligned} \text{Distance between Ribs [e]:} \\ &= \text{Web Length} / (\text{Nribs} - 1) \\ &= 2161.5994 / (5 - 1) \\ &= 540.400 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Baseplate Pressure Area [Ap]:} \\ &= e \cdot \text{Bpwid} / 2 \\ &= 540.3998 \cdot 314.0 / 2 \\ &= 848.428 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Bearing Pressure [Bp]:} \\ &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 37248.715 / (2350.0 \cdot 314.0) \\ &= 5.048 \text{ Kg/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Axial Load [P]:} \\ &= \text{Ap} \cdot \text{Bp} \\ &= 848.4 \cdot 5.05 \\ &= 4282.808 \text{ Kg} \end{aligned}$$

$$\begin{aligned} \text{Area of the Rib and Web [Ar]:} \\ &= \text{Rib Area} + \text{Web Area} \\ &= 33.77 + 42.894 \\ &= 76.664 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Compressive Stress [Sc]:} \\ &= P / \text{Ar} \\ &= 4282.8 / 76.6643 \\ &= 5.479 \text{ N./mm}^2 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	15.9	228.6	...	36.3	...	0.158E+04

$$\begin{aligned} \text{Rib dimension [D]:} \\ &= \text{Saddle Width} - \text{Web Thickness} \\ &= 228.6 - 15.875 \\ &= 212.725 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Distance to Centroid from Datum [ytot]:} \\ &= \text{AY} / \text{A} \\ &= 0.0 / 76.664 \\ &= 0.000 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Distance to Centroid [C1]:} \\ &= \text{Saddle Width} / 2 \\ &= 228.6 / 2 \\ &= 114.300 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Radius of Gyration [r]:} \\ &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\ &= \sqrt{(1580.4 / 76.664)} \\ &= 45.403 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Length of Outer Rib [L]:} \\ &= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\ &= 1905.0 - \cos(120.0/2) (1200.0 + 24.0 + 24.0) - 19.05 \\ &= 1261.950 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Intermediate Term [Cc]:} \\ &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \end{aligned}$$

149-D

Refrigerant receiver
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Saddle Calcs: Test Case: Step: 22 12:03pm Sep 16,2026



HEXAGON

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994E+09/260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1261.95/45.403$$

$$= 27.794$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 360.4 / (2 \cdot 2350.0) \cdot 540.4 \cdot 1261.95/2$$

$$= 26.144 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($27.7945 < 123.2106$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (27.79)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (27.79) / (8 \cdot 123.21) - (27.79^3) / (8 \cdot 123.21^3))$$

$$= 144.8 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 5.48/144.8 + (26.14 \cdot 114.3/15803786) / 173.33$$

$$= 0.049$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.9	212.7	0.0	33.8	0.0	0.158E+04
Web	540.4	15.9	0.0	85.8	0.0	18.0
Totals	...	...	...	119.6	...	0.160E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/119.559$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228.6/2$$

$$= 114.300 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1905.0 - \cos(1248.0/2) (19.05 + 0.0 + 0.0) - 0.0$$

$$= 637.950 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{1597.9/119.559}$$

$$= 36.558 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 637.95/36.558$$

$$= 17.450$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 360.352 / (2 \cdot 2350.0)$$

$$= 0.077 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.077 \cdot 540.4 \cdot 637.95$$

$$= 26.433 \text{ Kg-m.}$$



Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

= Bending Moment / Section Modulus

= 26.433/139795.812

= 1.854 N./mm²

Compressive Allowable, $KL/r < C_c$ (17.4505 < 123.2106) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)

= (1-(17.45)²/(2 • 123.21²))260/

(5/3+3•(17.45)/(8• 123.21)-(17.45³)/(8•123.21³)

= 149.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$

= 6.51/149.69 + (26.43 • 114.3/1597.865)/173.33

= 0.054

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	75.0 mm.
Nominal Bolt Diameter	Bnd	24.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	2.2577 cm ²
Saddle Load QO (Weight)	QO	35082.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	153.9 Kg
Maximum Transverse Force	Ft	819.3 Kg
Maximum Longitudinal Force	Fl	360.4 Kg
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [$\tau_{b,l}$]:

= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= 360.4/(2.258 • 8)

= 2.0 N./mm². Must be less than 29.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [$\tau_{b,t}$]:

= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= 819.3/(2.258 • 8)

= 4.4 N./mm². Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [$Bltarear_l$]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [$Bltarears$]:

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$

= 360.35/(28.98 • 8.0)

= 0.1524 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [R_{mom}]:

= $B \cdot F_t + \text{Sum of X Moments}$

= 1905.0 • 819.31 + 0.0

= 1560.82 Kg-m.

Eccentricity (e):

= R_{mom} / QO

= 1560.82/35082.56

= 44.49 mm. < $B_{plen}/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 0.1524, 0.0]
= 0.1524 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1212.00	mm.
Shell Thickness used in this Case	t	24.000	mm.
Stiffened Vessel Length per 4.15.6	L	7600.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	157.00	mm.
Saddle Width	b1 or b	228.60	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	315.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	611.50	mm.
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1905.00	mm.
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		58042.58	Kgf
Pressure used in Analysis	P	28.849	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	64.50	234.00
Long. Stress at Bottom of Midspan	81.19	234.00
Long. Stress at Top of Saddles	72.61	234.00
Long. Stress at Bottom of Saddles	73.09	234.00
Tangential Shear in Shell	17.22	138.58
Tangential Shear in Head	17.97	124.93
Circ. Stress at Horn of Saddle	15.69	350.99
Addl. Stress in Head as Stiffener	155.88	234.00
Circ. Compressive Stress in Shell	2.87	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3.0(1638.6/2 + 0) • 1905.0/2161.5994
= 2166.2 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
= max(Fl,wind; Sum of X Forces) • B / Ls
= max(360.35, 0) • 1905.0/4460.0
= 153.9 Kgf



Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 55876 + \max(154, 2166, 0, 0)$$

$$= 58042.6 \text{ Kg}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.0 \cdot 766.08(2.448 + \pi/4(2.448 \cdot 1.2)^2)$$

$$= 7067.590 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	58802.36	Kgf
Transverse Shear Load	819.31	Kgf
Longitudinal Shear Load	360.35	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(2400.0/2 + 24.0 + 24.0)\sin(120.0/2), 2 \cdot 1212.0)$$

$$= 2161.599 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0109			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is less than $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -58043 \cdot 157.0 [1 - (1 - 157.0/7600.0 + (1212.0^2 - 611.5^2)/(2 \cdot 157.0 \cdot 7600.0))/(1 + (4 \cdot 611.5)/(3 \cdot 7600.0))]$$

$$= 2723.4 \text{ Kg-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 58043 \cdot 7600/4((1 + 2(1212^2 - 612^2)/(7600^2))/(1 + (4 \cdot 612)/(3 \cdot 7600)) - 4 \cdot 157/7600)$$

$$= 94261.6 \text{ Kg-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 28.849 \cdot 1212.0/(2 \cdot 24.0) - 94261.6/(\pi \cdot 1212.0^2 \cdot 24.0)$$

$$= 64.50 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 28.849 \cdot 1212.0/(2 \cdot 24.0) + 94261.6/(\pi \cdot 1212.0^2 \cdot 24.0)$$

$$= 81.19 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m/(2t) - M1/(\pi \cdot R_m^2 \cdot t)$$

$$= 28.849 \cdot 1212.0/(2 \cdot 24.0) - 2723.4/(\pi \cdot 1212.0^2 \cdot 24.0)$$

$$= 72.61 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m/(2t) + M1/(\pi \cdot R_m^2 \cdot t)$$

$$= 28.849 \cdot 1212.0/(2 \cdot 24.0) + 2723.4/(\pi \cdot 1212.0^2 \cdot 24.0)$$

$$= 73.09 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

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Refrigerant receiver
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$$= 58043(7600.0 - 2 \cdot 157.0)/(7600.0 + (4 \cdot 611.5/3))$$

$$= 50253.3 \text{ Kg}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K_3 \cdot Q / (R_m \cdot t)$$

$$= 0.8799 \cdot 58043 / (1212.0 \cdot 24.0)$$

$$= 17.22 \text{ N./mm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_{3*}]:

$$= K_3 \cdot Q / (R_m \cdot t_h)$$

$$= 0.8799 \cdot 58043 / (1212.0 \cdot 23.0)$$

$$= 17.97 \text{ N./mm}^2$$

Membrane stress in the head as a stiffener (4.15.16) [σ_5]:

$$= K_4 \cdot Q / (R_m \cdot t_h) + ((P \cdot R_i) / (2 \cdot t_h)) \cdot (R_i / h_m)$$

$$= 0.4011 \cdot 58043 / (1212.0 \cdot 23.0) +$$

$$((28.85 \cdot 1200.0) / (2 \cdot 23.0)) \cdot (1200.0 / 611.5)$$

$$= 155.88 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1212.0 \cdot 24.0)}$$

$$= 133.031 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B_1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(228.6 + 1.56 \cdot \sqrt{(1212.0 \cdot 24.0)}, 2 \cdot 157.0)$$

$$= 314.00 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r))$$

$$= -0.7603 \cdot 58043 \cdot 0.1 / (314.0(24.0 + 1.0 \cdot 24.0))$$

$$= -2.87 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot t_r) b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot t_r)^2)$$

$$= -58043 / (4(24.0 + 1.0 \cdot 24.0) 314.0) -$$

$$12 \cdot 0.013 \cdot 58043 \cdot 1212.0 / (7600.0(24.0 + 1.0 \cdot 24.0)^2)$$

$$= -15.69 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm.
Baseplate Thickness	Bpthk	19.0500	mm.
Baseplate Width	Bpwid	314.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	15.8750	mm.
Web Thickness	Webtk	15.8750	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm2
Height of Web at Center	Hw,c	637.0	mm.
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1200.0 \cdot 24.0)} + 315.0) 24.0^3) / 12$$

$$= 66.786 \text{ cm}^4$$

Resolved Inertia for the Shell [I_{shell}]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 66.786 + 139.138(221.264 - 12.0)^2$$

$$= 60997.418 \text{ cm}^4$$



	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² (2) cm ⁴
Shell	579.7	24.0	12.0	139.1	166965.3	0.610E+05
Wearplate	315.0	24.0	36.0	75.6	272160.0	0.260E+05
Web	15.9	637.9	367.0	101.3	3716521.0	0.558E+05
BasePlate	314.0	19.0	695.5	59.8	4160121.0	0.135E+06
Totals	...	...	...	375.8	8315767.0	0.277E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 8315767/375.829$$

$$= 221.264 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 58042.582$$

$$= 11812.9248 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As)$$

=

$$4.8944 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str}$$

=

$$155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1905.0 - 1200.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 912.608 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d$$

=

$$10780.7832 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I)$$

=

$$8.4340 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str}$$

=

$$173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(58043 + 760)314.0 / (4 \cdot 2350.0 \cdot 173.33)}$$

$$= 18.259 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1200.0 + 24.0 + 24.0)$$

$$= 2161.599 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2161.5991 / (5 - 1)$$

$$= 540.400 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 540.3998 \cdot 314.0/2$$

$$= 848.428 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 58802.359 / (2350.0 \cdot 314.0)$$

$$= 7.969 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 848.4 \cdot 7.97$$

$$= 6761.018 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 33.77 + 42.894$$

$$= 76.664 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 6761.0 / 76.6643$$

$$= 8.649 \text{ N/mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	15.9	228.6	...	36.3	...	0.158E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 228.6 - 15.875$$

$$= 212.725 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 76.664$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228.6 / 2$$

$$= 114.300 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{1580.4 / 76.664}$$

$$= 45.403 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1905.0 - \cos(120.0/2) (1200.0 + 24.0 + 24.0) - 19.05$$

$$= 1261.950 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1261.95 / 45.403$$

$$= 27.794$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 360.4 / (2 \cdot 2350.0) \cdot 540.4 \cdot 1261.95 / 2$$

$$= 26.144 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($27.7945 < 123.2106$) per AISC E2-1 [Sca]:

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Refrigerant receiver
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$$\begin{aligned}
 &= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3)) \\
 &= (1 - (27.79)^2 / (2 \cdot 123.21^2)) 260 / \\
 &\quad (5/3 + 3 \cdot (27.79) / (8 \cdot 123.21) - (27.79^3) / (8 \cdot 123.21^3)) \\
 &= 144.8 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= Sc/Sca + (Rm \cdot C1 / I) / Sba \\
 &= 8.65/144.8 + (26.14 \cdot 114.3/15803786) / 173.33 \\
 &= 0.070
 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.9	212.7	0.0	33.8	0.0	0.158E+04
Web	540.4	15.9	0.0	85.8	0.0	18.0
Totals	...	...	...	119.6	...	0.160E+04

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0.0/119.559 \\
 &= 0.000 \text{ mm.}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 228.6/2 \\
 &= 114.300 \text{ mm.}
 \end{aligned}$$

Length of Inner Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \text{Outside Radius} - Bpthk \\
 &= 1905.0 - \cos(1248.0/2) (19.05 + 0.0 + 0.0) - 0.0 \\
 &= 637.950 \text{ mm.}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(1597.9/119.559)} \\
 &= 36.558 \text{ mm.}
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 637.95/36.558 \\
 &= 17.450
 \end{aligned}$$

Unit Force [Force,u]:

$$\begin{aligned}
 &= FI / (2 \cdot \text{Baseplate Length}) \\
 &= 360.352 / (2 \cdot 2350.0) \\
 &= 0.077 \text{ Kg/mm.}
 \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 0.077 \cdot 540.4 \cdot 637.95 \\
 &= 26.433 \text{ Kg-m.}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 26.433/139795.812 \\
 &= 1.854 \text{ N./mm}^2
 \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ (17.4505 < 123.2106) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3)) \\
 &= (1 - (17.45)^2 / (2 \cdot 123.21^2)) 260 / \\
 &\quad (5/3 + 3 \cdot (17.45) / (8 \cdot 123.21) - (17.45^3) / (8 \cdot 123.21^3)) \\
 &= 149.7 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$



$$= 10.27/149.69 + (26.43 \cdot 114.3/1597.865)/173.33$$

$$= 0.079$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	75.0 mm.
Nominal Bolt Diameter	Bnd	24.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	2.2577 cm ²
Saddle Load QO (Weight)	QO	56636.2 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	153.9 Kg
Maximum Transverse Force	Ft	819.3 Kg
Maximum Longitudinal Force	Fl	360.4 Kg
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 360.4 / (2.258 \cdot 8)$$

$$= 2.0 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 819.3 / (2.258 \cdot 8)$$

$$= 4.4 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 360.35 / (82.8 \cdot 8.0)$$

$$= 0.0534 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1905.0 \cdot 819.31 + 0.0$$

$$= 1560.82 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 1560.82/56636.2$$

$$= 27.56 \text{ mm.} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.0534, 0.0]$$

$$= 0.0534 \text{ cm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: M124 inch

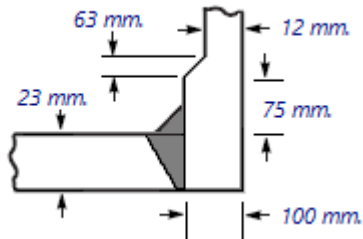
From: 10

Pressure for Reinforcement Calculations	P	22.184	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	2400.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	23.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	63.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		23.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: M124 inch

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 22.18 \cdot 2406.0 \cdot 0.8979 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 22.18)$
 $= 17.3927 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 22.18 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 22.18)$
 $= 4.8688 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 88.0 / 0.5773$
 $= 152.4205 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1183.2001 mm.
Parallel to Vessel Wall, opening length	d	591.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	50.0000 mm.

Weld Strength Reduction Factor [fr1]:

$= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

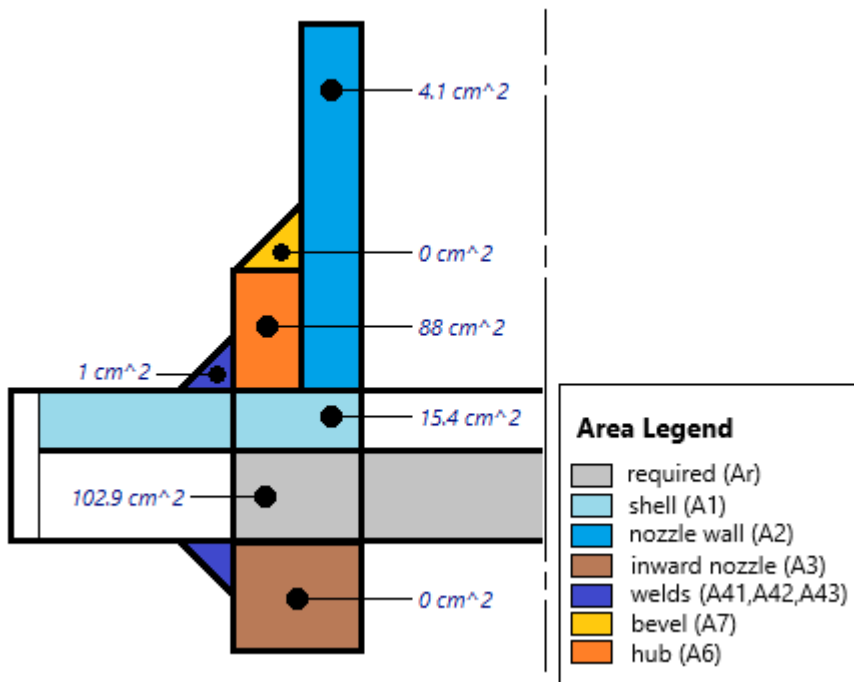


Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1.0, 1.0)$$

$$= 1.000$$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	102.895	NA	NA
Area in Shell	A1	15.425	NA	NA
Area in Nozzle Wall	A2	4.131	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	88.000	NA	NA
TOTAL AREA AVAILABLE	Atot	108.556	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (591.6 \cdot 17.3927 \cdot 1.0 + 2 \cdot 97.0 \cdot 17.3927 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 102.895 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 591.6(1.0 \cdot 20.0 - 1.0 \cdot 17.393) - 2 \cdot 97.0$$

$$(1.0 \cdot 20.0 - 1.0 \cdot 17.3927) \cdot (1 - 1.0)$$

$$= 15.425 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

149-D

Refrigerant receiver
QUEP

PV Elite 28 U1

Licensee: SL User

FileName : 149-D_Refrigerant receiver_7 Per

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Nozzle Calcs.: M1_24 inch

Noz1: 33 12:03pm Sep 16,2026



HEXAGON

$$= (2 \cdot 50.0)(9.0 - 4.87)1.0$$

$$= 4.131 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{hp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(50.0, 250.0, 75.0)) \cdot (100.0 - 12.0) \cdot 1.0$$

$$= 88.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:Wall Thickness for Internal/External pressures $t_a = 7.8688 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 7.8688 \text{ mm}.$$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld (Impact tested) :**[Note: This Element/Detail was specified as being Impact Tested.](#)**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :**[Note: This Element/Detail was specified as being Impact Tested.](#)Gov. MDMT of the nozzle to shell joint welded assembly : $-40 \text{ }^\circ\text{C}$ **Flange MDMT including Temperature reduction per UCS-66.1:**MDMT of ASME B16.5/47 flange per Matl. Specification $-18 \text{ }^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(i)(2) $-61 \text{ }^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 22.18/51.00 = 0.435$$

Weld Size Calculations, Description: M124 inchIntermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$.**Results Per UW-16.1:**

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr))S_v)$$

$$= \max(0, (102.8953 - 15.4247 + 2 \cdot 97.0 \cdot 1.0 \cdot (1.0 \cdot 20.0 - 17.3927))138)$$

$$= 130204.87 \text{ Kg}$$

[For hub type nozzles, A2 includes the area of the hub.](#)[Note: F is always set to 1.0 throughout the calculation.](#)

Weld Load [W1]:

$$= (A2 + A4 - (W_i - Can/0.707)^2 \cdot fr2) \cdot S_v$$

$$= (92.1312 + 1.0 - 0.0 \cdot 1.0) \cdot 138$$



$$= 131052.58 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (92.1312 + 0.0 + 1. + (38.8)) \cdot 138$$

$$= 185651.27 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (92.1312 + 0.0 + 1. + 0.0 + (38.8)) \cdot 138$$

$$= 185651.27 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 66025. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 300.3) \cdot (100.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 901417. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (23.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 199424. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (66025 + 901417) = 967442 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (66025 + 0 + 199424 + 0) = 265449 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (66025 + 199424 + 0) = 265449 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 967442 Kgf, must exceed W = 130204 Kgf	or W1 = 131052 Kgf
Path 2-2 = 265449 Kgf, must exceed W = 130204 Kgf	or W2 = 185651 Kgf
Path 3-3 = 265449 Kgf, must exceed W = 130204 Kgf	or W3 = 185651 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 23 bars

The Drop for this Nozzle is : 21.4291 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 294.4291 mm.

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Input, Nozzle Desc: K42"

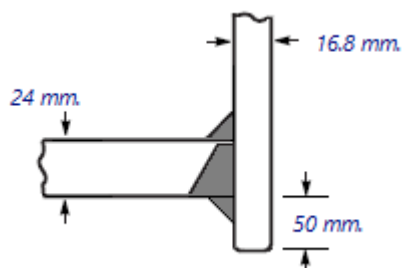
From: 20

Pressure for Reinforcement Calculations	P	22.254	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2400.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		350.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	50.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $22.25 \cdot 1203.0 / (138 \cdot 1.0 - 0.6 \cdot 22.25)$
= 19.5906 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $22.25 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 22.25)$
= 0.4625 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	126.4000 mm.
Parallel to Vessel Wall	Rn+tn+t	63.2000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000 mm.
Normal to Vessel Wall, Inward		27.0000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K42".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.4625 mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 22.5906 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 22.5906 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm.
Wall Thickness per table UG-45	tb3	= 7.8000 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[7.8, max(22.5906, 4.5)]
= 7.8000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.4625, 7.8)
= 7.8000 mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :**

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $22.25/51.00 = 0.436$

Weld Size Calculations, Description: K42"

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.7422 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 280.3750 mm.

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Input, Nozzle Desc: K32"

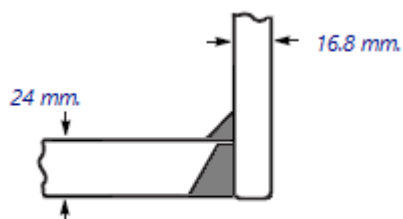
From: 20

Pressure for Reinforcement Calculations	P	22.115	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2400.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		350.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	477.6480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $22.11 \cdot 1203.0 / (138 \cdot 1.0 - 0.6 \cdot 22.11)$
= 19.4672 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $22.11 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 22.11)$
= 0.4596 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	126.4000 mm.
Parallel to Vessel Wall	Rn+tn+t	63.2000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K32".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.4596 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 22.4672 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 22.4672 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 7.8000 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]
= min[7.8, max(22.4672, 4.5)]
= 7.8000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(3.4596, 7.8)
= 7.8000 mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B**

Govrn. thk, tg = 16.8, tr = 0.46, c = 3.0 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.033$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

149-D

Refrigerant receiver
QUEP

PV Elite 28 U1 Licensee: SL User

FileName : 149-D_Refrigerant receiver_7 Per

Nozzle Calcs.: K3_2"

Noz1: 35 12:03pm Sep 16,2026

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HEXAGON

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $22.11/51.00 = 0.434$

Weld Size Calculations, Description: K32"

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.7422 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 502.3902 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 63.20 mm.

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Input, Nozzle Desc: K52"

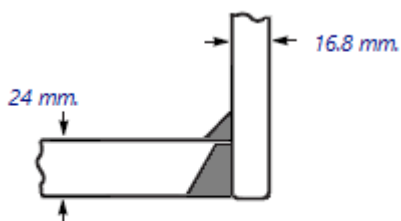
From: 20

Pressure for Reinforcement Calculations	P	22.115	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2400.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1200.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	477.6480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K52"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $22.11 \cdot 1203.0 / (138 \cdot 1.0 - 0.6 \cdot 22.11)$
= 19.4672 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $22.11 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 22.11)$
= 0.4596 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	126.4000 mm.
Parallel to Vessel Wall	Rn+tn+t	63.2000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K52".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.4596 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 22.4672 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 22.4672 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 7.8000 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]
= min[7.8, max(22.4672, 4.5)]
= 7.8000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(3.4596, 7.8)
= 7.8000 mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 0.46, c = 3.0 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c)$ = 0.033, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = 22.11/51.00 = 0.434

Weld Size Calculations, Description: K52"

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.7422 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 502.3902 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 63.20 mm.

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Input, Nozzle Desc: K62"

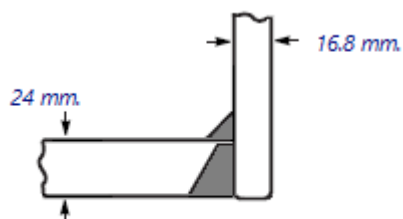
From: 20

Pressure for Reinforcement Calculations	P	22.115	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2400.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1500.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	477.6480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K62"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.661 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $22.11 \cdot 1203.0 / (138 \cdot 1.0 - 0.6 \cdot 22.11)$
= 19.4672 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $22.11 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 22.11)$
= 0.4596 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	126.4000 mm.
Parallel to Vessel Wall	Rn+tn+t	63.2000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.5000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K62".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.4596 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 22.4672 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 22.4672 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 7.8000 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[7.8, max(22.4672, 4.5)]
= 7.8000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.4596, 7.8)
= 7.8000 mm.

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 0.46, c = 3.0 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c)$ = 0.033, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = 22.11/51.00 = 0.434

Weld Size Calculations, Description: K62"

Intermediate Calc. for nozzle/shell Welds Tmin 13.8000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.7422 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 502.3902 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 63.20 mm.

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Input, Nozzle Desc: N112"

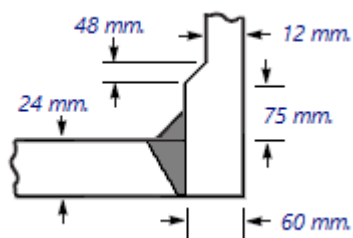
From: 20

Pressure for Reinforcement Calculations	P	22.115	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2400.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2400.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	48.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	449.2000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N112"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	12.000 in.
Actual Thickness Used in Calculation	0.472 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 22.11 \cdot 1203.0 / (138 \cdot 1.0 - 0.6 \cdot 22.11)$
 $= 19.4672 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 22.11 \cdot 152.4 / (138 \cdot 1.0 + 0.4 \cdot 22.11)$
 $= 2.4268 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 48.0 / 0.5773$
 $= 83.1385 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

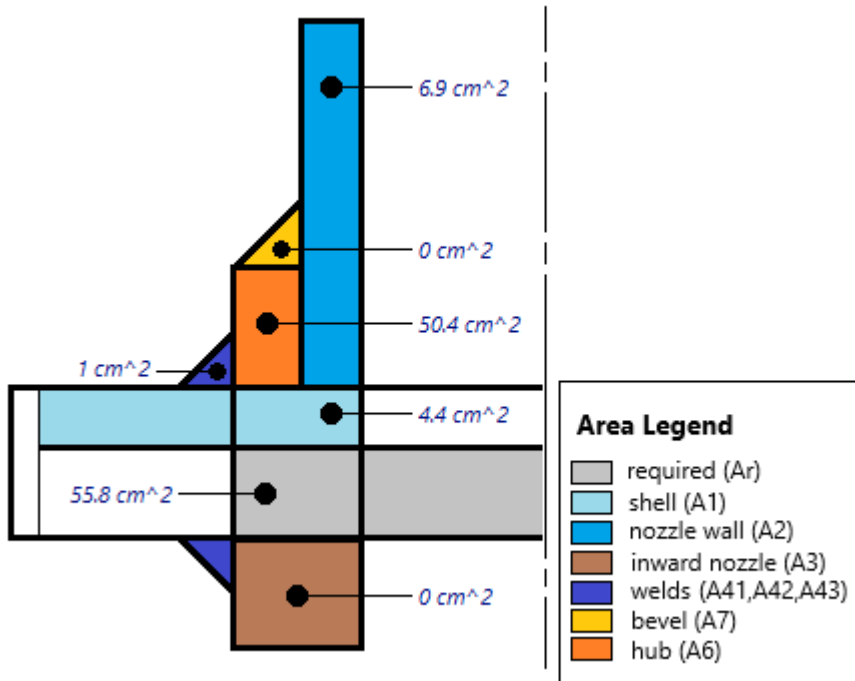
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	573.6000 mm.
Parallel to Vessel Wall, opening length	d	286.8000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	52.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	55.832	NA	NA
Area in Shell	A1	4.396	NA	NA
Area in Nozzle Wall	A2	6.902	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	50.400	NA	NA
TOTAL AREA AVAILABLE	Atot	62.698	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (286.8 \cdot 19.4672 \cdot 1.0 + 2 \cdot 57.0 \cdot 19.4672 \cdot 1.0 \cdot (1 - 1.0))$
 $= 55.832 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 286.8(1.0 \cdot 21.0 - 1.0 \cdot 19.467) - 2 \cdot 57.0$
 $(1.0 \cdot 21.0 - 1.0 \cdot 19.4672) \cdot (1 - 1.0)$
 $= 4.396 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 52.5)(9.0 - 2.43)1.0$
 $= 6.902 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(52.5, 449.2, 75.0)) \cdot (60.0 - 12.0) \cdot 1.0$$

$$= 50.400 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.4268 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 22.4672 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 22.4672 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(22.4672, 4.5)]$$

$$= 11.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.4268, 11.3312)$$

$$= 11.3312 \text{ mm.}$$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 44.0,	Allowable	: 138.0 N./mm2	Passed
Expansion	: 0.0,	Allowable	: 301.0 N./mm2	Passed
Occasional	: 17.1,	Allowable	: 183.5 N./mm2	Passed
Shear	: 20.7,	Allowable	: 96.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-18 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-61 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 22.11/51.00 = 0.434$$

Weld Size Calculations, Description: N112"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------

Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * \text{Wo mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (55.832 - 4.396 + 2 \cdot 57.0 \cdot 1.0 \cdot (1.0 \cdot 21.0 - 19.4672)) \cdot 138)$$

$$= 74837.17 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (57.3018 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 82039.55 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (57.3018 + 0.0 + 1. + (23.94)) \cdot 138$$

$$= 115726.79 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (57.3018 + 0.0 + 1. + 0.0 + (23.94)) \cdot 138$$

$$= 115726.79 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 304.8 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 33012. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 147.9) \cdot (60.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 260880. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 304.8 \cdot (24.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 104698. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (33012 + 260880) = 293892 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (33012 + 0 + 104698 + 0) = 137710 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (33012 + 104698 + 0) = 137710 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 293892 Kgf, must exceed W = 74837 Kgf	or W1 = 82039 Kgf
Path 2-2 = 137709 Kgf, must exceed W = 74837 Kgf	or W2 = 115726 Kgf
Path 3-3 = 137709 Kgf, must exceed W = 74837 Kgf	or W3 = 115726 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 23 bars

The Drop for this Nozzle is : 9.7168 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 482.9167 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 286.80 mm.

Input Echo, WRC297 Item 1, Description: N112" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2400.000	mm.
Shell Thickness	24.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm2
Vessel Hot S.I. Allowable	Smh 138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	400.800	mm.
Nozzle Thickness	60.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	22.11	bars
Radial Load	P	-2294.51	Kgf
Circumferential Shear	Vc	1529.68	Kgf
Longitudinal Shear	Vl	1529.68	Kgf
Circumferential Moment	Mc	1284.44	Kg-m.
Longitudinal Moment	Ml	1284.44	Kg-m.
Torsional Moment	Mt	1834.90	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (6.680) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 400.8 + 2 \cdot 1.65 \cdot \sqrt{(1213.5(24.0 - 3.0))}$$

$$= 927.597 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2427.000 mm.
Nozzle Outside Diameter	(d) =	400.800 mm.
Vessel Thickness used	(T) =	21.000 mm.

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Refrigerant receiver
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Nozzle Calcs.: N1_12"

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Nozzle Thickness used (t) = 57.000 mm.
 T / t = 0.368
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.775$$

Nr/P = 0.074
 Mr/P = 0.156
 M0/P = 0.045
 N0/P = 0.095
 MrD/Mc = 0.291
 NrDL/Mc = 0.094
 M0d/Mc = 0.090
 N0DL/Mc = 0.068
 MrD/M1 = 0.124
 NrDL/M1 = 0.080
 M0D/M1 = 0.039
 N0DL/M1 = 0.119

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	125	127	125	127
Outplane Stress Summary	119	127	169	110

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-52	52	52	-52
Inplane Pressure Stress	62	62	62	62
Inplane Stress Summary	55	66	172	-28

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	119	127	174	138

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Nozzle Calcs.: N1_12"

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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-38	38	38	-38
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	62	62	62	62
Outplane Stress Summary	38	87	125	20

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-124	124	124	-124
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	125	127	125	127
Inplane Stress Summary	45	201	308	-33

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	46	201	308	54

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-8	-8	8	8
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	8	5	8
Hoop Stress Summary	1	4	18	21

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

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Nozzle Calcs.: N1_12"

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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-4	4	4	-4
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	0	0	15	-3
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	0	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	3	5	19	25
------------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	8	5	8

Hoop Stress Summary	5	8	15	17
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-14	14	14	-14
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-8	9	24	-13
----------------------	----	---	----	-----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	0	0	1	1
----------------------	---	---	---	---

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Nozzle Calcs.: N1_12"

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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	14	10	25	31

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	125	127	125	127	125	127	125	127
Circ. Pl (SUS)	-4	-4	12	12	-3	-3	9	9
Circ. Q (SUS)	-3	3	29	-29	-77	77	171	-171
Long. Pm (SUS)	62	62	62	62	62	62	62	62
Long. Pl (SUS)	-2	-2	8	8	0	0	8	8
Long. Q (SUS)	-5	5	99	-99	-25	25	51	-51
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	125.0	127.0	125.0	127.0	125.0	127.0	125.0	127.0
Pm+Pl (SUS)	121.0	123.0	137.0	139.0	122.0	124.0	134.0	136.0
Pm+Pl+Q (Total)	118.3	126.3	170.0	139.1	45.5	201.0	305.1	54.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	127.00	138.00	Passed
Pm+Pl (SUS)	139.01	207.00	Passed
Pm+Pl+Q (TOTAL)	305.09	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	8	5	8	5	8	5	8
Circ. Pl (SUS)	-4	-4	12	12	0	0	8	8
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	0	0	8	-8	-10	10	18	-18
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	5.0	8.0	5.0	8.0	5.0	8.0	5.0	8.0
Pm+Pl (SUS)	1.0	4.0	17.0	20.0	5.0	8.0	13.0	16.0

Pm+Pl+Q (Total)	2.2	4.5	17.2	24.1	15.1	10.4	22.1	30.1
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Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.00	138.00	Passed
Pm+Pl (SUS)	20.00	207.00	Passed
Pm+Pl+Q (TOTAL)	30.07	414.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N26"

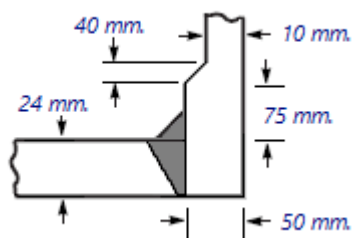
From: 20

Pressure for Reinforcement Calculations	P	22.115	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2400.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		800.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	+IT
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	40.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	449.2000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N26"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.000 in.
Actual Thickness Used in Calculation	0.394 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 22.11 \cdot 1203.0 / (138 \cdot 1.0 - 0.6 \cdot 22.11)$
 $= 19.4672 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 22.11 \cdot 76.2 / (138 \cdot 1.0 + 0.4 \cdot 22.11)$
 $= 1.2134 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40.0 / 0.5773$
 $= 69.2821 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

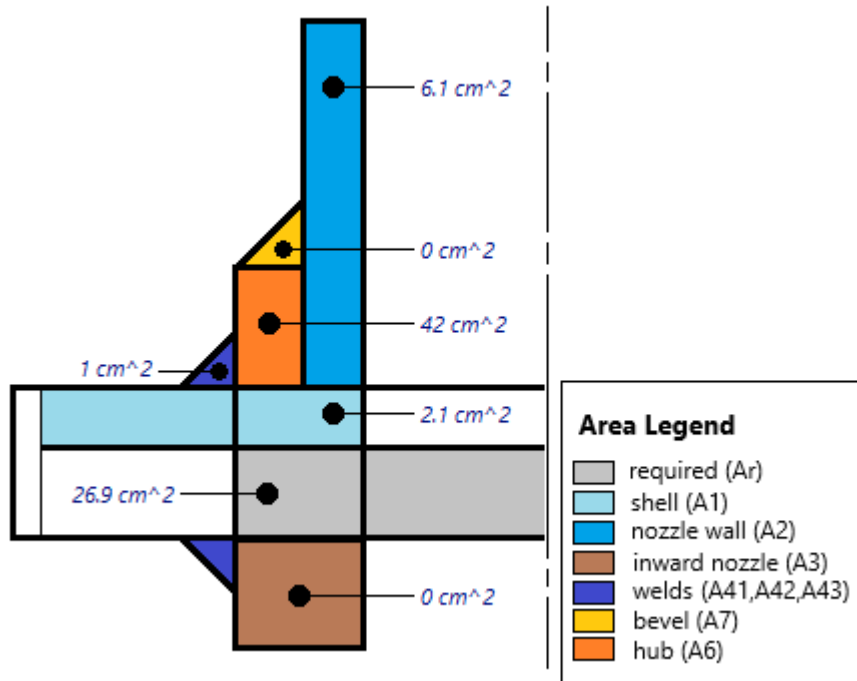
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	276.8000 mm.
Parallel to Vessel Wall, opening length	d	138.4000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	52.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	26.943	NA	NA
Area in Shell	A1	2.121	NA	NA
Area in Nozzle Wall	A2	6.076	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	42.000	NA	NA
TOTAL AREA AVAILABLE	Atot	51.197	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (138.4 \cdot 19.4672 \cdot 1.0 + 2 \cdot 47.0 \cdot 19.4672 \cdot 1.0 \cdot (1 - 1.0))$
 $= 26.943 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 138.4 (1.0 \cdot 21.0 - 1.0 \cdot 19.467) - 2 \cdot 47.0$
 $(1.0 \cdot 21.0 - 1.0 \cdot 19.4672) \cdot (1 - 1.0)$
 $= 2.121 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 52.5) (7.0 - 1.21) 1.0$
 $= 6.076 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(52.5, 449.2, 75.0)) \cdot (50.0 - 10.0) \cdot 1.0$$

$$= 42.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.2134 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 22.4672 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 22.4672 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 9.2200 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.22, \max(22.4672, 4.5)]$$

$$= 9.2200 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.2134, 9.22)$$

$$= 9.2200 \text{ mm.}$$

Available Nozzle Neck Thickness = 10.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 47.0,	Allowable	: 138.0 N./mm2	Passed
Expansion	: 0.0,	Allowable	: 298.0 N./mm2	Passed
Occasional	: 10.4,	Allowable	: 183.5 N./mm2	Passed
Shear	: 28.3,	Allowable	: 96.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-18 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-61 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 22.11/51.00 = 0.434$$

Weld Size Calculations, Description: N26"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------

Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * \text{Wo mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (26.9427 - 2.1213 + 2 \cdot 47.0 \cdot 1.0 \cdot (1.0 \cdot 21.0 - 19.4672)) \cdot 138)$$

$$= 36954.79 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (48.0759 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 69057.29 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (48.0759 + 0.0 + 1. + (19.74)) \cdot 138$$

$$= 96834.49 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (48.0759 + 0.0 + 1. + 0.0 + (19.74)) \cdot 138$$

$$= 96834.49 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 152.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 16506. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 72.7) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 105738. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 152.4 \cdot (24.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 52349. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (16506 + 105738) = 122244 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (16506 + 0 + 52349 + 0) = 68855 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (16506 + 52349 + 0) = 68855 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 122243 Kgf, must exceed W = 36954 Kgf	or W1 = 69057 Kgf
Path 2-2 = 68854 Kgf, must exceed W = 36954 Kgf	or W2 = 96834 Kgf
Path 3-3 = 68854 Kgf, must exceed W = 36954 Kgf	or W3 = 96834 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

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Nozzle Calcs.: N2_6"

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Converged Maximum Allowable Pressure in the Operating case: 24 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 2.4218 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 475.6217 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 138.40 mm.

Input Echo, WRC297 Item 1, Description: N26" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2400.000	mm.
Shell Thickness	24.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm2
Vessel Hot S.I. Allowable	Smh 138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	232.400	mm.
Nozzle Thickness	50.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	22.11	bars
Radial Load	P	-1147.26	Kgf
Circumferential Shear	Vc	764.84	Kgf
Longitudinal Shear	Vl	764.84	Kgf
Circumferential Moment	Mc	321.11	Kg-m.
Longitudinal Moment	Ml	321.11	Kg-m.
Torsional Moment	Mt	458.73	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (4.648) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 232.4 + 2 \cdot 1.65 \cdot \sqrt{(1213.5(24.0 - 3.0))}$$

$$= 759.197 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2427.000 mm.

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Nozzle Calcs.: N2_6"

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Nozzle Outside Diameter (d) = 232.400 mm.
 Vessel Thickness used (T) = 21.000 mm.
 Nozzle Thickness used (t) = 47.000 mm.
 T / t = 0.447
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.029$$

Nr/P = 0.103
 Mr/P = 0.220
 M0/P = 0.065
 N0/P = 0.129
 MrD/Mc = 0.307
 NrDL/Mc = 0.088
 M0d/Mc = 0.096
 N0DL/Mc = 0.041
 MrD/Ml = 0.192
 NrDL/Ml = 0.088
 M0D/Ml = 0.062
 N0DL/Ml = 0.119

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-11	11	11	-11
Normal Pressure Stress	125	127	125	127
Outplane Stress Summary	123	128	153	113

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	33	-33	33	-33
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-35	35	35	-35
Inplane Pressure Stress	62	62	62	62
Inplane Stress Summary	61	64	137	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Nozzle Calcs.: N2_6"

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Two * Max Shear Stress	123	129	153	114
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-17	17	17	-17
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	62	62	62	62

Outplane Stress Summary	57	72	94	39
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	33	-33	33	-33
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-56	56	56	-56
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	125	127	125	127

Inplane Stress Summary	102	150	221	42
------------------------	-----	-----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	102	150	221	45
------------------------	-----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	5	2	5

Hoop Stress Summary	2	5	9	12
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
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(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
--	----------------	---------------	-------------------	------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-5	5	5	-5
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-1	0	15	-5
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	0	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	4	6	15	18
------------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	5	2	5

Hoop Stress Summary	4	7	7	10
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-9	9	9	-9
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-5	3	19	-10
----------------------	----	---	----	-----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1



Shear Stress Summary 0 0 1 1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE Cu Cl Du D1
(Stress Intensities Outside Inside Outside Inside

Two * Max Shear Stress 10 7 19 20

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	125	127	125	127	125	127	125	127
Circ. Pl (SUS)	0	0	6	6	0	0	4	4
Circ. Q (SUS)	-2	2	20	-20	-23	23	89	-89
Long. Pm (SUS)	62	62	62	62	62	62	62	62
Long. Pl (SUS)	0	0	4	4	2	2	4	4
Long. Q (SUS)	-2	2	68	-68	-8	8	26	-26
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	125.0	127.0	125.0	127.0	125.0	127.0	125.0	127.0
Pm+Pl (SUS)	125.0	127.0	131.0	133.0	125.0	127.0	129.0	131.0
Pm+Pl+Q (Total)	123.1	129.1	151.2	115.1	102.1	150.1	218.0	43.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	127.00	138.00	Passed
Pm+Pl (SUS)	133.00	207.00	Passed
Pm+Pl+Q (TOTAL)	218.03	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	5	2	5	2	5	2	5
Circ. Pl (SUS)	0	0	6	6	2	2	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	0	0	10	-10	-4	4	14	-14
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	2.0	5.0	2.0	5.0	2.0	5.0	2.0	5.0

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Pm+Pl (SUS)	3.0	6.0	8.0	11.0	5.0	8.0	6.0	9.0
Pm+Pl+Q (Total)	3.6	6.3	13.2	18.1	9.2	7.2	17.1	20.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	138.00	Passed
Pm+Pl (SUS)	11.00	207.00	Passed
Pm+Pl+Q (TOTAL)	20.10	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N36"

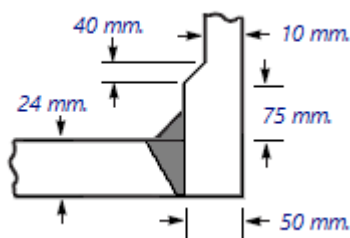
From: 20

Pressure for Reinforcement Calculations	P	22.254	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2400.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		800.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	40.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	201.4480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	150.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N36"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.000 in.
Actual Thickness Used in Calculation	0.394 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 22.25 \cdot 1203.0 / (138 \cdot 1.0 - 0.6 \cdot 22.25)$$

$$= 19.5906 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= 22.25 \cdot 76.2 / (138 \cdot 1.0 + 0.4 \cdot 22.25)$$

$$= 1.2210 \text{ mm.}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 40.0 / 0.5773$$

$$= 69.2821 \text{ mm.}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	276.8000	mm.
--	----	----------	-----

Parallel to Vessel Wall, opening length	d	138.4000	mm.
---	---	----------	-----

Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	52.5000	mm.
---	------	---------	-----

Normal to Vessel Wall, Inward		10.0000	mm.
-------------------------------	--	---------	-----

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138.0 / 138.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138.0 / 138.0)$$

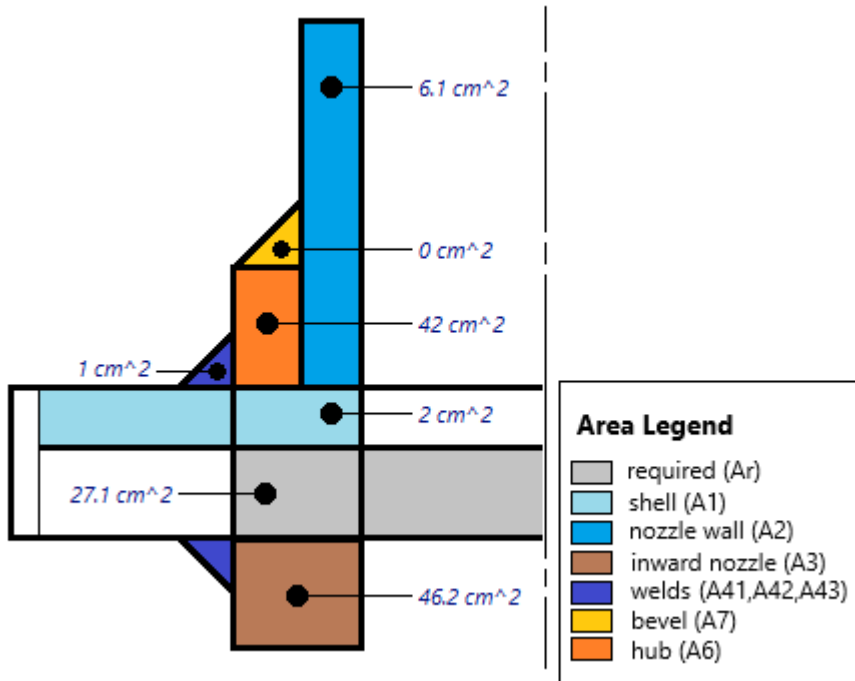
$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1.0, 1.0)$$

$$= 1.000$$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
Area Required	Ar	27.113	NA
Area in Shell	A1	1.951	NA
Area in Nozzle Wall	A2	6.068	NA
Area in Inward Nozzle	A3	46.200	NA
Area in Welds	A41+A42+A43	1.000	NA
Area in Element	A5	0.000	NA
Area in Hub	A6	42.000	NA
TOTAL AREA AVAILABLE	Atot	97.219	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (138.4 \cdot 19.5906 \cdot 1.0 + 2 \cdot 47.0 \cdot 19.5906 \cdot 1.0 \cdot (1 - 1.0))$
 $= 27.113 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 138.4 (1.0 \cdot 21.0 - 1.0 \cdot 19.591) - 2 \cdot 47.0$
 $(1.0 \cdot 21.0 - 1.0 \cdot 19.5906) \cdot (1 - 1.0)$
 $= 1.951 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 52.5) (7.0 - 1.22) 1.0$
 $= 6.068 \text{ cm}^2$

Area Available in Inward Nozzle [A3]:
 $= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$

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Nozzle Calcs.: N3_6"

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$$= 2 \cdot 44.0 \cdot (52.5) \cdot 1.0$$

$$= 46.200 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{hp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(52.5, 201.4, 75.0)) \cdot (50.0 - 10.0) \cdot 1.0$$

$$= 42.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.2210 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 22.5906 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 22.5906 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 9.2200 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.22, \max(22.5906, 4.5)]$$

$$= 9.2200 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.221, 9.22)$$

$$= 9.2200 \text{ mm.}$$

Available Nozzle Neck Thickness = 10.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 47.0,	Allowable	: 138.0 N./mm2	Passed
Expansion	: 0.0,	Allowable	: 298.0 N./mm2	Passed
Occasional	: 10.5,	Allowable	: 183.5 N./mm2	Passed
Shear	: 28.3,	Allowable	: 96.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, Curve: B**

Govrn. thk, tg = 10.0, tr = 1.221, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.174, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = 22.25/51.00 = 0.436

Weld Size Calculations, Description: N36"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (27.1134 - 1.9506 + 2 \cdot 47.0 \cdot 1.0 \cdot (1.0 \cdot 21.0 - 19.5906)) \cdot 138)$$

$$= 37272.16 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (48.068 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 69046.10 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (48.068 + 46.2 + 1. + (19.74)) \cdot 138$$

$$= 161833.78 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (48.068 + 46.2 + 1. + 0.0 + (19.74)) \cdot 138$$

$$= 161833.78 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 152.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 16506. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 72.7) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 105738. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 152.4 \cdot (24.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 52349. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (16506 + 105738) = 122244 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (16506 + 0 + 52349 + 0) = 68855 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

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Nozzle Calcs.: N3_6"

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$$= (16506 + 52349 + 0) = 68855 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 122243 Kgf, must exceed W = 37272 Kgf or W1 = 69046 Kgf
 Path 2-2 = 68854 Kgf, must exceed W = 37272 Kgf or W2 = 161833 Kgf
 Path 3-3 = 68854 Kgf, must exceed W = 37272 Kgf or W3 = 161833 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 2.4218 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 375.4480 mm.

Input Echo, WRC297 Item 1, Description: N36" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2400.000	mm.
Shell Thickness	24.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	232.400	mm.
Nozzle Thickness	50.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	22.25	bars
Radial Load	P	-1147.26	Kgf
Circumferential Shear	Vc	764.84	Kgf
Longitudinal Shear	Vl	764.84	Kgf
Circumferential Moment	Mc	321.11	Kg-m.
Longitudinal Moment	Ml	321.11	Kg-m.
Torsional Moment	Mt	458.73	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (4.648) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 232.4 + 2 \cdot 1.65 \cdot \sqrt{(1213.5(24.0 - 3.0))}$$

$$= 759.197 \text{ mm.}$$



WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2427.000 mm.
 Nozzle Outside Diameter (d) = 232.400 mm.
 Vessel Thickness used (T) = 21.000 mm.
 Nozzle Thickness used (t) = 47.000 mm.
 T / t = 0.447
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.029$$

Nr/P = 0.103
 Mr/P = 0.220
 M0/P = 0.065
 N0/P = 0.129
 MrD/Mc = 0.307
 NrDL/Mc = 0.088
 M0d/Mc = 0.096
 N0DL/Mc = 0.041
 MrD/M1 = 0.192
 NrDL/M1 = 0.088
 M0D/M1 = 0.062
 N0DL/M1 = 0.119

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-11	11	11	-11
Normal Pressure Stress	126	128	126	128
Outplane Stress Summary	124	129	154	114

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	33	-33	33	-33
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-35	35	35	-35
Inplane Pressure Stress	63	63	63	63
Inplane Stress Summary	61	64	137	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

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LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	124	129	154	114

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-17	17	17	-17
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	63	63	63
Outplane Stress Summary	57	73	95	40

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	33	-33	33	-33
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-56	56	56	-56
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	126	128	126	128
Inplane Stress Summary	103	151	222	43

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Two * Max Shear Stress	103	151	222	45

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	5	2	5

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Hoop Stress Summary	2	5	9	12
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-5	5	5	-5
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-1	0	15	-5
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	4	6	15	18
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	5	2	5

Hoop Stress Summary	4	7	7	10
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-9	9	9	-9
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-5	3	19	-10
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	10	7	19	20

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	126	128	126	128	126	128	126	128
Circ. Pl (SUS)	0	0	6	6	0	0	4	4
Circ. Q (SUS)	-2	2	20	-20	-23	23	89	-89
Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. Pl (SUS)	0	0	4	4	2	2	4	4
Long. Q (SUS)	-2	2	68	-68	-8	8	26	-26
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	126.0	128.0	126.0	128.0	126.0	128.0	126.0	128.0
Pm+Pl (SUS)	126.0	128.0	132.0	134.0	126.0	128.0	130.0	132.0
Pm+Pl+Q (Total)	124.1	130.1	152.2	115.1	103.1	151.1	219.0	44.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	128.00	138.00	Passed
Pm+Pl (SUS)	134.00	207.00	Passed
Pm+Pl+Q (TOTAL)	219.03	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	5	2	5	2	5	2	5
Circ. Pl (SUS)	0	0	6	6	2	2	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	0	0	10	-10	-4	4	14	-14
Shear Pm (SUS)	0	0	0	0	0	0	0	0

Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	2.0	5.0	2.0	5.0	2.0	5.0	2.0	5.0
Pm+P1 (SUS)	3.0	6.0	8.0	11.0	5.0	8.0	6.0	9.0
Pm+P1+Q (Total)	3.6	6.3	13.2	18.1	9.2	7.2	17.1	20.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	138.00	Passed
Pm+P1 (SUS)	11.00	207.00	Passed
Pm+P1+Q (TOTAL)	20.10	414.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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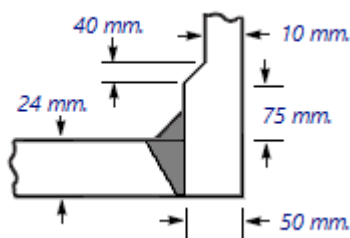
**Input, Nozzle Desc: N93"****From: 20**

Pressure for Reinforcement Calculations	P	22.254	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2400.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1350.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	40.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	201.4480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N93"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.394 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 22.25 \cdot 1203.0 / (138 \cdot 1.0 - 0.6 \cdot 22.25)$
 $= 19.5906 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 22.25 \cdot 38.1 / (138 \cdot 1.0 + 0.4 \cdot 22.25)$
 $= 0.6105 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 24.1 / 0.5773$
 $= 41.7424 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

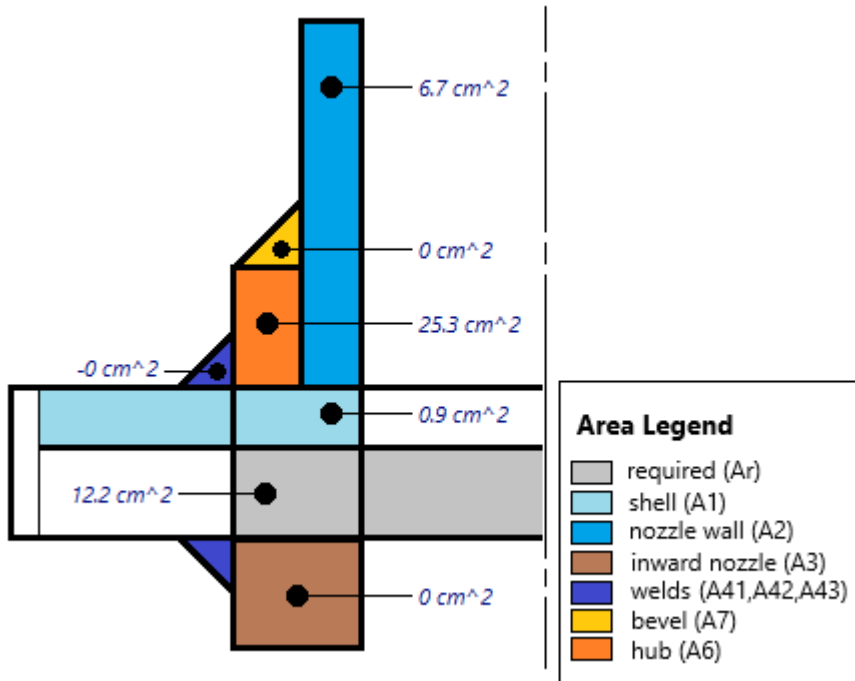
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	124.4000 mm.
Parallel to Vessel Wall, opening length	d	62.2000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	52.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	12.185	NA	NA
Area in Shell	A1	0.877	NA	NA
Area in Nozzle Wall	A2	6.709	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	-0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	25.305	NA	NA
TOTAL AREA AVAILABLE	Atot	32.891	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (62.2 \cdot 19.5906 \cdot 1.0 + 2 \cdot 31.1 \cdot 19.5906 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 12.185 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 62.2 (1.0 \cdot 21.0 - 1.0 \cdot 19.591) - 2 \cdot 31.1$$

$$(1.0 \cdot 21.0 - 1.0 \cdot 19.5906) \cdot (1 - 1.0)$$

$$= 0.877 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 52.5) (7.0 - 0.61) 1.0$$

$$= 6.709 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (Wo^2 - \text{Area Lost}) \cdot fr2 + ((Wi - can / 0.707)^2 - \text{Area Lost}) \cdot fr2$$

$$= (10.0^2 - 1.0) \cdot 1.0 + (0.0^2 - 0.0) \cdot 1.0$$

$$= -0.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(52.5, 201.4, 75.0)) \cdot (34.1 - 10.0) \cdot 1.0$$

$$= 25.305 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.6105 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 22.5906 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 22.5906 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 7.8000 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.8, \max(22.5906, 4.5)]$$

$$= 7.8000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.6105, 7.8)$$

$$= 7.8000 \text{ mm.}$$

Available Nozzle Neck Thickness = 10.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 52.9,	Allowable	: 138.0 N./mm2	Passed
Expansion	: 0.0,	Allowable	: 292.1 N./mm2	Passed
Occasional	: 4.4,	Allowable	: 183.5 N./mm2	Passed
Shear	: 38.4,	Allowable	: 96.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 10.0, tr = 0.61, c = 3.0 mm., E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.087, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 22.25/102.00 = 0.218$$

Weld Size Calculations, Description: N93"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (12.1854 - 0.8766 + 2 \cdot 31.1 \cdot 1.0 \cdot (1.0 \cdot 21.0 - 19.5906)) \cdot 138)$
 $= 17146.67 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (32.014 + -0.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 45048.55 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (32.014 + 0.0 + -0.0 + (13.062)) \cdot 138$
 $= 63428.78 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (32.014 + 0.0 + -0.0 + 0.0 + (13.062)) \cdot 138$
 $= 63428.78 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 76.2 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 8253. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 34.6) \cdot (34.1 - 3.0) \cdot 0.7 \cdot 138$
 $= 33299. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 76.2 \cdot (24.0 - 3.0) \cdot 0.74 \cdot 138$
 $= 26174. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (8253 + 33299) = 41552 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (8253 + 0 + 26174 + 0) = 34427 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (8253 + 26174 + 0) = 34427 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 41552	Kgf, must exceed W = 17146 Kgf	or W1 = 45048 Kgf
Path 2-2 = 34427	Kgf, must exceed W = 17146 Kgf	or W2 = 63428 Kgf

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Path 3-3 = 34427 Kgf, must exceed W = 17146 Kgf or W3 = 63428 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.6050 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.0530 mm.

Input Echo, WRC297 Item 1, Description: N93" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2400.000	mm.
Shell Thickness	24.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	156.200	mm.
Nozzle Thickness	50.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:*External Forces and Moments in WRC 107/537 Convention:**These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	22.25	bars
Radial Load	P	-611.87	Kgf
Circumferential Shear	Vc	407.91	Kgf
Longitudinal Shear	Vl	407.91	Kgf
Circumferential Moment	Mc	91.34	Kg-m.
Longitudinal Moment	Ml	91.34	Kg-m.
Torsional Moment	Mt	130.48	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_n (3.124) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 156.2 + 2 \cdot 1.65 \cdot \sqrt{1213.5(24.0 - 3.0)}$$

$$= 682.997 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2427.000 mm.
Nozzle Outside Diameter	(d) =	156.200 mm.

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Vessel Thickness used (T) = 21.000 mm.
 Nozzle Thickness used (t) = 47.000 mm.
 T / t = 0.447
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.692$$

Nr/P = 0.124
 Mr/P = 0.278
 M0/P = 0.082
 N0/P = 0.151
 MrD/Mc = 0.312
 NrDL/Mc = 0.079
 M0d/Mc = 0.098
 N0DL/Mc = 0.021
 MrD/Ml = 0.241
 NrDL/Ml = 0.088
 M0D/Ml = 0.077
 N0DL/Ml = 0.086

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	126	128	126	128
Outplane Stress Summary	127	128	142	119

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	22	-22	22	-22
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-18	18	18	-18
Inplane Pressure Stress	63	63	63	63
Inplane Stress Summary	67	59	107	24

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	128	128	142	119

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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-7	7	7	-7
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	63	63	63
Outplane Stress Summary	63	65	79	51

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	22	-22	22	-22
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-24	24	24	-24
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	126	128	126	128
Inplane Stress Summary	125	130	176	84

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	125	130	176	84

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	3	0	3
Hoop Stress Summary	1	4	4	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom

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pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-1	-2	9	-3

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	3	6	9	10

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	3	0	3
Hoop Stress Summary	2	5	3	5

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-4	4	4	-4
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-2	0	11	-4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	1	1

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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	5	6	11	10

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	126	128	126	128	126	128	126	128
Circ. Pl (SUS)	1	1	3	3	0	0	2	2
Circ. Q (SUS)	0	0	12	-12	-2	2	46	-46
Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. Pl (SUS)	0	0	2	2	2	2	2	2
Long. Q (SUS)	4	-4	40	-40	-1	1	13	-13
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	126.0	128.0	126.0	128.0	126.0	128.0	126.0	128.0
Pm+Pl (SUS)	127.0	129.0	129.0	131.0	126.0	128.0	128.0	130.0
Pm+Pl+Q (Total)	127.0	129.0	141.0	119.0	124.0	130.0	174.0	84.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	128.00	138.00	Passed
Pm+Pl (SUS)	131.00	207.00	Passed
Pm+Pl+Q (TOTAL)	174.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	0	3	0	3	0	3	0	3
Circ. Pl (SUS)	1	1	3	3	2	2	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	0	0	6	-6	-1	1	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	0.0	3.0	0.0	3.0	0.0	3.0	0.0	3.0
Pm+Pl (SUS)	3.0	6.0	3.0	6.0	4.0	7.0	2.0	5.0

Pm+Pl+Q (Total)	3.0	6.0	8.0	10.0	5.0	6.0	9.0	10.0
-----------------	-----	-----	-----	------	-----	-----	-----	------

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	138.00	Passed
Pm+Pl (SUS)	7.00	207.00	Passed
Pm+Pl+Q (TOTAL)	10.00	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N74"

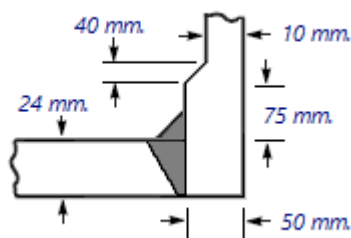
From: 50

Pressure for Reinforcement Calculations	P	22.115	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2400.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		7200.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	40.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	449.2000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N74"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.000 in.
Actual Thickness Used in Calculation	0.394 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 22.11 \cdot 1203.0 / (138 \cdot 1.0 - 0.6 \cdot 22.11)$
 $= 19.4672 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 22.11 \cdot 50.8 / (138 \cdot 1.0 + 0.4 \cdot 22.11)$
 $= 0.8089 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 36.8 / 0.5773$
 $= 63.7395 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	175.2000 mm.
Parallel to Vessel Wall, opening length	d	87.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	52.5000 mm.

Weld Strength Reduction Factor [fr1]:

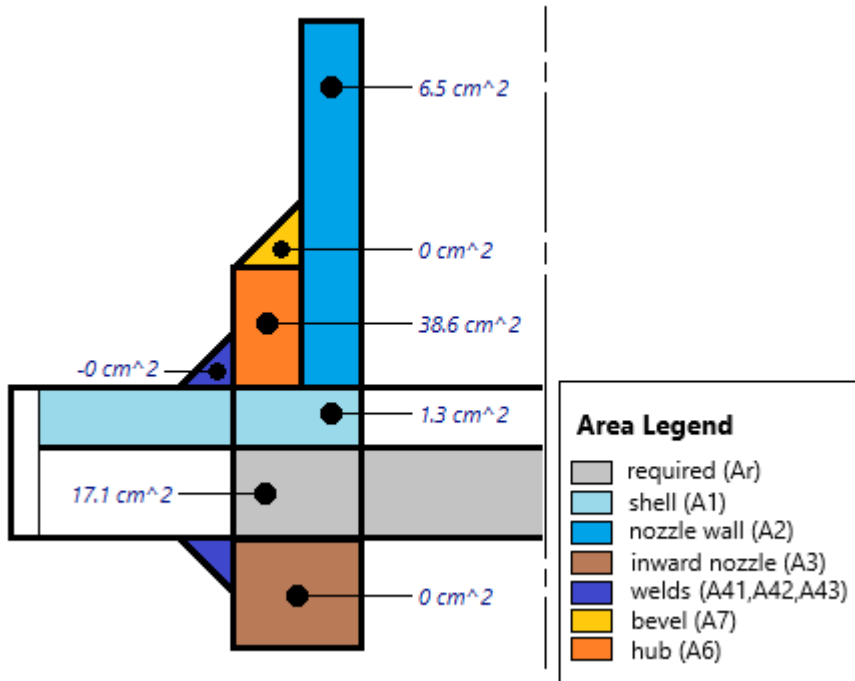
$= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	17.053	NA	NA
Area in Shell	A1	1.343	NA	NA
Area in Nozzle Wall	A2	6.501	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	-0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	38.640	NA	NA
TOTAL AREA AVAILABLE	Atot	46.483	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (87.6 \cdot 19.4672 \cdot 1.0 + 2 \cdot 43.8 \cdot 19.4672 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 17.053 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 87.6 (1.0 \cdot 21.0 - 1.0 \cdot 19.467) - 2 \cdot 43.8$$

$$(1.0 \cdot 21.0 - 1.0 \cdot 19.4672) \cdot (1 - 1.0)$$

$$= 1.343 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 52.5) (7.0 - 0.81) 1.0$$

$$= 6.501 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (Wo^2 - \text{Area Lost}) \cdot fr2 + ((Wi - can/0.707)^2 - \text{Area Lost}) \cdot fr2$$

$$= (10.0^2 - 1.0) \cdot 1.0 + (0.0^2 - 0.0) \cdot 1.0$$

$$= -0.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(52.5, 449.2, 75.0)) \cdot (46.8 - 10.0) \cdot 1.0$$

$$= 38.640 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.8089 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 22.4672 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 22.4672 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 8.0200 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.02, \max(22.4672, 4.5)]$$

$$= 8.0200 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.8089, 8.02)$$

$$= 8.0200 \text{ mm.}$$

Available Nozzle Neck Thickness = 10.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 45.8,	Allowable	: 138.0 N./mm2	Passed
Expansion	: 0.0,	Allowable	: 299.2 N./mm2	Passed
Occasional	: 6.4,	Allowable	: 183.5 N./mm2	Passed
Shear	: 31.0,	Allowable	: 96.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 10.0, tr = 0.809, c = 3.0 mm., E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.116, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 22.11/51.00 = 0.434$$

Weld Size Calculations, Description: N74"

Intermediate Calc. for nozzle/shell welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (17.0533 - 1.3427 + 2 \cdot 43.8 \cdot 1.0 \cdot (1.0 \cdot 21.0 - 19.4672)) \cdot 138)$$

$$= 23996.59 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (45.1406 + -0.0 - 0.0 \cdot 1.0) \cdot 138$$

$$= 63519.71 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (45.1406 + 0.0 + -0.0 + (18.396)) \cdot 138$$

$$= 89405.70 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (45.1406 + 0.0 + -0.0 + 0.0 + (18.396)) \cdot 138$$

$$= 89405.70 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 101.6 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 11004. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 47.3) \cdot (46.8 - 3.0) \cdot 0.7 \cdot 138$$

$$= 64111. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 101.6 \cdot (24.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 34899. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (11004 + 64111) = 75115 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (11004 + 0 + 34899 + 0) = 45903 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (11004 + 34899 + 0) = 45903 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 75115	Kgf, must exceed W = 23996 Kgf	or W1 = 63519 Kgf
Path 2-2 = 45903	Kgf, must exceed W = 23996 Kgf	or W2 = 89405 Kgf

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Refrigerant receiver
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Nozzle Calcs.: N7_4"

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Path 3-3 = 45903 Kgf, must exceed W = 23996 Kgf or W3 = 89405 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.0758 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 474.2758 mm.

Input Echo, WRC297 Item 1, Description: N74" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2400.000	mm.
Shell Thickness	24.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	181.600	mm.
Nozzle Thickness	50.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:*External Forces and Moments in WRC 107/537 Convention:**These loads are assumed to be SUStained loads.*

Design Internal Pressure	Dp	22.11	bars
Radial Load	P	-764.84	Kgf
Circumferential Shear	Vc	509.89	Kgf
Longitudinal Shear	Vl	509.89	Kgf
Circumferential Moment	Mc	142.71	Kg-m.
Longitudinal Moment	Ml	142.71	Kg-m.
Torsional Moment	Mt	203.88	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_n (3.632) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 181.6 + 2 \cdot 1.65 \cdot \sqrt{1213.5(24.0 - 3.0)}$$

$$= 708.397 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2427.000 mm.
Nozzle Outside Diameter	(d) =	181.600 mm.

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Vessel Thickness used (T) = 21.000 mm.
 Nozzle Thickness used (t) = 47.000 mm.
 T / t = 0.447
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.804$$

Nr/P = 0.116
 Mr/P = 0.255
 M0/P = 0.075
 N0/P = 0.143
 MrD/Mc = 0.310
 NrDL/Mc = 0.082
 M0d/Mc = 0.097
 N0DL/Mc = 0.028
 MrD/Ml = 0.222
 NrDL/Ml = 0.090
 M0D/Ml = 0.072
 N0DL/Ml = 0.101

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-7	7	7	-7
Normal Pressure Stress	125	127	125	127
Outplane Stress Summary	126	128	144	116

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	26	-26	26	-26
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-23	23	23	-23
Inplane Pressure Stress	62	62	62	62
Inplane Stress Summary	65	60	115	16

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	126	128	144	116

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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-10	10	10	-10
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	62	62	62	62
Outplane Stress Summary	62	67	83	47

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	26	-26	26	-26
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-32	32	32	-32
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	125	127	125	127
Inplane Stress Summary	119	134	187	72

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	119	134	187	72

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	4	1	4
Hoop Stress Summary	2	4	5	8

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom

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pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-1	-1	11	-4
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	1	1	0	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	4	7	11	12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	1	4	1	4

Hoop Stress Summary	3	6	4	7
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-5	5	5	-5
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-2	0	13	-5
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	1	1
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	6	6	13	13

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	125	127	125	127	125	127	125	127
Circ. Pl (SUS)	1	1	3	3	0	0	2	2
Circ. Q (SUS)	0	0	14	-14	-6	6	58	-58
Long. Pm (SUS)	62	62	62	62	62	62	62	62
Long. Pl (SUS)	0	0	2	2	2	2	2	2
Long. Q (SUS)	3	-3	49	-49	-3	3	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	125.0	127.0	125.0	127.0	125.0	127.0	125.0	127.0
Pm+Pl (SUS)	126.0	128.0	128.0	130.0	125.0	127.0	127.0	129.0
Pm+Pl+Q (Total)	126.0	128.0	142.0	116.0	119.0	133.0	185.0	71.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	127.00	138.00	Passed
Pm+Pl (SUS)	130.00	207.00	Passed
Pm+Pl+Q (TOTAL)	185.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1	4	1	4	1	4	1	4
Circ. Pl (SUS)	1	1	3	3	2	2	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	1	-1	7	-7	-1	1	9	-9
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1.0	4.0	1.0	4.0	1.0	4.0	1.0	4.0
Pm+Pl (SUS)	4.0	7.0	4.0	7.0	5.0	8.0	3.0	6.0

Pm+Pl+Q (Total)	3.0	8.0	9.0	12.0	6.0	7.0	11.0	13.0
-----------------	-----	-----	-----	------	-----	-----	------	------

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	4.00	138.00	Passed
Pm+Pl (SUS)	8.00	207.00	Passed
Pm+Pl+Q (TOTAL)	13.00	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: D13"

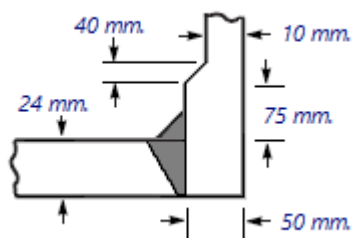
From: 50

Pressure for Reinforcement Calculations	P	22.254	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2400.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4850.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	40.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	201.4480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: D13"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.394 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 22.25 \cdot 1203.0 / (138 \cdot 1.0 - 0.6 \cdot 22.25)$
 $= 19.5906 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 22.25 \cdot 38.1 / (138 \cdot 1.0 + 0.4 \cdot 22.25)$
 $= 0.6105 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 24.1 / 0.5773$
 $= 41.7424 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	124.4000 mm.
Parallel to Vessel Wall, opening length	d	62.2000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	52.5000 mm.

Weld Strength Reduction Factor [fr1]:

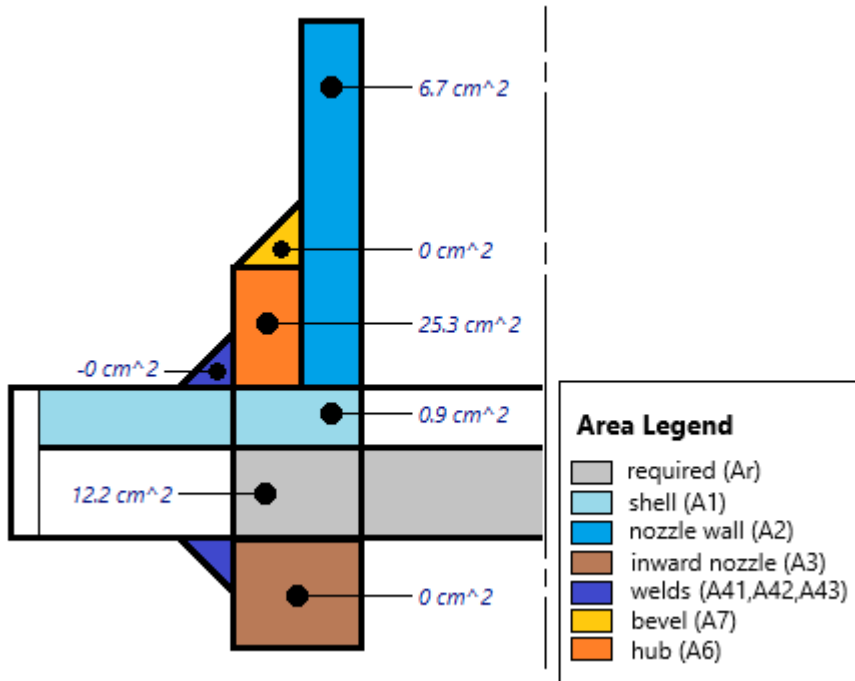
$= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	12.185	NA	NA
Area in Shell	A1	0.877	NA	NA
Area in Nozzle Wall	A2	6.709	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	-0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	25.305	NA	NA
TOTAL AREA AVAILABLE	Atot	32.891	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (62.2 \cdot 19.5906 \cdot 1.0 + 2 \cdot 31.1 \cdot 19.5906 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 12.185 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 62.2 (1.0 \cdot 21.0 - 1.0 \cdot 19.591) - 2 \cdot 31.1$$

$$(1.0 \cdot 21.0 - 1.0 \cdot 19.5906) \cdot (1 - 1.0)$$

$$= 0.877 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 52.5) (7.0 - 0.61) 1.0$$

$$= 6.709 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (Wo^2 - \text{Area Lost}) \cdot fr2 + (Wi\text{-can}/0.707)^2 - \text{Area Lost}) \cdot fr2$$

$$= (10.0^2 - 1.0) \cdot 1.0 + (0.0^2 - 0.0) \cdot 1.0$$

$$= -0.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(52.5, 201.4, 75.0)) \cdot (34.1 - 10.0) \cdot 1.0$$

$$= 25.305 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.6105 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 22.5906 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 22.5906 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 7.8000 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.8, \max(22.5906, 4.5)]$$

$$= 7.8000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.6105, 7.8)$$

$$= 7.8000 \text{ mm.}$$

Available Nozzle Neck Thickness = 10.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 52.9,	Allowable	: 138.0 N./mm2	Passed
Expansion	: 0.0,	Allowable	: 292.1 N./mm2	Passed
Occasional	: 4.4,	Allowable	: 183.5 N./mm2	Passed
Shear	: 38.4,	Allowable	: 96.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 10.0, tr = 0.61, c = 3.0 mm., E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.087, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 22.25/102.00 = 0.218$$

Weld Size Calculations, Description: D13"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

**Results Per UW-16.1:**

Required Thickness Actual Thickness
Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (12.1854 - 0.8766 + 2 \cdot 31.1 \cdot 1.0 \cdot (1.0 \cdot 21.0 - 19.5906)) \cdot 138)$$

$$= 17146.66 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$$

$$= (32.014 + -0.0 - 0.0 \cdot 1.0) \cdot 138$$

$$= 45048.55 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$$

$$= (32.014 + 0.0 + -0.0 + (13.062)) \cdot 138$$

$$= 63428.78 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$

$$= (32.014 + 0.0 + -0.0 + 0.0 + (13.062)) \cdot 138$$

$$= 63428.78 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 76.2 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 8253. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 34.6) \cdot (34.1 - 3.0) \cdot 0.7 \cdot 138$$

$$= 33299. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 76.2 \cdot (24.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 26174. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (8253 + 33299) = 41552 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (8253 + 0 + 26174 + 0) = 34427 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (8253 + 26174 + 0) = 34427 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 41552 Kgf, must exceed W = 17146 Kgf or W1 = 45048 Kgf
Path 2-2 = 34427 Kgf, must exceed W = 17146 Kgf or W2 = 63428 Kgf

149-D

Refrigerant receiver
QUEP

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Nozzle Calcs.: D1_3"

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Path 3-3 = 34427 Kgf, must exceed W = 17146 Kgf or W3 = 63428 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.6050 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.0530 mm.

Input Echo, WRC297 Item 1, Description: D13" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2400.000	mm.
Shell Thickness	24.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	156.200	mm.
Nozzle Thickness	50.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:*External Forces and Moments in WRC 107/537 Convention:**These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	22.25	bars
Radial Load	P	-611.87	Kgf
Circumferential Shear	Vc	407.91	Kgf
Longitudinal Shear	Vl	407.91	Kgf
Circumferential Moment	Mc	91.34	Kg-m.
Longitudinal Moment	Ml	91.34	Kg-m.
Torsional Moment	Mt	130.48	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_n (3.124) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 156.2 + 2 \cdot 1.65 \cdot \sqrt{1213.5(24.0 - 3.0)}$$

$$= 682.997 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2427.000 mm.
Nozzle Outside Diameter	(d) =	156.200 mm.

149-D

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Nozzle Calcs.: D1_3"

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Vessel Thickness used (T) = 21.000 mm.
 Nozzle Thickness used (t) = 47.000 mm.
 T / t = 0.447
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.692$$

Nr/P = 0.124
 Mr/P = 0.278
 M0/P = 0.082
 N0/P = 0.151
 MrD/Mc = 0.312
 NrDL/Mc = 0.079
 M0d/Mc = 0.098
 N0DL/Mc = 0.021
 MrD/M1 = 0.241
 NrDL/M1 = 0.088
 M0D/M1 = 0.077
 N0DL/M1 = 0.086

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	126	128	126	128
Outplane Stress Summary	127	128	142	119

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	22	-22	22	-22
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-18	18	18	-18
Inplane Pressure Stress	63	63	63	63
Inplane Stress Summary	67	59	107	24

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	128	128	142	119

149-D

Refrigerant receiver
QUEP

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Nozzle Calcs.: D1_3"

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HEXAGON

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-7	7	7	-7
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	63	63	63
Outplane Stress Summary	63	65	79	51

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	22	-22	22	-22
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-24	24	24	-24
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	126	128	126	128
Inplane Stress Summary	125	130	176	84

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	125	130	176	84

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	3	0	3
Hoop Stress Summary	1	4	4	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom

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Nozzle Calcs.: D1_3"

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pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-1	-2	9	-3

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	3	6	9	10

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	3	0	3
Hoop Stress Summary	2	5	3	5

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-4	4	4	-4
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-2	0	11	-4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	1	1

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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	5	6	11	10

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	126	128	126	128	126	128	126	128
Circ. Pl (SUS)	1	1	3	3	0	0	2	2
Circ. Q (SUS)	0	0	12	-12	-2	2	46	-46
Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. Pl (SUS)	0	0	2	2	2	2	2	2
Long. Q (SUS)	4	-4	40	-40	-1	1	13	-13
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	126.0	128.0	126.0	128.0	126.0	128.0	126.0	128.0
Pm+Pl (SUS)	127.0	129.0	129.0	131.0	126.0	128.0	128.0	130.0
Pm+Pl+Q (Total)	127.0	129.0	141.0	119.0	124.0	130.0	174.0	84.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	128.00	138.00	Passed
Pm+Pl (SUS)	131.00	207.00	Passed
Pm+Pl+Q (TOTAL)	174.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	0	3	0	3	0	3	0	3
Circ. Pl (SUS)	1	1	3	3	2	2	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	0	0	6	-6	-1	1	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	0.0	3.0	0.0	3.0	0.0	3.0	0.0	3.0
Pm+Pl (SUS)	3.0	6.0	3.0	6.0	4.0	7.0	2.0	5.0

Pm+Pl+Q (Total)	3.0	6.0	8.0	10.0	5.0	6.0	9.0	10.0
-----------------	-----	-----	-----	------	-----	-----	-----	------

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	138.00	Passed
Pm+Pl (SUS)	7.00	207.00	Passed
Pm+Pl+Q (TOTAL)	10.00	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N410"

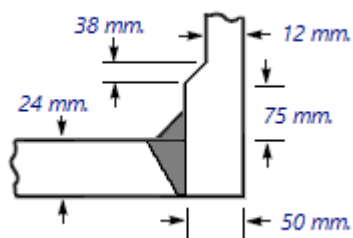
From: 50

Pressure for Reinforcement Calculations	P	22.254	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2400.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		6500.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	38.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	201.4480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	150.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N410"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.000 in.
Actual Thickness Used in Calculation	0.472 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $22.25 \cdot 1203.0 / (138 \cdot 1.0 - 0.6 \cdot 22.25)$

= 19.5906 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $22.25 \cdot 127.0 / (138 \cdot 1.0 + 0.4 \cdot 22.25)$

= 2.0350 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

= (Hub Thickness - Neck Thickness) / tan(30)

= 38.0/0.5773

= 65.8180 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	472.0000	mm.
--	----	----------	-----

Parallel to Vessel Wall, opening length	d	236.0000	mm.
---	---	----------	-----

Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	52.5000	mm.
---	------	---------	-----

Normal to Vessel Wall, Inward		15.0000	mm.
-------------------------------	--	---------	-----

Weld Strength Reduction Factor [fr1]:

= min(1, S_n / S_v)

= min(1, $138.0 / 138.0$)

= 1.000

Weld Strength Reduction Factor [fr2]:

= min(1, S_n / S_v)

= min(1, $138.0 / 138.0$)

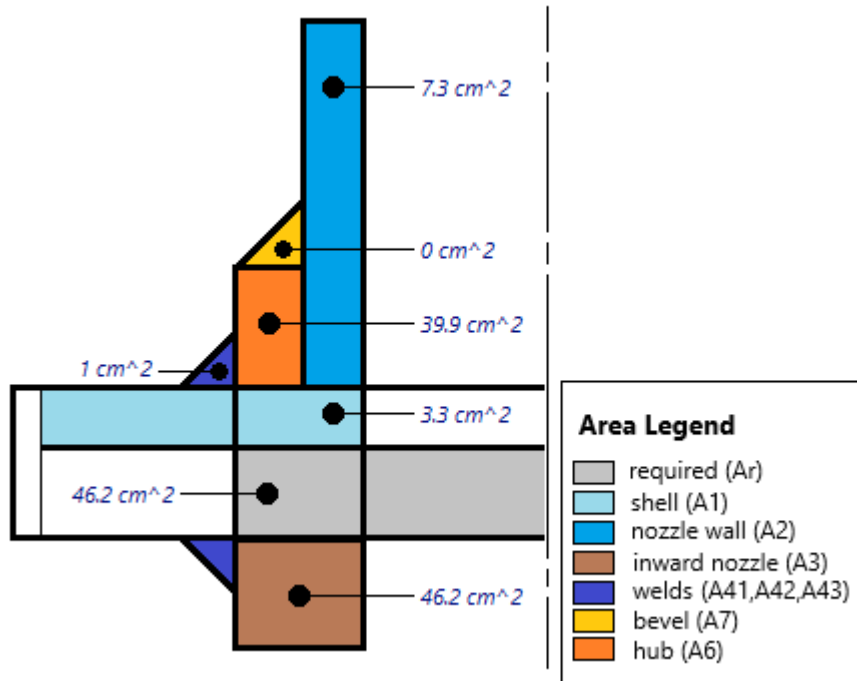
= 1.000

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(1.0, 1.0)

= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	46.234	NA	NA
Area in Shell	A1	3.326	NA	NA
Area in Nozzle Wall	A2	7.313	NA	NA
Area in Inward Nozzle	A3	46.200	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	39.900	NA	NA
TOTAL AREA AVAILABLE	Atot	97.739	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (236.0 \cdot 19.5906 \cdot 1.0 + 2 \cdot 47.0 \cdot 19.5906 \cdot 1.0 \cdot (1 - 1.0))$
 $= 46.234 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 236.0 (1.0 \cdot 21.0 - 1.0 \cdot 19.591) - 2 \cdot 47.0$
 $(1.0 \cdot 21.0 - 1.0 \cdot 19.5906) \cdot (1 - 1.0)$
 $= 3.326 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 52.5) (9.0 - 2.03) 1.0$
 $= 7.313 \text{ cm}^2$

Area Available in Inward Nozzle [A3]:
 $= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$

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Nozzle Calcs.: N4_10"

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$$= 2 \cdot 44.0 \cdot (52.5) \cdot 1.0$$

$$= 46.200 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{hp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(52.5, 201.4, 75.0)) \cdot (50.0 - 12.0) \cdot 1.0$$

$$= 39.900 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.0350 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 22.5906 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 22.5906 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.1026 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.103, \max(22.5906, 4.5)]$$

$$= 11.1026 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.035, 11.1026)$$

$$= 11.1026 \text{ mm.}$$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 54.3,	Allowable	: 138.0 N./mm2	Passed
Expansion	: 0.0,	Allowable	: 290.7 N./mm2	Passed
Occasional	: 14.1,	Allowable	: 183.5 N./mm2	Passed
Shear	: 29.2,	Allowable	: 96.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, Curve: B**

Govrn. thk, tg = 12.0, tr = 2.035, c = 3.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.226, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $22.25/51.00 = 0.436$

Weld Size Calculations, Description: N410"

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (46.2338 - 3.3262 + 2 \cdot 47.0 \cdot 1.0 \cdot (1.0 \cdot 21.0 - 19.5906)) \cdot 138)$
 $= 62241.91 \text{ Kg}$

For hub type nozzles, A₂ includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (47.2133 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 67843.43 \text{ Kg}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (47.2133 + 46.2 + 1. + (19.74)) \cdot 138$
 $= 160631.06 \text{ Kg}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (47.2133 + 46.2 + 1. + 0.0 + (19.74)) \cdot 138$
 $= 160631.06 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 254.0 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 27510. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 122.5) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 178169. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 254.0 \cdot (24.0 - 3.0) \cdot 0.74 \cdot 138$
 $= 87248. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (27510 + 178169) = 205679 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (27510 + 0 + 87248 + 0) = 114758 \text{ Kg}$

Failure Path [Path 3-3]:

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$$= (Sonw + Tngw + Sinw)$$

$$= (27510 + 87248 + 0) = 114758 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 205679 Kgf, must exceed W = 62241 Kgf or W1 = 67843 Kgf
 Path 2-2 = 114757 Kgf, must exceed W = 62241 Kgf or W2 = 160631 Kgf
 Path 3-3 = 114757 Kgf, must exceed W = 62241 Kgf or W3 = 160631 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 6.7394 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 375.4480 mm.

Input Echo, WRC297 Item 1, Description: N410" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2400.000	mm.
Shell Thickness	24.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	330.000	mm.
Nozzle Thickness	50.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	22.25	bars
Radial Load	P	-2294.51	Kgf
Circumferential Shear	Vc	1529.68	Kgf
Longitudinal Shear	Vl	1529.68	Kgf
Circumferential Moment	Mc	1284.44	Kg-m.
Longitudinal Moment	Ml	1284.44	Kg-m.
Torsional Moment	Mt	1834.90	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (6.600) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 330.0 + 2 \cdot 1.65 \cdot \sqrt{(1213.5(24.0 - 3.0))}$$

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= 856.797 mm.

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2427.000 mm.
 Nozzle Outside Diameter (d) = 330.000 mm.
 Vessel Thickness used (T) = 21.000 mm.
 Nozzle Thickness used (t) = 47.000 mm.
 T / t = 0.447
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.462$$

Nr/P = 0.083
 Mr/P = 0.176
 M0/P = 0.051
 N0/P = 0.110
 MrD/Mc = 0.298
 NrDL/Mc = 0.095
 M0d/Mc = 0.092
 N0DL/Mc = 0.058
 MrD/Ml = 0.147
 NrDL/Ml = 0.081
 M0D/Ml = 0.047
 N0DL/Ml = 0.128

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-11	-11	11	11
Outplane Bending (ML)	-24	24	24	-24
Normal Pressure Stress	126	128	126	128
Outplane Stress Summary	112	131	183	105

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	54	-54	54	-54
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-76	76	76	-76
Inplane Pressure Stress	63	63	63	63
Inplane Stress Summary	37	82	205	-56

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	6	6	3	3

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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	112	132	205	161
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-47	47	47	-47
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	63	63	63

Outplane Stress Summary	31	95	137	10
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	54	-54	54	-54
Inplane Membrane (Mc)	-8	-8	8	8
Inplane Bending (Mc)	-154	154	154	-154
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	126	128	126	128

Inplane Stress Summary	21	225	347	-67
------------------------	----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	3	3	6	6
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	32	225	347	79
------------------------	----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-11	-11	11	11
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	8	5	8

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Hoop Stress Summary	0	2	22	24
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	8	-8	8	-8
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-4	-4	4	4
Axial Bending (ML)	-11	11	11	-11
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-4	1	27	-11
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	2	2	1	1
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	7	5	27	36
------------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	8	5	8

Hoop Stress Summary	6	8	16	18
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	8	-8	8	-8
Axial Membrane (Mc)	-4	-4	4	4
Axial Bending (Mc)	-26	26	26	-26
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-19	16	42	-26
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	1	1	2	2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	25	16	42	45

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	126	128	126	128	126	128	126	128
Circ. Pl (SUS)	-6	-6	16	16	-4	-4	12	12
Circ. Q (SUS)	-9	9	39	-39	-100	100	208	-208
Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. Pl (SUS)	-3	-3	11	11	0	0	10	10
Long. Q (SUS)	-22	22	130	-130	-32	32	62	-62
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	126.0	128.0	126.0	128.0	126.0	128.0	126.0	128.0
Pm+Pl (SUS)	120.0	122.0	142.0	144.0	122.0	124.0	138.0	140.0
Pm+Pl+Q (Total)	111.5	131.7	204.7	161.2	32.5	224.1	346.2	79.9

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	128.00	138.00	Passed
Pm+Pl (SUS)	144.01	207.00	Passed
Pm+Pl+Q (TOTAL)	346.17	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	8	5	8	5	8	5	8
Circ. Pl (SUS)	-6	-6	16	16	0	0	10	10
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-4	-4	4	4	-4	-4	4	4
Long. Q (SUS)	-3	3	19	-19	-18	18	34	-34

Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	5.0	8.0	5.0	8.0	5.0	8.0	5.0	8.0
Pm+Pl (SUS)	2.0	4.0	21.0	24.0	7.0	10.0	15.0	18.0
Pm+Pl+Q (Total)	5.8	4.1	25.8	37.2	25.3	16.5	40.2	46.2

Nozzle Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.00	138.00	Passed
Pm+Pl (SUS)	24.00	207.00	Passed
Pm+Pl+Q (TOTAL)	46.17	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: Scrubber40"

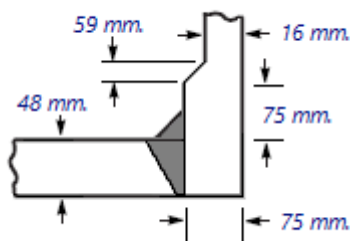
From: 50

Pressure for Reinforcement Calculations	P	22.115	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	2400.00	mm.
Shell Finished (Minimum) Thickness	t	48.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		5300.00	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		40.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm.
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	59.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		75.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Scrubber40"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

149-D

Refrigerant receiver

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Actual Inside Diameter Used in Calculation 40.000 in.
 Actual Thickness Used in Calculation 0.630 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 22.11 \cdot 1203.0 / (138 \cdot 1.0 - 0.6 \cdot 22.11)$
 $= 19.4672$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 22.11 \cdot 511.0 / (138 \cdot 1.0 - 0.6 \cdot 22.11)$
 $= 8.2689$ mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 59.0 / 0.5773$
 $= 102.1910$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

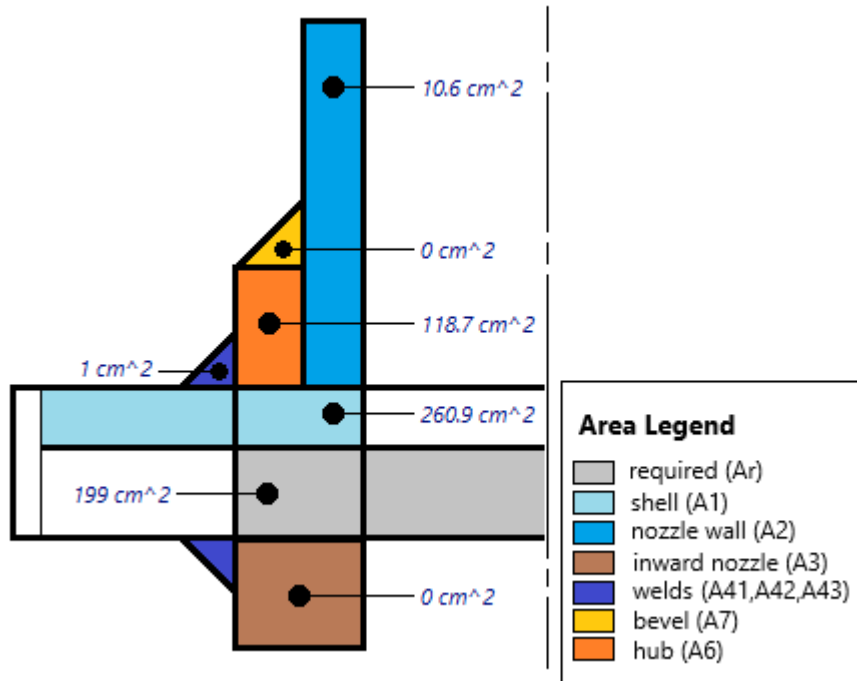
UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	2044.0000	mm.
Parallel to Vessel Wall, opening length	d	1022.0000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	112.5000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	198.955	NA	NA
Area in Shell	A1	260.945	NA	NA
Area in Nozzle Wall	A2	10.645	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	118.688	NA	NA
TOTAL AREA AVAILABLE	Atot	391.277	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (1022.0 \cdot 19.4672 \cdot 1.0 + 2 \cdot 72.0 \cdot 19.4672 \cdot 1.0 \cdot (1 - 1.0))$
 $= 198.955 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 1022.0(1.0 \cdot 45.0 - 1.0 \cdot 19.467) - 2 \cdot 72.0$
 $(1.0 \cdot 45.0 - 1.0 \cdot 19.4672) \cdot (1 - 1.0)$
 $= 260.945 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 112.5)(13.0 - 8.27)1.0$
 $= 10.645 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$

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$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(112.5, 300.0, 75.0)) \cdot (75.0 - 16.0) \cdot 1.0$$

$$= 118.688 \text{ cm}^2 \text{ Includes Bevel Area of } 30.188 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 11.2689 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 22.4672 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 22.4672 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(22.4672, 4.5)]$$

$$= 11.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(11.2689, 11.3312)$$

$$= 11.3312 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	171.3,	Allowable	:	138.0 N./mm2	Failed
Expansion	:	0.0,	Allowable	:	207.0 N./mm2	Passed
Occasional	:	42.9,	Allowable	:	183.5 N./mm2	Passed
Shear	:	12.8,	Allowable	:	96.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :**

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -76 °C

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

$$\text{Height value from sketch (e-1) [te]:}$$

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 59.0/0.5773$$

$$= 102.1910 \text{ mm.}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1533.0000 mm.
Parallel to Vessel Wall, opening length	d	766.5000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	112.5000 mm.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm2)

AREA AVAILABLE, A1 to A6	No Pad	With Pad
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Area Required [Internal Pressure]	AR	132.643	132.643
Area Available in Shell	A1	130.472	130.472
Area Available in Nozzle Wall	A2	10.645	10.645
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	1.000	1.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	118.688	118.688
TOTAL AREA AVAILABLE	Atot	260.805	260.805

The area available without a pad is Sufficient.

The following Calculation is performed using the internal design pressure (P) in the corroded condition.

Warning: The flange could not be considered in the moment of inertia calculations.

Area/Inertia Contributions from Fig.1-7-2 (for Bending stress only):

	Area	Distance	Inertia	d	I+Ad ²
Nozzle	32.8900	126.5000	1754.3785	91.241	4492.4204
Hub	61.6550	53.6380	649.3104	18.379	857.5637
Shell	324.0001	22.5000	546.7496	12.759	1074.2260
Sum	418.5451				6424.21

Membrane stress per Appendix 1-7 (1 or 2) Sm	117.78	N./mm ²
Allowable Membrane Stress S	138.00	N./mm ²
Bending stress at the inside of the Nozzle Sb	36.52	N./mm ²
Combined Membrane plus Bending Stress Sm + Sb	154.30	N./mm ²
Allowable Membrane + Bending Stress 1.5 * S	207.00	N./mm ²
M.A.W.P for the Appendix 1-7 Stress calc	25.91	bars

Area contributions from Fig. 1-7-1 for Membrane stress per 1-7(b)(4):

Area from Nozzle =	16.513 cm ²
Area from Shell =	105.676 cm ²
Total Area, As =	183.600 cm ²

Membrane Stress Calc for Large Nozzles per 1-7(b)(4) [Sm]:

$$= P [R (R_n + t_n + \sqrt{(R_m \cdot t)}) + R_n (t + \sqrt{(R_m \cdot t_n)})] / A_s$$

$$= 22.115 [1203.0 (511.0 + 13.0 + 234.835) + 511.0 (127.021)] / 183.6$$

$$= 117.783 \text{ N./mm}^2$$

The area As was computed per figure 1-7-1 for the Sm calculation.

Moment [M]:

$$= (R_n^3 / 6 + R \cdot R_n \cdot e) P$$

$$= (511.0^3 / 6 + 1203.0 \cdot 511.0 \cdot 12.759) 22.115$$

$$= 6784.294 \text{ Kg-m.}$$

Bending Stress Calc for Large Nozzles per 1-7(b)(4)(5) [Sb]:

$$= (M (e + t/2)) / I$$

$$= (6784.294 (12.759 + 45.0/2)) / 6424.21$$

$$= 36.516 \text{ N./mm}^2$$

Weld Size Calculations, Description: Scrubber40"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:



$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (198.9551 - 260.9449 + 2 \cdot 72.0 \cdot 1.0 \cdot (1.0 \cdot 45.0 - 19.4672)) \cdot 138) \\
 &= \max(0, -35492.04) \text{ Kg}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (129.3324 + 1. - 0.0 \cdot 1.0) \cdot 138 \\
 &= 183397.56 \text{ Kg}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (129.3324 + 0.0 + 1. + (64.8)) \cdot 138 \\
 &= 274581.09 \text{ Kg}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (129.3324 + 0.0 + 1. + 0.0 + (64.8)) \cdot 138 \\
 &= 274581.09 \text{ Kg}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 1166.0 \cdot 10.0 \cdot 0.49 \cdot 138 \\
 &= 126286. \text{ Kg}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 547.0) \cdot (75.0 - 3.0) \cdot 0.7 \cdot 138 \\
 &= 1218760. \text{ Kg}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 1166.0 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138 \\
 &= 457733. \text{ Kg}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (126286 + 1218760) = 1345046 \text{ Kg}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (126286 + 0 + 457733 + 0) = 584019 \text{ Kg}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (126286 + 457733 + 0) = 584019 \text{ Kg}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1345046 Kg, must exceed W = 0 Kg	or W1 = 183397 Kg
Path 2-2 = 584018 Kg, must exceed W = 0 Kg	or W2 = 274581 Kg
Path 3-3 = 584018 Kg, must exceed W = 0 Kg	or W3 = 274581 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 24 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 151.1383 mm.

149-D

Refrigerant receiver
QUEP

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Nozzle Calcs.: Scrubber_40"

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The Cut Length for this Nozzle is, Drop + Ho + H + T : 499.1382 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 944.82 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 1022.00 mm.

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: Scrubber40" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2400.000	mm.
Shell Thickness	48.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm2
Vessel Hot S.I. Allowable	Smh 138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	1016.000	mm.
Nozzle Thickness	75.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	22.11	bars
Radial Load	P	-7648.38	Kgf
Circumferential Shear	Vc	5098.92	Kgf
Longitudinal Shear	Vl	5098.92	Kgf
Circumferential Moment	Mc	14271.47	Kg-m.
Longitudinal Moment	Ml	142714.75	Kg-m.
Torsional Moment	Mt	20387.82	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 1166.0 + 2 \cdot 1.65 \cdot \sqrt{(1225.5(48.0 - 3.0))}$$

$$= 1940.956 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2451.000	mm.
Nozzle Outside Diameter	(d) =	1166.000	mm.
Vessel Thickness used	(T) =	45.000	mm.
Nozzle Thickness used	(t) =	72.000	mm.

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Nozzle Calcs.: Scrubber_40"

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T / t = 0.625
d / t = 16.194

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 3.511$$

Nr/P = 0.054
Mr/P = 0.092
M0/P = 0.030
N0/P = 0.093
MrD/Mc = 0.251
NrDL/Mc = 0.089
M0d/Mc = 0.080
N0DL/Mc = 0.093
MrD/Ml = 0.091
NrDL/Ml = 0.045
M0D/Ml = 0.028
N0DL/Ml = 0.114

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-67	-67	67	67
Outplane Bending (ML)	-99	99	99	-99
Normal Pressure Stress	58	60	58	60
Outplane Stress Summary	-99	89	235	24

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	20	-20	20	-20
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-26	-26	26	26
Inplane Bending (ML)	-322	322	322	-322
Inplane Pressure Stress	29	29	29	29
Inplane Stress Summary	-297	306	399	-284

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	297	306	399	309

Vessel Stresses (N./mm²):

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Nozzle Calcs.: Scrubber_40"

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CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-28	28	28	-28
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	29	29	29	29
Outplane Stress Summary	5	48	73	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	20	-20	20	-20
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-89	89	89	-89
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	58	60	58	60
Inplane Stress Summary	-14	126	175	-42

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	19	126	175	45

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-67	-67	67	67
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	20	17	20
Hoop Stress Summary	-46	-43	88	91

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

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Nozzle Calcs.: Scrubber_40"

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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-21	-21	21	21
Axial Bending (ML)	-94	94	94	-94
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-102	76	131	-69
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	102	120	131	160
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	20	17	20

Hoop Stress Summary	15	18	26	29
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-28	28	28	-28
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-16	30	45	-23
----------------------	-----	----	----	-----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	0	0	1	1
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

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Nozzle Calcs.: Scrubber_40"

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CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	31	30	45	52

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	58	60	58	60	58	60	58	60
Circ. Pl (SUS)	-64	-64	70	70	-3	-3	7	7
Circ. Q (SUS)	-93	93	105	-105	-69	69	109	-109
Long. Pm (SUS)	29	29	29	29	29	29	29	29
Long. Pl (SUS)	-24	-24	28	28	-2	-2	8	8
Long. Q (SUS)	-302	302	342	-342	-22	22	34	-34
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	58.0	60.0	58.0	60.0	58.0	60.0	58.0	60.0
Pm+Pl (SUS)	11.0	9.0	128.0	130.0	55.0	57.0	65.0	67.0
Pm+Pl+Q (Total)	297.0	307.0	399.0	310.0	19.4	126.1	174.0	45.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	60.00	138.00	Passed
Pm+Pl (SUS)	130.00	207.00	Passed
Pm+Pl+Q (TOTAL)	399.02	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	17	20	17	20	17	20	17	20
Circ. Pl (SUS)	-64	-64	70	70	-2	-2	8	8
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-21	-21	21	21	-2	-2	2	2
Long. Q (SUS)	-89	89	99	-99	-23	23	33	-33
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	17.0	20.0	17.0	20.0	17.0	20.0	17.0	20.0
Pm+Pl (SUS)	47.0	44.0	87.0	90.0	15.0	18.0	25.0	28.0
Pm+Pl+Q (Total)	102.0	120.0	128.0	160.0	32.1	29.1	43.1	51.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	138.00	Passed
Pm+Pl (SUS)	90.00	207.00	Passed
Pm+Pl+Q (TOTAL)	160.01	414.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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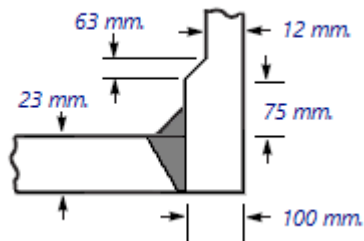
**Input, Nozzle Desc: M224"****From: 60**

Pressure for Reinforcement Calculations	P	22.184	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	2400.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	23.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	+IT
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	63.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		23.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: M224"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 22.18 \cdot 2406.0 \cdot 0.8979 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 22.18)$
 $= 17.3931 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 22.18 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 22.18)$
 $= 4.8688 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 88.0 / 0.5773$
 $= 152.4205 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1183.2001 mm.
Parallel to Vessel Wall, opening length	d	591.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	50.0000 mm.

Weld Strength Reduction Factor [fr1]:

$= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

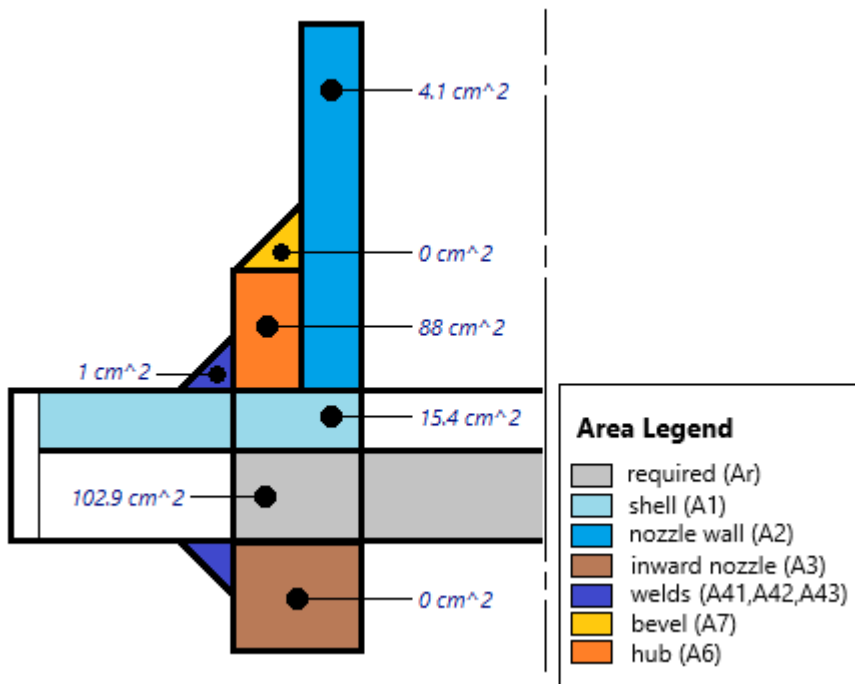
$= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1.0, 1.0)$$

$$= 1.000$$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	102.898	NA	NA
Area in Shell	A1	15.422	NA	NA
Area in Nozzle Wall	A2	4.131	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	88.000	NA	NA
TOTAL AREA AVAILABLE	Atot	108.554	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (591.6 \cdot 17.3931 \cdot 1.0 + 2 \cdot 97.0 \cdot 17.3931 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 102.898 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 591.6(1.0 \cdot 20.0 - 1.0 \cdot 17.393) - 2 \cdot 97.0$$

$$(1.0 \cdot 20.0 - 1.0 \cdot 17.3931) \cdot (1 - 1.0)$$

$$= 15.422 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn) fr2$$

$$= (2 \cdot 50.0)(9.0 - 4.87)1.0$$

$$= 4.131 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{hp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(50.0, 250.0, 75.0)) \cdot (100.0 - 12.0) \cdot 1.0$$

$$= 88.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 7.8688 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 7.8688 \text{ mm}.$$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 4.869$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.541$, Temp. Reduction = $28 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-48 $^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	-29 $^\circ\text{C}$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : $-40 \text{ }^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $22.18/51.00 = 0.435$

Weld Size Calculations, Description: M224"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - t_r)) S_v)$$

$$= \max(0, (102.8975 - 15.4225 + 2 \cdot 97.0 \cdot 1.0 \cdot (1.0 \cdot 20.0 - 17.3931)) 138)$$

$$= 130207.27 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (92.1312 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 131049.77 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (92.1312 + 0.0 + 1. + (38.8)) \cdot 138$$

$$= 185647.30 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (92.1312 + 0.0 + 1. + 0.0 + (38.8)) \cdot 138$$

$$= 185647.30 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 66024. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 300.3) \cdot (100.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 901417. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (23.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 199424. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (66024 + 901417) = 967441 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (66024 + 0 + 199424 + 0) = 265448 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (66024 + 199424 + 0) = 265448 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 967440 Kgf, must exceed W = 130207 Kgf or W1 = 131049 Kgf
 Path 2-2 = 265447 Kgf, must exceed W = 130207 Kgf or W2 = 185647 Kgf
 Path 3-3 = 265447 Kgf, must exceed W = 130207 Kgf or W3 = 185647 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 23 bars

The Drop for this Nozzle is : 21.4291 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 294.4291 mm.

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Input, Nozzle Desc: K13"

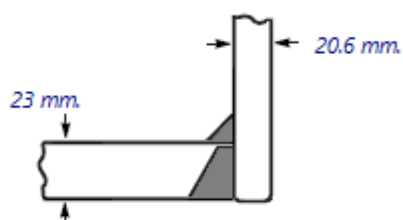
From: 60

Pressure for Reinforcement Calculations	P	22.146	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	2400.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	23.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	670.0000	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.6000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	171.5520	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	23.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

149-D

Refrigerant receiver
QUEP

PV Elite 28 U1

Licensee: SL User

FileName : 149-D_Refrigerant receiver_7 Per

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Nozzle Calcs.: K1_3"

Noz1: 47 12:03pm Sep 16,2026



HEXAGON

Reinforcement CALCULATION, Description: K13"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.811 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 22.15 \cdot 2406.0 \cdot 0.8979 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 22.15)$
 $= 17.3627 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 22.15 \cdot 41.1 / (138 \cdot 1.0 - 0.6 \cdot 22.15)$
 $= 0.6660 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	173.4460 mm.
Parallel to Vessel Wall, opening length	d	86.7230 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	44.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

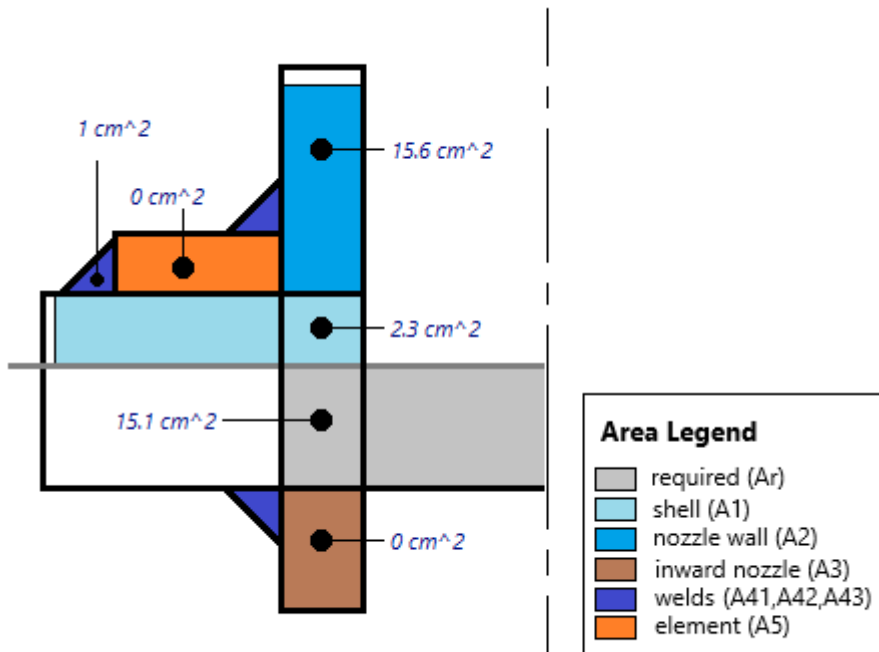


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	15.057	NA	NA
Area in Shell	A1	2.287	NA	NA
Area in Nozzle Wall	A2	15.557	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	18.844	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 71.41 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (86.723 \cdot 17.3627 \cdot 1.0 + 2 \cdot 17.6 \cdot 17.3627 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 15.057 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 86.723(1.0 \cdot 20.0 - 1.0 \cdot 17.363) - 2 \cdot 17.6$$

$$(1.0 \cdot 20.0 - 1.0 \cdot 17.3627) \cdot (1 - 1.0)$$

$$= 2.287 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2/\sin(\alpha3)$$

$$= (2 \cdot 44.0)(17.6 - 0.67)1.0/\sin(73.3)$$

$$= 15.557 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 20.6, tr = 0.666, c = 3.0 mm., E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.038, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 22.15/51.00 = 0.434$$

Weld Size Calculations, Description: K13"

Intermediate Calc. for nozzle/shell Welds Tmin 17.6000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr_1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (15.0574 - 2.2872 + 2 \cdot 17.6 \cdot 1.0 \cdot (1.0 \cdot 20.0 - 17.3627)) \cdot 138)$$

$$= 19276.00 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - Can / 0.707)^2 \cdot fr_2) \cdot Sv$$

$$= (15.5571 + 0.0 + 1.0 - 0.0 \cdot 1.0) \cdot 138$$

$$= 23298.42 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr_1)) \cdot Sv$$

$$= (15.5571 + 0.0 + 1.0 + (7.04)) \cdot 138$$

$$= 33204.78 \text{ Kg}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr_1)) \cdot S$$

$$= (15.5571 + 0.0 + 1.0 + 0.0 + (7.04)) \cdot 138$$

$$= 33204.78 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 123.8599 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 13415. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 52.6457) \cdot (20.6 - 3.0) \cdot 0.7 \cdot 138$$

$$= 28673. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 123.8599 \cdot (23.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 40519. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (13415 + 28673) = 42088 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (13415 + 0 + 40519 + 0) = 53934 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (13415 + 40519 + 0) = 53934 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 42087	Kgf, must exceed W = 19275 Kg	or W1 = 23298 Kg
Path 2-2 = 53934	Kgf, must exceed W = 19275 Kg	or W2 = 33204 Kg
Path 3-3 = 53934	Kgf, must exceed W = 19275 Kg	or W3 = 33204 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 23 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K13"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.811 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 22.15 \cdot 2406.0 \cdot 0.8979 / (2 \cdot 138.0 \cdot 1.0 \cdot 0.2 \cdot 22.15)$$

$$= 17.3627 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 22.15 \cdot 41.1 / (138 \cdot 1.0 \cdot 0.6 \cdot 22.15)$$

$$= 0.6660 \text{ mm.}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	164.4000 mm.
Parallel to Vessel Wall, opening length	d	82.2000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	44.0000 mm.

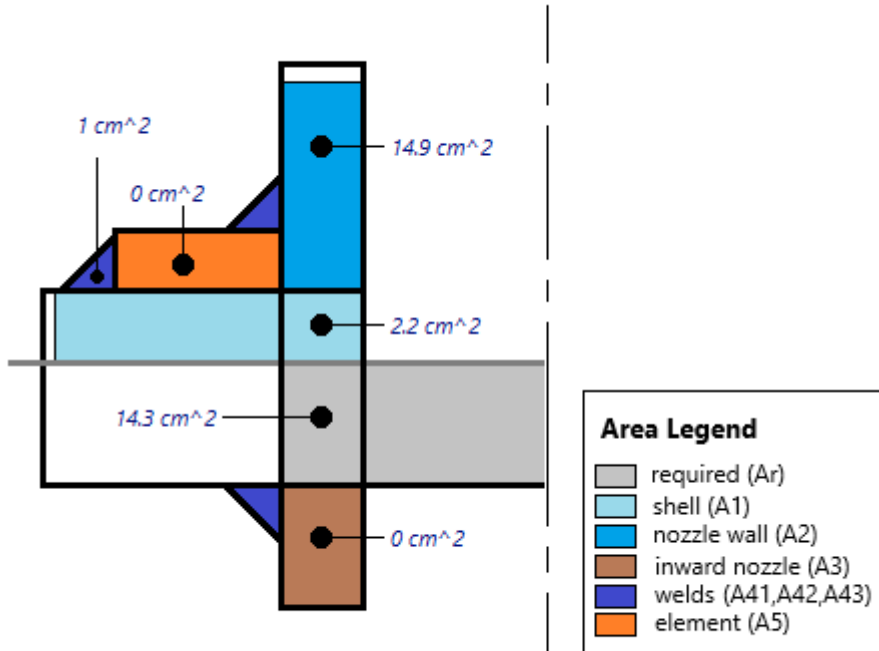


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	14.272	NA	NA
Area in Shell	A1	2.168	NA	NA
Area in Nozzle Wall	A2	14.902	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	18.070	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (82.2 \cdot 17.3627 \cdot 1.0 + 2 \cdot 17.6 \cdot 17.3627 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 14.272 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 82.2(1.0 \cdot 20.0 - 1.0 \cdot 17.363) - 2 \cdot 17.6$$

$$(1.0 \cdot 20.0 - 1.0 \cdot 17.3627) \cdot (1 - 1.0)$$

$$= 2.168 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 44.0)(17.6 - 0.67)1.0$$

$$= 14.902 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:



$$\begin{aligned}
 &= W_o^2 \cdot fr2 + (W_i\text{-can}/0.707)^2 \cdot fr2 \\
 &= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 \\
 &= 1.000 \text{ cm}^2
 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.6660 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 22.2737 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 22.2737 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 8.7300 mm.

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned}
 &= \min[tb3, \max(tb1, tb2)] \\
 &= \min[8.73, \max(22.2737, 4.5)] \\
 &= 8.7300 \text{ mm.}
 \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned}
 &= \max(ta, tb) \\
 &= \max(3.666, 8.73) \\
 &= 8.7300 \text{ mm.}
 \end{aligned}$$

Available Nozzle Neck Thickness = 20.6000 mm. -> OK

Weld Size Calculations, Description: K13"

Intermediate Calc. for nozzle/shell welds Tmin 17.6000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t \cdot tr)) \cdot Sv) \\
 &= \max(0, (14.2721 - 2.1679 + 2 \cdot 17.6 \cdot 1.0 \cdot (1.0 \cdot 20.0 - 17.3627)) \cdot 138) \\
 &= 18338.80 \text{ Kgf}
 \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A5+A4-(W_i\text{-Can}/.707)^2 \cdot fr2) \cdot Sv \\
 &= (14.9019 + 0.0 + 1. - 0.0 \cdot 1.0) \cdot 138 \\
 &= 22376.42 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\
 &= (14.9019 + 0.0 + 1. + (7.04)) \cdot 138 \\
 &= 32282.78 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S \\
 &= (14.9019 + 0.0 + 1. + 0.0 + (7.04)) \cdot 138 \\
 &= 32282.78 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 117.4 \cdot 10.0 \cdot 0.49 \cdot 138 \\
 &= 12715. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 49.9) \cdot (20.6 - 3.0) \cdot 0.7 \cdot 138
 \end{aligned}$$

= 27178. Kgf

Tension, Shell Groove Weld [Tngw]:

= $(\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$

= $(3.1416/2.0) \cdot 117.4 \cdot (23.0 - 3.0) \cdot 0.74 \cdot 138$

= 38406. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= (SONW + SNW) = (12715 + 27178) = 39893 Kgf

Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (12715 + 0 + 38406 + 0) = 51121 Kgf

Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (12715 + 38406 + 0) = 51121 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 39892 Kgf, must exceed W = 18338 Kgf or W1 = 22376 Kgf

Path 2-2 = 51121 Kgf, must exceed W = 18338 Kgf or W2 = 32282 Kgf

Path 3-3 = 51121 Kgf, must exceed W = 18338 Kgf or W3 = 32282 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 23 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 19.9005 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 215.6109 mm.

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Input, Nozzle Desc: K23"

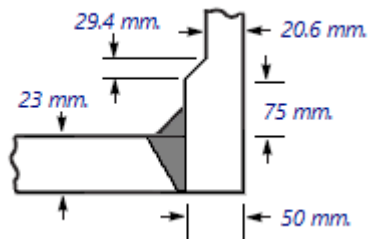
From: 60

Pressure for Reinforcement Calculations	P	22.240	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	2400.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	23.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	960.0000	mm.
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.6000	mm.
Flange Material		SA-105	+IT
Flange Type		N/A	
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	29.4260	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Style of FVC Nozzle		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0250	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	23.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K23"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.811 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 22.24 \cdot 2406.0 \cdot 0.997 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 22.24)$
 $= 19.3559 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 22.24 \cdot 41.1 / (138 \cdot 1.0 - 0.6 \cdot 22.24)$
 $= 0.6689 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 29.4 / 0.5773$
 $= 50.9223 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	197.5455 mm.
Parallel to Vessel Wall, opening length	d	98.7728 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	50.0000 mm.

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n/S_v)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

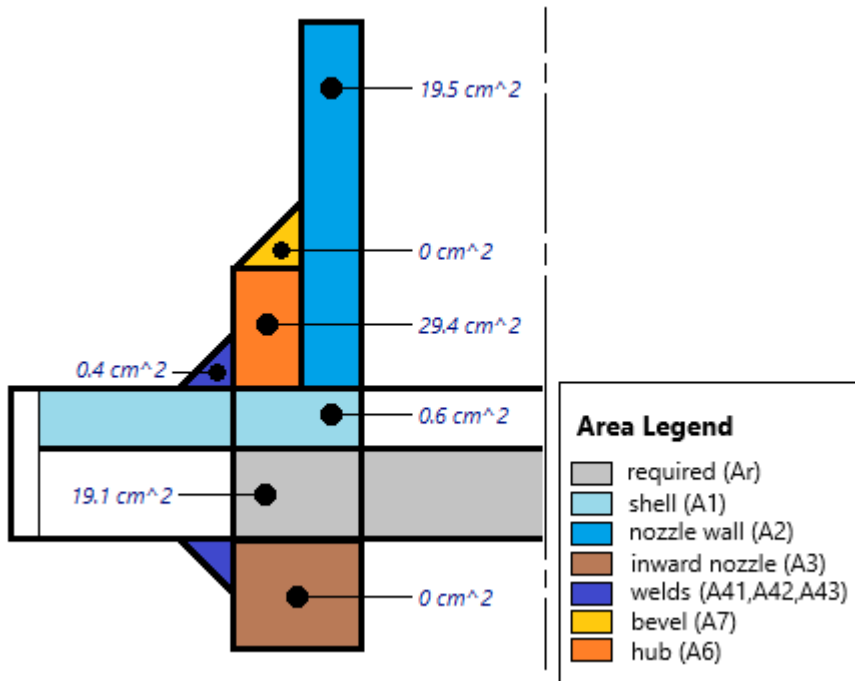
Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n/S_v)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$

= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	19.118	NA	NA
Area in Shell	A1	0.636	NA	NA
Area in Nozzle Wall	A2	19.482	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.420	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	29.400	NA	NA
TOTAL AREA AVAILABLE	Atot	49.939	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 56.33 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (98.7728 \cdot 19.3559 \cdot 1.0 + 2 \cdot 47.0 \cdot 19.3559 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 19.118 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 98.773 (1.0 \cdot 20.0 - 1.0 \cdot 19.356) - 2 \cdot 47.0$$

$$(1.0 \cdot 20.0 - 1.0 \cdot 19.3559) \cdot (1 - 1.0)$$

$$= 0.636 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$$

$$= (2 \cdot 50.0) (17.6 - 0.67) 1.0 / \sin(60.3)$$

$$= 19.482 \text{ cm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= (W_o^2 - \text{Area Lost}) \cdot fr2 + ((W_i - \text{can}/0.707)^2 - \text{Area Lost}) \cdot fr2$
 $= (10.0^2 - 0.5797) \cdot 1.0 + (0.0^2 - 0.0) \cdot 1.0$
 $= 0.420 \text{ cm}^2$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{hp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$
 $= (2 \cdot \min(50.0, 200.0, 75.0)) \cdot (50.0 - 20.6) \cdot 1.0$
 $= 29.400 \text{ cm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -35 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $22.24/51.00 = 0.436$

Weld Size Calculations, Description: K23"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (19.1184 - 0.6362 + 2 \cdot 47.0 \cdot 1.0 \cdot (1.0 \cdot 20.0 - 19.3559)) \cdot 138)$
 $= 26859.21 \text{ Kg}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A4 - (W_i - \text{Can}/.707)^2 \cdot fr2) \cdot Sv$
 $= (48.8825 + 0.4203 - 0.0 \cdot 1.0) \cdot 138$
 $= 69376.55 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (48.8825 + 0.0 + 0.4203 + (18.8)) \cdot 138$
 $= 95831.03 \text{ Kg}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (48.8825 + 0.0 + 0.4203 + 0.0 + (18.8)) \cdot 138$
 $= 95831.03 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 211.7246 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 22931. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 77.6243) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 112900. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 211.7246 \cdot (23.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 69263. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (22931 + 112900) = 135831 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (22931 + 0 + 69263 + 0) = 92195 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (22931 + 69263 + 0) = 92195 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 135831	Kgf, must exceed W = 26859	Kgf	or W1 = 69376	Kgf
Path 2-2 = 92194	Kgf, must exceed W = 26859	Kgf	or W2 = 95831	Kgf
Path 3-3 = 92194	Kgf, must exceed W = 26859	Kgf	or W3 = 95831	Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 23 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K23"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000	in.
Actual Thickness Used in Calculation	0.811	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 22.24 \cdot 2406.0 \cdot 0.997 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 22.24)$$

$$= 19.3559 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 22.24 \cdot 41.1 / (138 \cdot 1.0 - 0.6 \cdot 22.24)$$

$$= 0.6689 \text{ mm.}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (Hub \text{ Thickness} - Neck \text{ Thickness}) / \tan(30)$$

149-D

Refrigerant receiver
QUEP

PV Elite 28 U1 Licensee: SL User

FileName : 149-D_Refrigerant receiver_7 Per

Nozzle Calcs.: K2_3"

Noz1: 48 12:03pm Sep 16,2026

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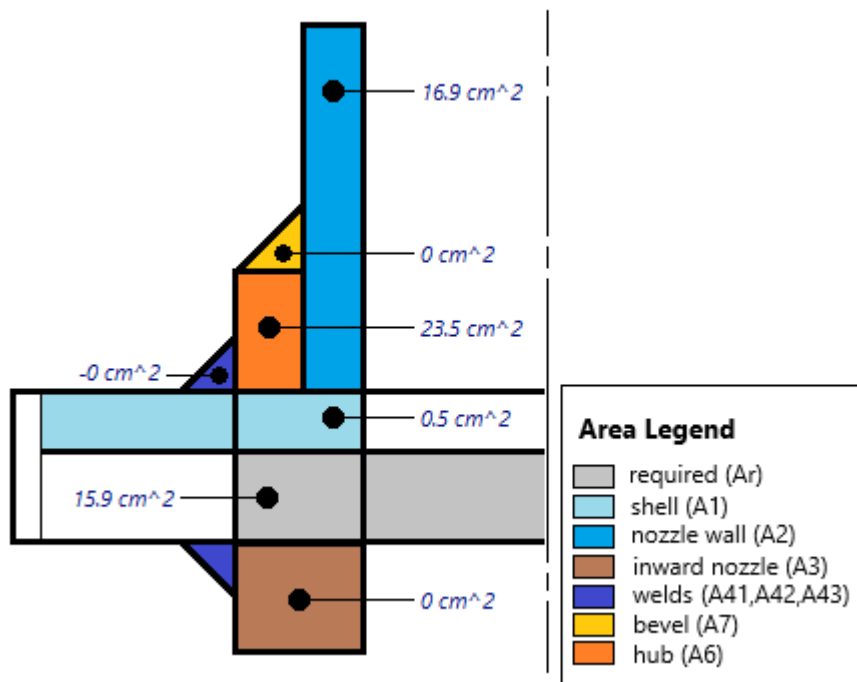
HEXAGON

= 23.5/0.5773
= 40.7032 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 164.4000 mm.
Parallel to Vessel Wall, opening length d 82.2000 mm.
Normal to Vessel Wall (Thickness Limit), no pad Tlnp 50.0000 mm.



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	15.911	NA	NA
Area in Shell	A1	0.529	NA	NA
Area in Nozzle Wall	A2	16.931	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		-0.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	23.500	NA	NA
TOTAL AREA AVAILABLE	Atot	40.961	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (82.2 \cdot 19.3559 \cdot 1.0 + 2 \cdot 41.1 \cdot 19.3559 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 15.911 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 82.2(1.0 \cdot 20.0 - 1.0 \cdot 19.356) - 2 \cdot 41.1$$

149-D

Refrigerant receiver
QUEP

PV Elite 28 U1

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FileName : 149-D_Refrigerant receiver_7 Per

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Nozzle Calcs.: K2_3"

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HEXAGON

$$(1.0 \cdot 20.0 - 1.0 \cdot 19.3559) \cdot (1 - 1.0) \\ = 0.529 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2} \\ = (2 \cdot 50.0) (17.6 - 0.67) 1.0 \\ = 16.931 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (W_o^2 - \text{Area Lost}) \cdot f_{r2} + ((W_i - \text{can}/0.707)^2 - \text{Area Lost}) \cdot f_{r2} \\ = (10.0^2 - 1.0) \cdot 1.0 + (0.0^2 - 0.0) \cdot 1.0 \\ = -0.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2} \\ = (2 \cdot \min(50.0, 200.0, 75.0)) \cdot (44.1 - 20.6) \cdot 1.0 \\ = 23.500 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.6689 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 22.3559 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 22.3559 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 8.7300 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)] \\ = \min[8.73, \max(22.3559, 4.5)] \\ = 8.7300 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb) \\ = \max(3.6689, 8.73) \\ = 8.7300 \text{ mm.}$$

Available Nozzle Neck Thickness = 20.6000 mm. -> OK

Weld Size Calculations, Description: K23"

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) S_v) \\ = \max(0, (15.9106 - 0.5294 + 2 \cdot 41.1 \cdot 1.0 \cdot (1.0 \cdot 20.0 - 19.3559)) 138) \\ = 22388.59 \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can}/0.707)^2 \cdot f_{r2}) \cdot S_v \\ = (40.4311 + -0.0 - 0.0 \cdot 1.0) \cdot 138 \\ = 56892.77 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ = (40.4311 + 0.0 + -0.0 + (16.44)) \cdot 138 \\ = 80026.37 \text{ Kg}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (40.4311 + 0.0 + -0.0 + 0.0 + (16.44)) \cdot 138$$

$$= 80026.37 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 164.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 17806. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 61.65) \cdot (44.1 - 3.0) \cdot 0.7 \cdot 138$$

$$= 78410. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 164.4 \cdot (23.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 53782. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (17806 + 78410) = 96216 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (17806 + 0 + 53782 + 0) = 71587 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (17806 + 53782 + 0) = 71587 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 96216	Kgf, must exceed W = 22388 Kgf	or W1 = 56892 Kgf
Path 2-2 = 71587	Kgf, must exceed W = 22388 Kgf	or W2 = 80026 Kgf
Path 3-3 = 71587	Kgf, must exceed W = 22388 Kgf	or W3 = 80026 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 23 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 42.5453 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 268.1602 mm.

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: